2 - Theory of Flight

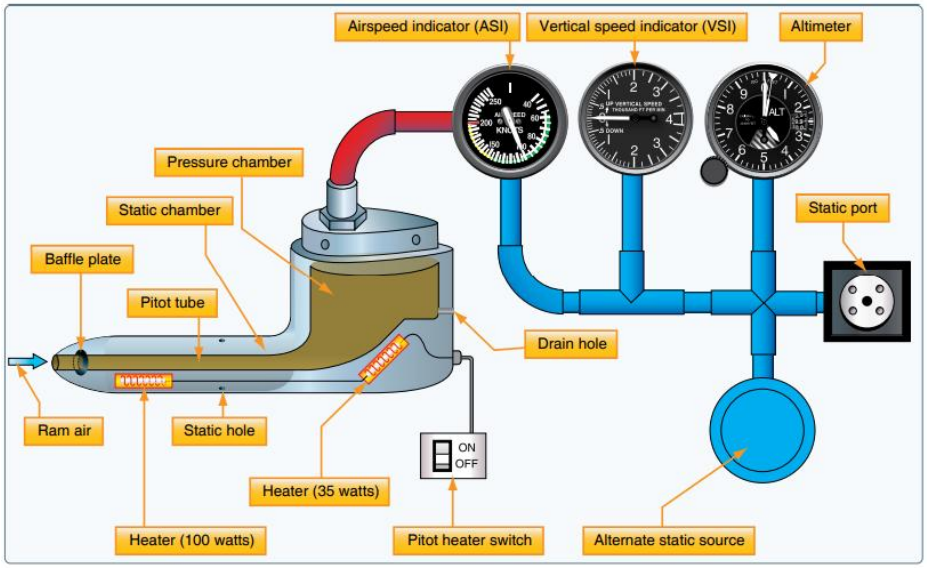
Flight Instruments
https://www.youtube.com/playlist?list=PLJ-0SVdtegU-SS5Kp12Yw70iJbx4ApHhD

2.2 Flight Instruments

2.2.1 Pitot Static Instruments

The pitot-static system measures an aircraft's airspeed, altitude, and vertical speed using air pressure differences. The respective instruments for showing these readings are:

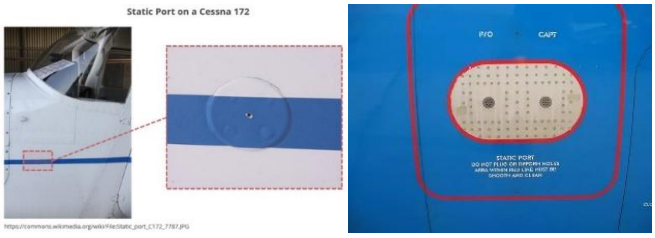
- The Airspeed Indicator (ASI)
- The Altimeter
- The Vertical Speed Indicator (VSI)



Key Definitions

Static Pressure: Static pressure is the ambient (meaning: immediate surroundings of something) air pressure surrounding the aircraft, existing whether the aircraft is moving or stationary.

Measurement: Static pressure is typically sensed by static ports on the aircraft, which are designed to be relatively insensitive to airflow direction.



Dynamic Pressure: Dynamic pressure is the pressure generated by the aircraft's movement through the air. It's the pressure that results when the airflow is brought to rest relative to the aircraft.

Measurement: Dynamic pressure is calculated by subtracting the static pressure from the total pressure measured by the pitot tube.

Example: When a gust of wind hits the aircraft, the dynamic pressure increases.

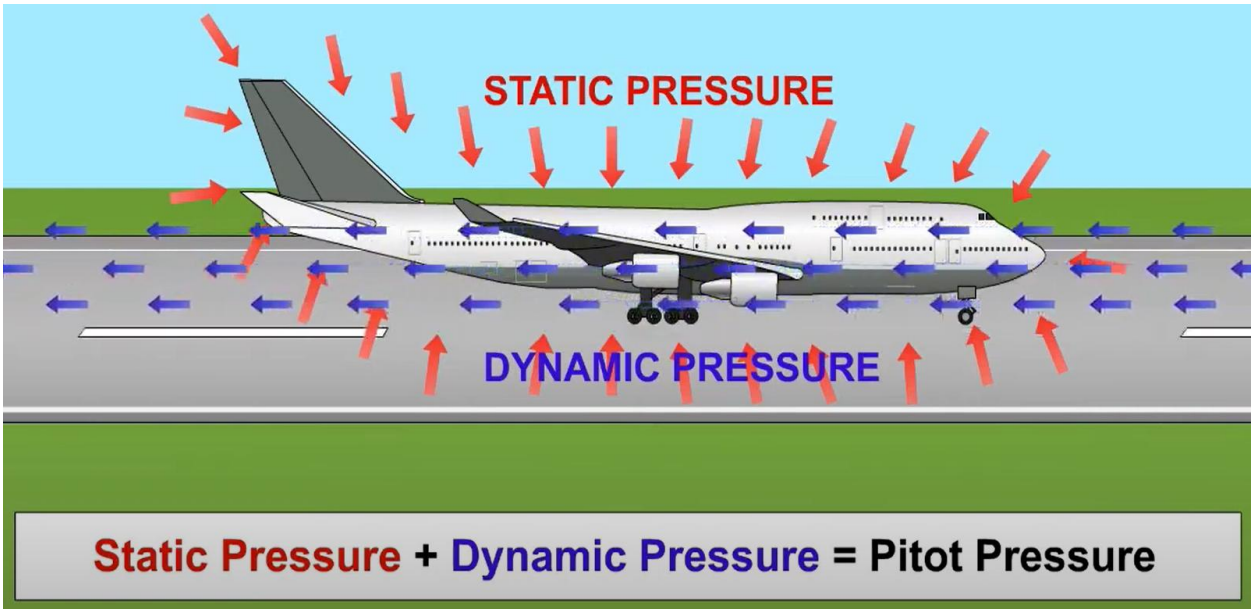
Key Role: Dynamic pressure is a key factor in determining airspeed and lift generation.

Pitot Tube and Total Pressure: The pitot tube is a device that measures total pressure (also known as stagnation pressure).



Total Pressure aka RAM Air Pressure: Total pressure is the sum of static pressure and dynamic pressure

Relationship: The pitot tube captures both static and dynamic pressure, and the dynamic pressure is then calculated by subtracting the static pressure from the total pressure.



The pitot pressure source, usually on the wing's leading edge and clear of slipstream, airflow disturbances, faces the line of flight to capture undisturbed air. In flight, forward motion increases

pressure in the pitot system through dynamic pressure. The airspeed indicator is the only instrument connected to this source.

The airspeed indicator, altimeter, and vertical speed indicator connect to the static pressure source, which vents them to outside barometric pressure as the aircraft changes altitude. Static ports, usually two, are placed on opposite sides of the fuselage to avoid turbulence and ram air effects. Dual vents also help correct pressure errors that can occur during attitude changes, such as in steep turns.

The pitot pressure system is prone to blockage from dirt, water, or ice, causing inaccurate instrument readings. Complete blockage, with drain holes also blocked, traps pressure in the airspeed indicator, can results in incorrect readings; partial blockage or leaks cause the reading to drop to zero. Many modern aircraft have electrically heated pitot tubes to prevent icing. Failure to turn on pitot heat or failure of the electrical element does, however, happen. Ice buildup restricts airflow, causing falling airspeed readings. In IFR conditions, a pilot relying solely on the faulty indicator may reduce angle of attack to maintain indicated speed, risking overspeed beyond the never-exceed limit, potentially leading to structural failure or in-flight breakup.

Helicopters are especially prone to pitot tube freezing in winter when landing or taking off in powder snow, as rotor wash can blow snow into the tube, where it may freeze and block the pressure system.

A partially clogged static pressure system causes instrument errors: during climb, the altimeter, airspeed, and vertical speed indicators under-read and only correct after leveling off; during descent, the airspeed over-reads, the vertical speed under-reads, and the altimeter over-reads.

Before flight, ensure pitot and static sources are clear, and always use pitot heat to prevent blockage of the dynamic pressure source. If in flight you suspect static system blockage due to conflicting instrument readings, open the alternate static source or remove the static line; significant needle movement confirms the blockage.

Static pressure sources are positioned to avoid ram air, but maneuvers like sideslips can let ram air enter, causing instrument errors. On aircraft with a single static source, sideslipping toward the port affects readings, while sideslipping the opposite way does not.

Summary of the above 7 paragraphs:

Pitot-Static System (Memorization Notes)

Pitot Pressure Source

- Located usually on the **wing leading edge**.
- Positioned clear of **slipstream and airflow disturbances**.
- Faces directly into the **line of flight** to capture undisturbed air.
- Aircraft forward motion creates **dynamic pressure** in the pitot system.
- The **airspeed indicator (ASI)** is the **only instrument** connected to the pitot source.

Static Pressure Source

- Supplies **barometric (static) pressure** to:
  - Airspeed Indicator (ASI)
  - Altimeter
  - Vertical Speed Indicator (VSI)
- Static ports vent instruments to outside atmospheric pressure as altitude changes.
- Usually has **two static ports** located on opposite sides of the fuselage.
- Placement minimizes:
  - Turbulence effects
  - Ram air effects
- Dual ports help reduce pressure errors during attitude changes such as:
  - Steep turns

Pitot System Blockage Effects

Complete Pitot Blockage

(Drain hole also blocked)

- Pressure becomes trapped in the ASI.
- Causes incorrect airspeed indications.

Partial Blockage or Leak

- Airspeed indication gradually drops toward **zero**.

Causes of Blockage

- Dirt
- Water
- Ice

Pitot Heat

- Many modern aircraft use **electrically heated pitot tubes**.
- Prevents ice formation and blockage.
- Problems occur when:
  - Pilot forgets to turn pitot heat on.
  - Heating element fails.

Ice Formation Consequences

- Restricts airflow into pitot tube.

- Causes airspeed readings to fall.

IFR Danger from Pitot Icing

- Pilot may trust the falsely low airspeed indication.
- Pilot may lower the **angle of attack** to maintain indicated airspeed.
- Aircraft could accelerate excessively.
- May exceed **VNE (Never Exceed Speed)**.
- Risk:
    - Structural damage
    - Structural failure
    - In-flight breakup

Memory Chain:
Pitot Ice → Low ASI → Nose Down → Overspeed → Possible Breakup

Helicopter Pitot Freezing

- Particularly vulnerable during winter operations.
- Most common during:
    - Takeoff
    - Landing
    - Operations in powder snow
- Rotor wash can blow snow into the pitot tube.
- Snow can freeze and block the pitot system.

Static System Partial Blockage

During Climb

- Altimeter **under-reads**
- Airspeed Indicator **under-reads**
- VSI **under-reads**
- Instruments return to normal after leveling off.

During Descent

- Airspeed Indicator **over-reads**
- VSI **under-reads**
- Altimeter **over-reads**

Pre-Flight and In-Flight Checks

Before Flight

- Ensure:

    - Pitot tube is clear.
    - Static ports are clear.
- Use pitot heat whenever required to prevent blockage.

Suspected Static Blockage in Flight

- Suspect blockage when instrument indications conflict.
- Open the **alternate static source** or remove the static line.
- Significant instrument needle movement confirms blockage.

Ram Air Effects on Static Sources

- Static ports are positioned to avoid ram air contamination.
- Certain maneuvers can still introduce ram air:
    - Sideslips

Aircraft with Single Static Source

- Sideslip **toward** the static port:
    - Causes instrument errors.
- Sideslip **away from** the static port:
    - Little or no effect on instrument readings.

Instruments

- **Pitot only:** ASI
- **Static:** ASI + Altimeter + VSI

Blockages

Situation	Effect
Pitot blocked + drain blocked	Pressure trapped, ASI incorrect
Pitot leak/partial blockage	ASI drops toward zero
Static blocked during climb	ASI ↓, Altimeter ↓, VSI ↓ only correct after leveling off
Static blocked during descent	ASI ↑, Altimeter ↑, VSI ↓

Memory Aids

- **Pitot = Dynamic = Speed**

- **Static = ASI + Altimeter + VSI**
- **Climb + Static Blockage = Everything Under**
- **Descent + Static Blockage = ASI & Altimeter Over, VSI Under**
- **Pitot Ice → Low ASI → Overspeed Risk**

THE ALTIMETER

How an Altimeter Work
https://www.youtube.com/watch?v=L1ml_vlibJc

Instrument used to measure the vertical distance between the aircraft and a reference level.

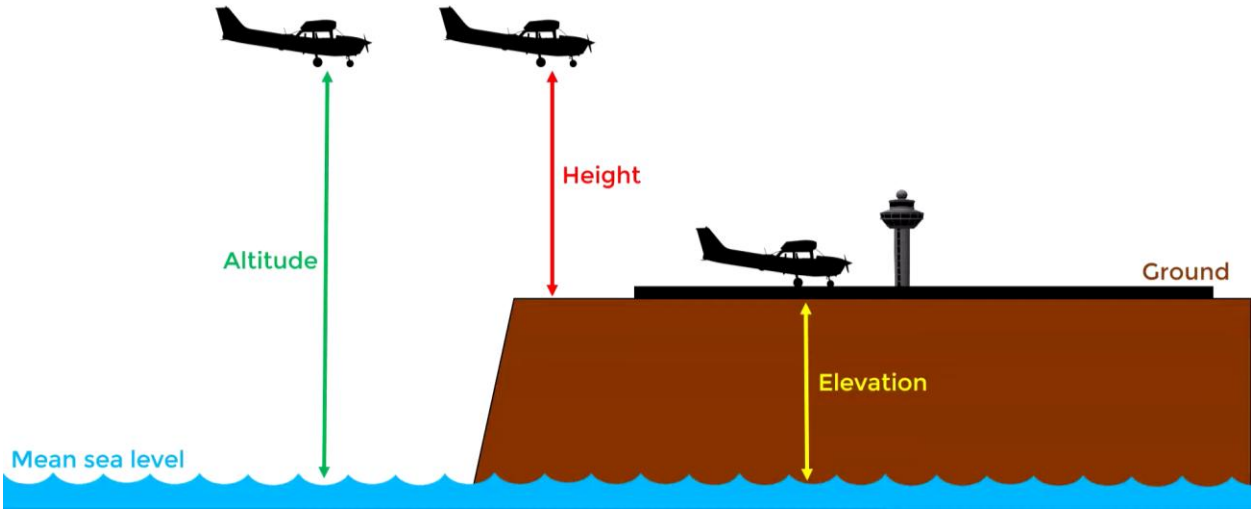


At this point we are intentionally stating "vertical distance" instead of altitude or height as the vertical distance have different names depending on the reference level we will be using.

Altitude: Vertical distance between mean sea level and a point in the air.

Height: Vertical distance between a point on the ground and a point in the air.

Elevation: Vertical distance between mean sea level and a point on the ground.



The altimeter, a type of aneroid barometer (a barometer without liquid), measures atmospheric (static) pressure, connected to the static pressure source through an outlet. **The outlet servs as a vent at the back allows static pressure to enter or exit the case as the aircraft climbs or descends**.

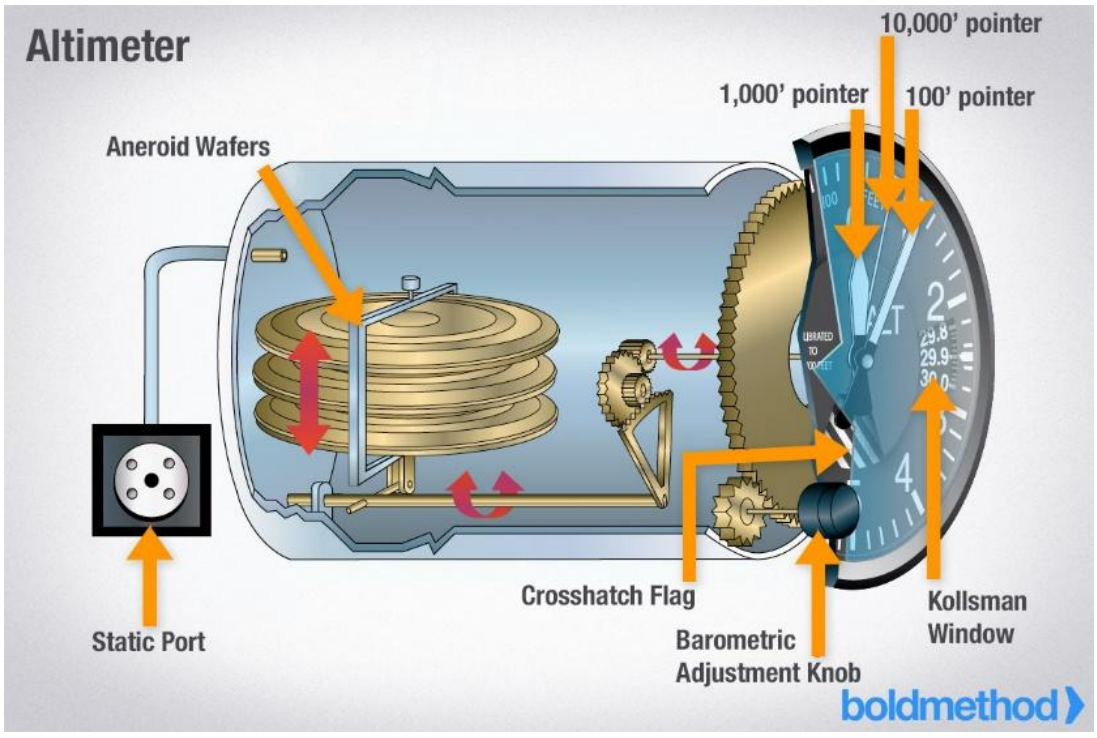
The atmospheric (static) pressure at any point is due to the weight of the overlying air above, which decreases as the height above sea level increases. Hence, the instrument can be calibrated to read in terms of height. It takes advantage of the fact that the static pressure decreases with altitude, therefore, if we measure the pressure at low altitudes, it will be higher than the pressure at higher altitudes. This means that if the altimeter senses a high static pressure, it will interpret that the aircraft is flying at a low altitude and in the same way if it senses a low static pressure it will interpret that the aircraft is flying at a high altitude. So, in other words the barometric altimeter interprets pressure changes as altitude changes.

Under standard air conditions of 15°C, the weight of a column of air, one square inch in area, is 14.7 lb. at sea level. It exerts a pressure of 14.7 lb. per sq. in. This pressure is recorded on a barometer as 29.92 inches of mercury (AKA InHg) or 1013.25 hPa (hectopascals or millibars (mb)). and by an altimeter as 0 feet.

At 10,000 feet the weight of a one square inch air column has decreased to 10.11 lb., the corresponding barometric pressure to 20.58 inches of mercury, and the altimeter records 10,000 feet. The decrease in pressure is sensed by the altimeter and registered as an increase in height.

A standard altimeter contains a stack of sealed aneroid wafers with an internal pressure of 29.92" Hg. These wafers expand and contract based on the static pressure inside the casing of the altimeter. This static air enters the casing through a tube attached to the static ports on your airplane. The chamber is otherwise sealed, so only static air from directly outside the airplane enters the chamber.

A higher static pressure presses down on the wafers and causes them to collapse. A lower static pressure (less than 29.92" Hg) allows the wafers to expand". Mechanical linkages connect the movement of these wafers to needles on the interior face of the altimeter. **Compression of the wafers translates to a decrease in altitude, while expansion translates to an increase in altitude.**



Summary of Altimeter

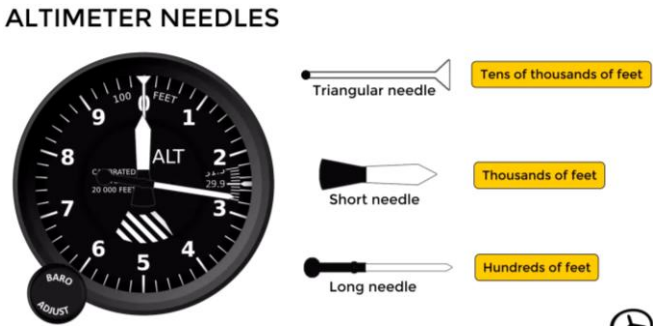
Point	Easy-to-remember explanation
What it measures	An altimeter is an **aneroid barometer** that measures **static atmospheric pressure** and displays it as altitude.
Static pressure	Air pressure comes from the **weight of the air above**. As altitude increases, there is less air above → **pressure decreases**.
Pressure–altitude relationship	**High pressure = Low altitude**; **Low pressure = High altitude**. The altimeter converts pressure changes into altitude changes.
Static source	Static air enters/exits the altimeter case through the **static pressure system/outlet**, allowing pressure inside the case to change as the aircraft climbs or descends.
Aneroid wafers	Inside are **sealed, evacuated aneroid wafers** with a reference pressure of **29.92 inHg**.
When pressure increases	Higher static pressure **compresses/collapses the wafers → indicates lower altitude**.
When pressure decreases	Lower static pressure allows the wafers to **expand → indicates higher altitude**.
How it displays altitude	Mechanical linkages transfer wafer movement to the **altimeter needles**, converting pressure changes into an altitude indication.

Standard Atmosphere Reference

- **Sea level:** 15°C
  29.92 inHg = 1013.25 hPa (mb) = 0 ft
- **10,000 ft:**
  20.58 inHg = 10,000 ft
- Therefore, as the aircraft climbs, **static pressure decreases and the altimeter indicates increasing altitude**.

Reading The Altimeter

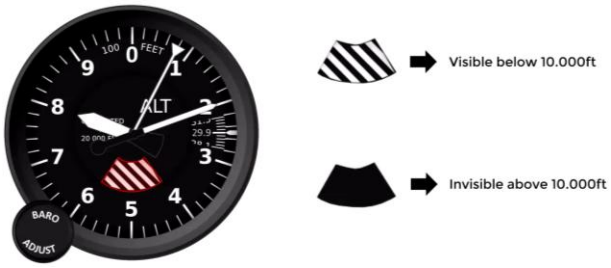
Reading a standard 3-hand altimeter is easy. The long pointer measures altitude in intervals of 10,000 feet (2 = 20,000 feet). The short, wide pointer measures altitude in intervals of 1,000 feet (2 = 2,000 feet). The medium, thin pointer measures altitude in intervals of 100 feet (2 = 200 feet).



Pressure altimeters are calibrated during manufacturing to indicate a true altitude in standard atmosphere conditions. The maximum allowable tolerance is plus or minus 20 feet at sea level. If, having set the current altimeter setting on the subscale and having compared the altimeter reading to the known airport elevation, there is an error of more than plus or minus 50 feet, the instrument should be checked by maintenance.

10,000 FT. Flag

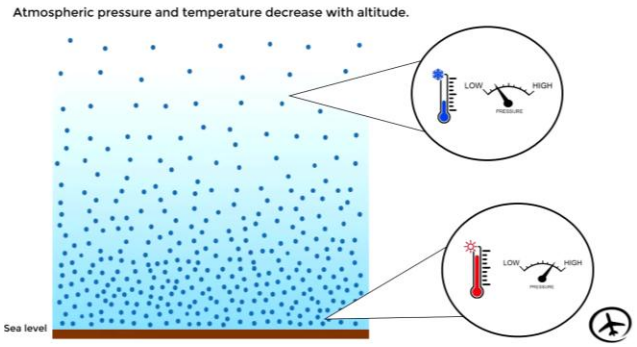
A 10,000 ft flag with black and white stripes is visible when flying below 10,000 feet but it disappears when flying above this altitude this way the pilot can easily determine if the aircraft is flying above or below 10,000 feet



ISA – International Standard Atmosphere

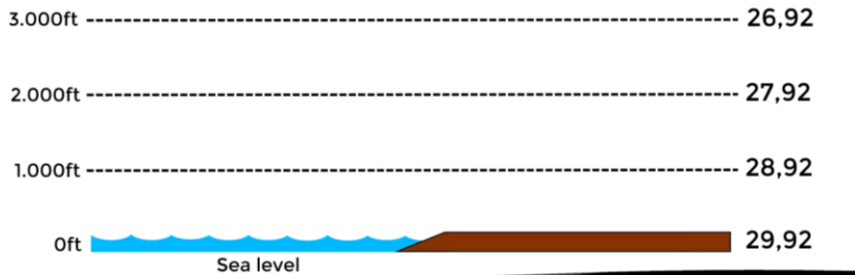
Standard Conditions - ISA
https://www.youtube.com/watch?v=S3TrKvHadMQ&t=777s

- Universal frame of reference for atmospheric conditions
- A worldwide standard reference against which different atmospheric conditions can be compared
- Refers mainly to air pressure and temperature values at different altitudes
- ICAO (International Civil Aviation Organization) defines ISA.
- ICAO determines certain reference values for air pressure and temperature at different altit**Pressure**
    - At high altitude the temperature is low and the pressure is low. While in low altitude, the temperature and pressure are high.

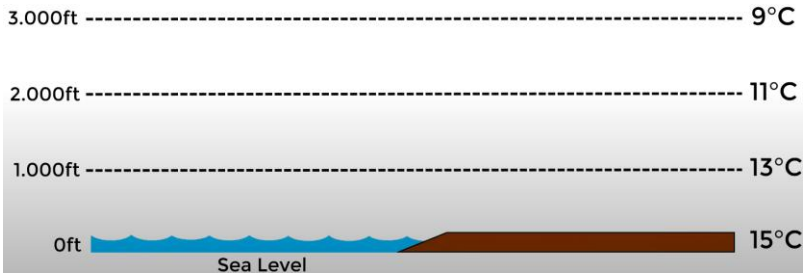


    - ISA assumes that the pressure at sea level is 29.92” Hg or 1013.25 hPa.

- pressure reduction for every 1" Hg = ~1000 feet or 1hPa = ~ 30 feet. These are approximate and may change with altitude.

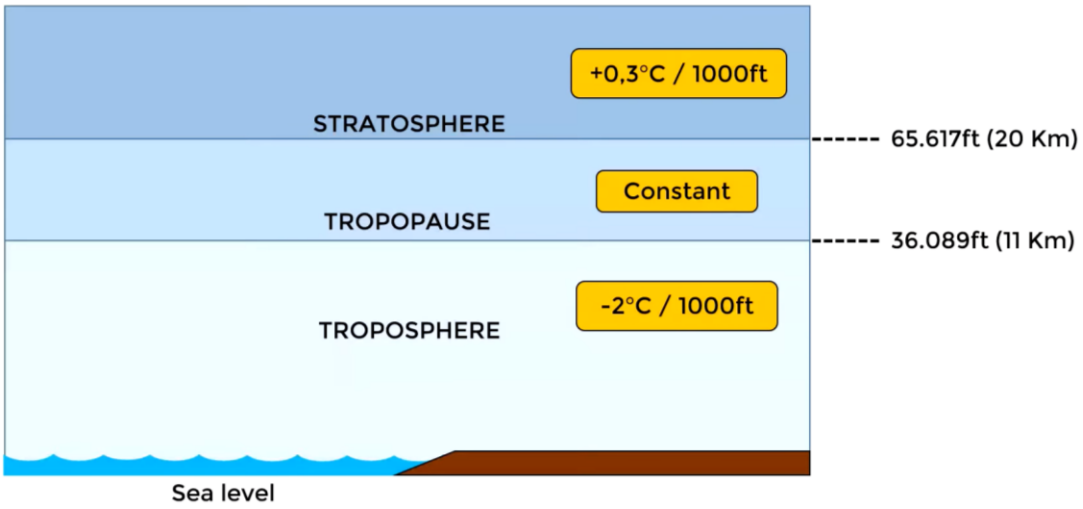


- Temperature
  - temperature is assumed to be 15° C at sea level
  - Reduced by 1.98° (~2°) C for each 1000 feet of altitude increase within the troposphere

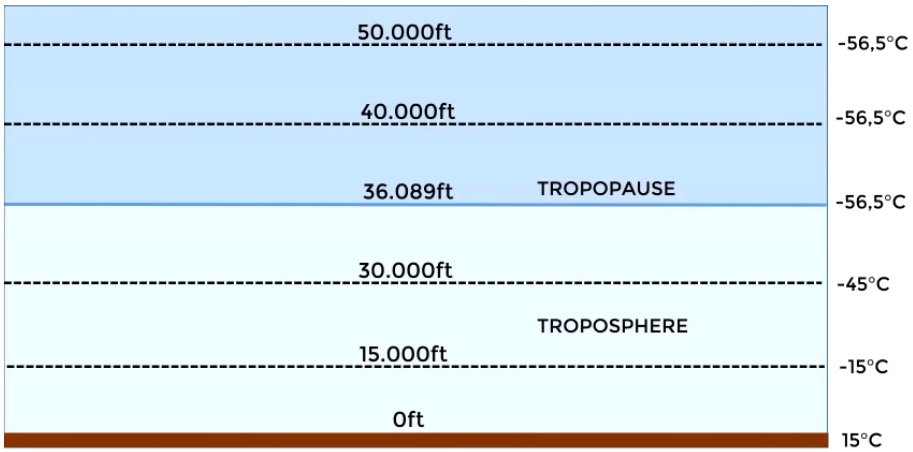


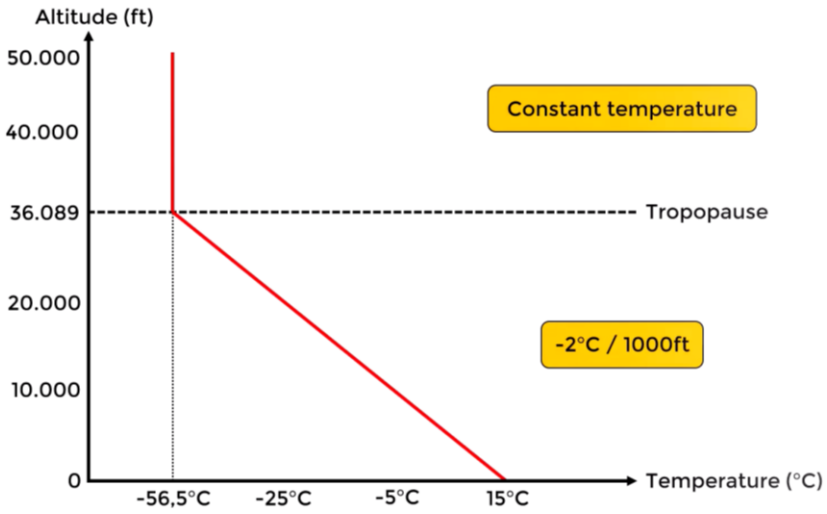
Earth's atmosphere

- Right above the surface is known as the troposphere
  - under standard conditions the temperature decreases by 2° C / 1000 ft until 36,089 ft
- Tropopause is the transition which is the transition layer between the troposphere and the stratosphere
  - the temperature is assumed to remain constant with between 36,089 - 65,617 ft
  - considering that most aircraft operate below 50,000 ft, the temperature increase in the stratosphere is not considered for our study.



- Temperature decreases with altitude until reaching the tropopause.





We must keep in mind that these values apply only within the ISA model. In practice, the actual atmospheric conditions are almost never standard (ISA).

The standard temperature at a certain altitude can be calculated with the following formula:

$$ISA = 15 - \frac{(2 \times ALT)}{1000}$$

NOTE: If the altitude is higher than 36,000ft, it is directly assumed that ISA temp is -56.5°C.

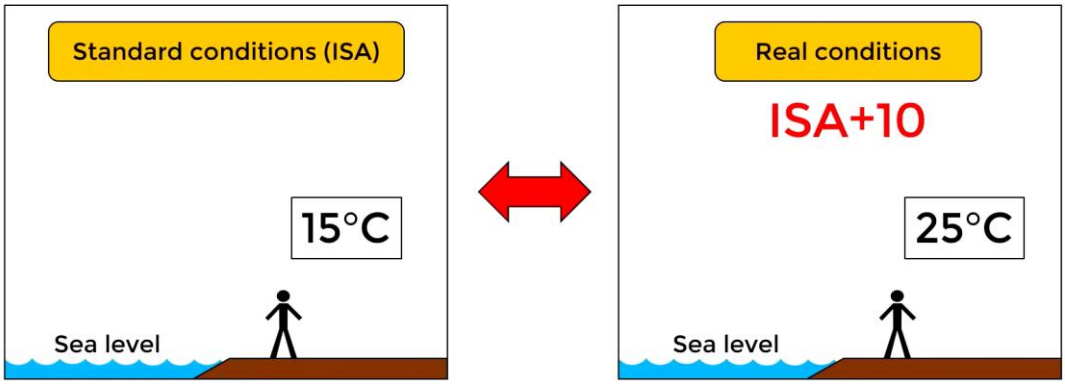
EXAMPLE:

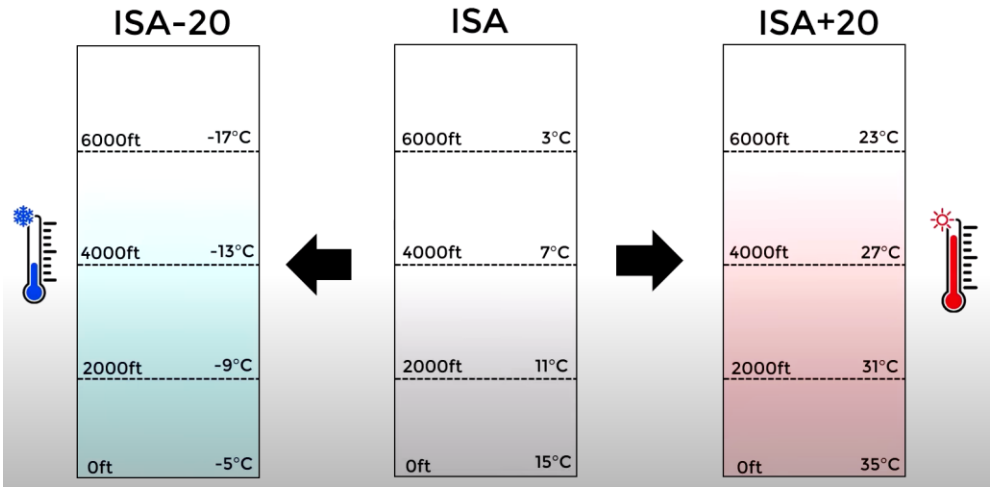
- Determine ISA temperature at 23,000ft.

$$ISA = 15 - \frac{(2 \times 23.000)}{1000} \Rightarrow ISA = 15 - (46) \Rightarrow ISA = -31°C$$

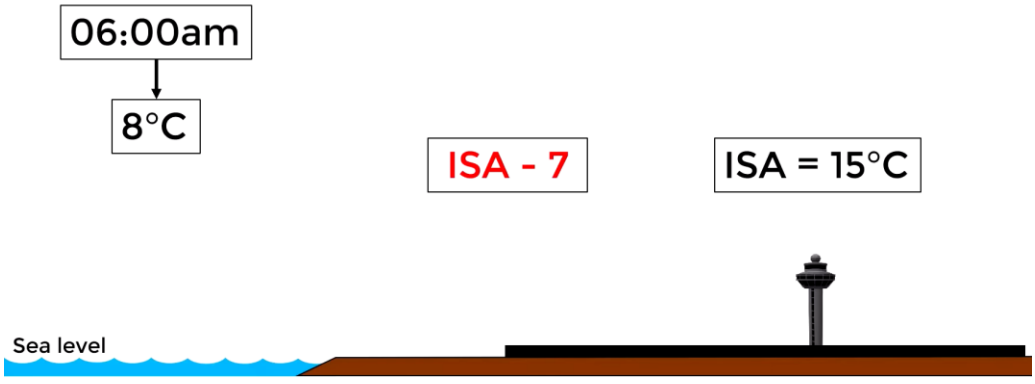
The ISA model is widely used in aviation, mainly in the calibration of flight instruments and determination of the nominal performance of an aircraft.

ISA Deviation aka ISA Dev: A way of expressing how different the current conditions are in relation to ISA in terms of temperature. Temperature plays a very important role in aircraft performance.





At the same airport the ISA deviation varies throughout the day.



At the same airport the ISA deviation varies throughout the day.

01:00pm

↓

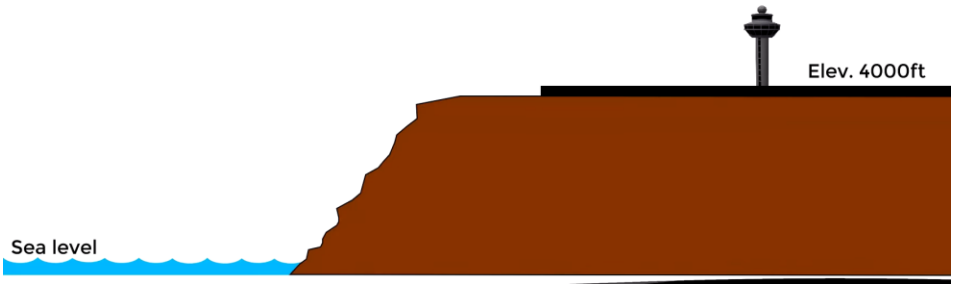
18°C

ISA + 3

ISA = 15°C



$$ISA = 15 - \frac{(2 \times 4.000)}{1000} \Rightarrow ISA = 15 - (8) \Rightarrow ISA = 7°C$$



ISA = 7°C

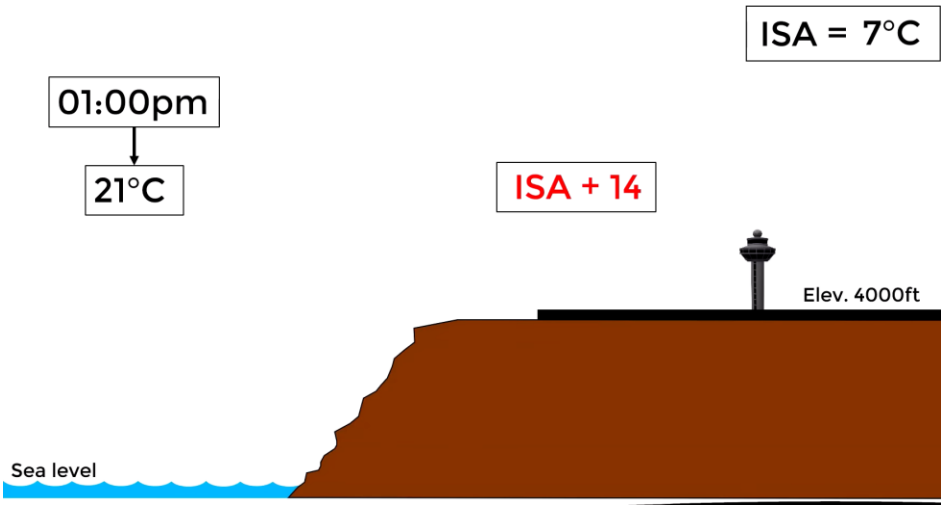
06:00am

↓

11°C

ISA + 4





ALTIMETER PRESSURE ERROR

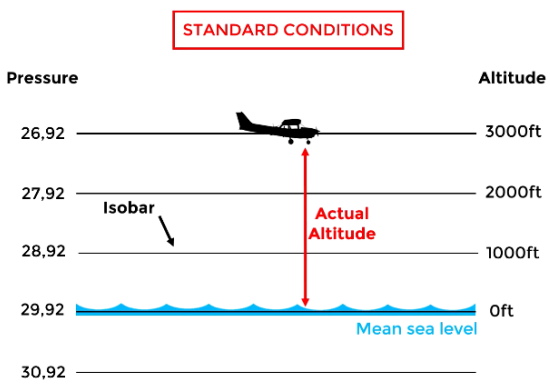
What Happens When Air Pressure Isn't Standard? (29.92" Hg)

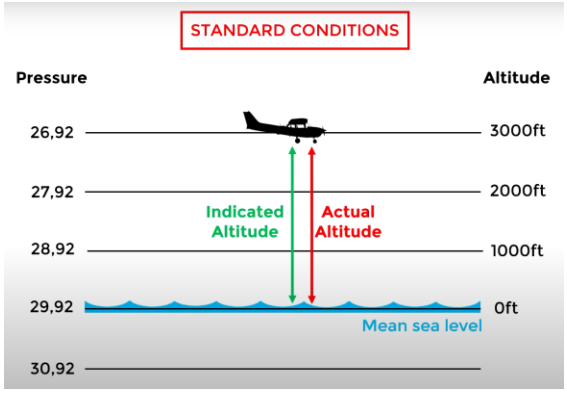
Across the globe, or even across a few miles, different air pressure can have a dramatic effect on altimeter settings. As you fly from high-pressure weather systems to low-pressure systems (or vice versa), you need to adjust your altimeter to get an accurate Mean Sea Level (MSL) altitude reading on your altimeter.

By default, the altimeter is calibrated to use 29.92" Hg as barometric reference. The altimeter can be adjusted to use another barometric reference. Let's look at different scenarios:

Standard conditions:

- Under standard conditions the pressure level that identifies the mean sea level is 29.92" Hg
- From the standard barometric reference static pressure decreases with altitude at a rate of approximately 1" / 1000 feet.
- Theoretically below the mean sea level the static pressure would increase at the same rate.
- Each of these lines that identify a pressure level is called an isobar.
- Here under standard conditions as the altimeter is calibrated by default to use 29.92, it will indicate the vertical distance between the 29.92 isobar and the aircraft.





As the 29.92 isobar and the mean sea level are in the same position, the indicated altitude will be the same as the actual altitude.

Indicated Altitude - Indicated altitude is the reading on the altimeter when the altimeter setting is set to the local barometric pressure. It's the most basic reading, but not always accurate due to atmospheric conditions.

True/Actual Altitude - The vertical distance of the aircraft above mean sea level (MSL), which is the indicated altitude corrected for non-standard temperature and pressure.

Airport, terrain, and obstacle elevations on aeronautical charts are true altitudes.

True altitude equals the indicated altitude when conditions are standard, which is rarely the case.

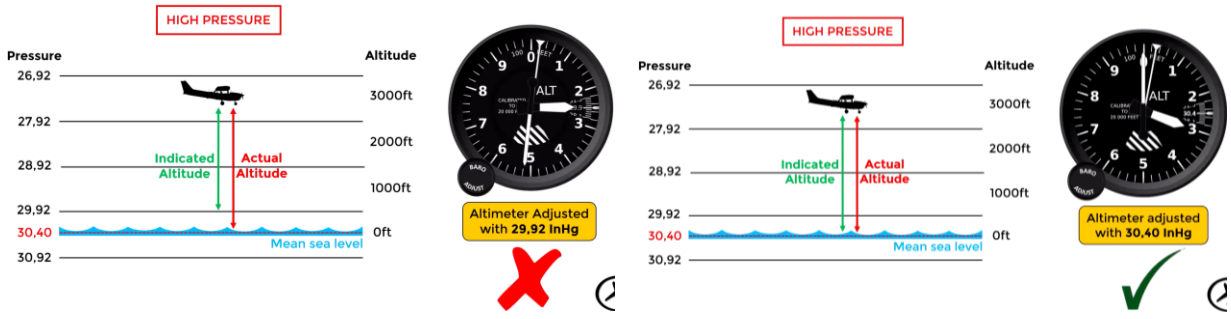
High Pressure: Air molecules are packed closer together, leading to higher air density.

Low Pressure: Air molecules are spread out, resulting in lower air density.

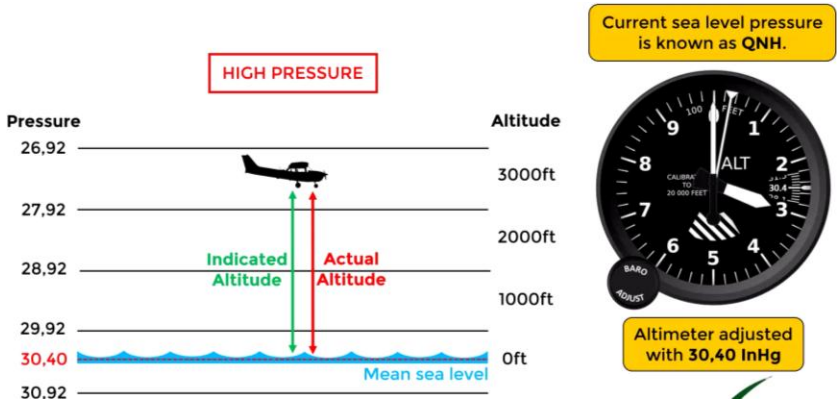
Temperature's Role: Warmer air is less dense than colder air. So, when air is heated, it expands and rises, leading to lower pressure at a given altitude.

Higher than standard pressure

- Isobars will move upwards causing the 29.92 isobar to no longer represent the mean sea level.
- Mean sea level is 30.40 isobar.
- In this situation if our altimeter is still using 29.92 as barometric reference, the indicated altitude will be lower than the actual altitude. This happens because the instrument is measuring the altitude in relation to the 29.92 isobar.
- The solution here is to readjust the altimeter to use 30.40 as the new barometric reference.

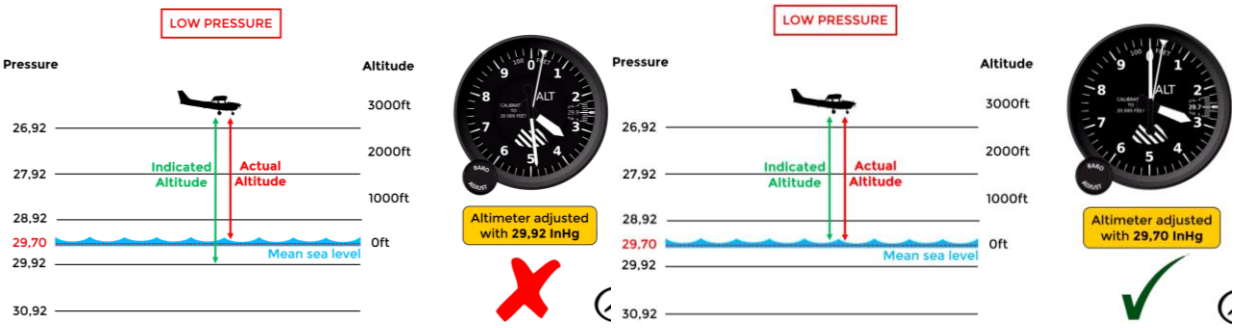


The current level pressure that identifies the mean sea level in a certain moment is known as QNH and it will vary depending on the atmospheric pressure changes. It is therefore important that the pilot receives updated QNH information at regular intervals while flying in order to adjust the altimeter to indicate the actual altitude precisely.



Lower than standard pressure

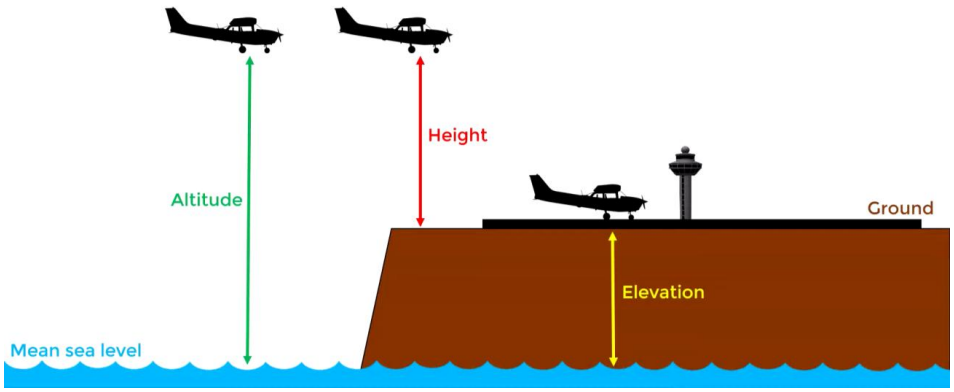
- In this situation the isobars will move downwards causing the 29.92 isobar to no longer represent the mean sea level.
- The new pressure level that identifies the mean sea level is the 29.70 isobar.
- The QNH is now 29.70.
- The indicated altitude will be higher than the actual altitude.
- the pilot will have to adjust 2970 as the new pressure level that identifies the mean sea level.



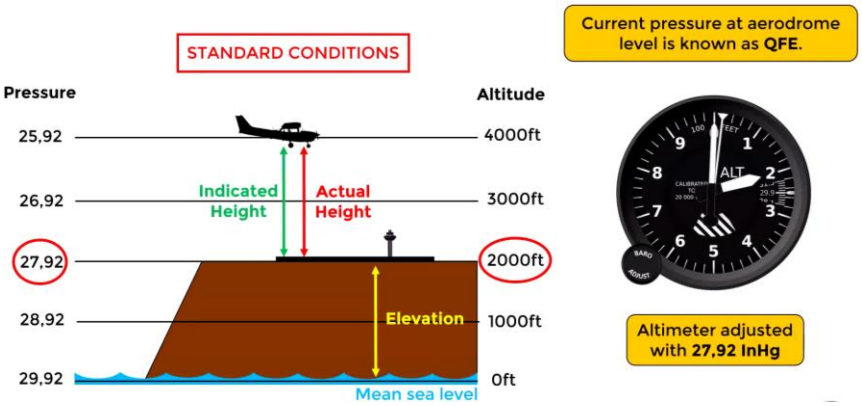
What if we want to measure the height in relation to the terrain

Quick revision of a few earlier stated concepts:

Altitude: Vertical distance between mean sea level and a point in the air.
Height: Vertical distance between a point on the ground and a point in the air.
Elevation: Vertical distance between mean sea level and a point on the ground.



In this particular case the airport elevation is two thousand feet and as we can see the pressure level that identifies two thousand feet is 27.92. So, to indicate height above that airport the altimeter must be set to use 27.92 as barometric reference.



- in this situation as we are no longer using the sea level pressure as a reference. The value used to adjust the altimeter is no longer the QNH.
- It is now the **pressure at the airport/aerodrome level which is known as QFE**.
- Like QNH if the pressure conditions are different from the standard, the QFE for a certain airport will vary. Therefore, it is important to obtain updated QFE information if we want to use heights instead of altitudes in our altimeter.

You reset your altimeter to match local, nonstandard station pressure readings, using the Kollsman window on your altimeter. This is usually done every 100NM for aircraft flying below FL180.

Take, for example, a situation in which a flight altitude of 6,000 feet has been selected to clear a 4,800 foot mountain ridge that lies across the course near the destination. The altimeter setting at the airport of departure was 29.80". The altimeter setting at the destination airport, however, is only 29.20". If the pilot does not reset his/her altimeter, the aeroplane will clear the mountain ridge by only 600 feet instead of the 1,200 feet expected.

Where do you get the altimeter settings from? If you're flying VFR, you can tune in to a nearby ASOS or AWOS weather station at an airport. If you're getting VFR flight following from ATC, or if you're on an IFR flight plan, ATC will give it to you periodically. And if you have ADS-B or Sirius XM weather, you can pull up the altimeter setting for a nearby airport.

Every .1" Hg is equivalent to 100 feet in altitude. So, let's say you took off with an altimeter setting of 29.96" Hg. 150 miles into your flight, the pressure dropped to 29.70" Hg. The altimeter would be off by approximately 260 feet in altitude if you didn't make any adjustments. (29.96 - 29.70 = +.26).

If you kept flying at your indicated altitude (let's say 3,500' MSL) without adjusting your altimeter, you'd be 260 feet low. (Remember the saying "high to low, look out below).

But by dialing the new altimeter setting in the Kollsman window, or the altimeter setting of a glass-panel aircraft, you'll read out an accurate MSL altitude. When an aeroplane is flying from an area of high pressure into an area of low pressure (or from warm air into colder air) if a corrected altimeter setting has not been obtained by radio, the altimeter will read high.

In the Northern Hemisphere a drift to the right or starboard indicates that an aeroplane is flying towards an area of low pressure. Continued drift to starboard over a long period should act as a warning that an uncorrected altimeter may be reading high.

Abnormally high-pressure factors can result in altimeter pressure errors. Cold, dry air masses can produce barometric pressures higher than 31.00" Hg. On most standard altimeters, the barometric setting scale does not go higher than 31.00" Hg and the instrument will not indicate altitude accurately in conditions when the barometric pressure exceeds this figure. In such conditions, the aircraft's true altitude will be higher than the indicated altitude. It is possible to estimate the error in the indicated altitude for pressures above 31.00" Hg by adding 100 feet in aircraft altitude for each .10" Hg. Aircraft operating IFR and VFR, and especially those operating VFR at night, must exercise extra diligence both in flight planning and during the flight. When such conditions of abnormally high pressure exist, ATC will issue the actual altimeter setting and pilots are advised to set 31.00" Hg on their altimeters throughout the duration of the flight. To determine the suitability of an aerodrome from which or to which you intend to operate, increase the ceiling requirements by 100 feet and the visibility requirements by 1/4 s.m. for each .10" hg over 31.00" Hg.

A visibility requirement of "1/4 s.m." refers to one-quarter of a statute mile, which is a unit of distance used in aviation to measure visibility. In the context of aviation, it specifically refers to Runway Visual Range (RVR) and is a critical factor for safe aircraft operations, particularly during takeoff and landing. A visibility requirement of 1/4 s.m. means the pilot can see a distance of one-quarter of a statute mile (approximately 1,320 feet or 402 meters) down the runway.

Summary of the above 10 paragraphs

✈ Altimeter Setting — Short Summary & Memory Tips

1. Altimeter Setting

- Adjust the **Kollsman window** to match local pressure for accurate MSL altitude.
- Typically reset every **100 NM below FL180**.
- Obtain settings from **ASOS, AWOS, ATC, ADS-B or SiriusXM weather**.

2. Pressure Difference and Altitude Error

Every 0.10" Hg ≈ 100 ft.

Example: Departure pressure = 29.96"; destination pressure = 29.70".

- Difference = 0.26" Hg.
- Altitude error ≈ **260 ft**.
- At an indicated 3,500 ft, the aircraft would be approximately **3,240 ft MSL** without adjustment.

Memory tip: *High to Low — Look Out Below!* Flying from high to low pressure without resetting means your true altitude is lower than indicated.

3. Mountain Clearance Example

- Planned altitude: 6,000 ft.
- Mountain ridge: 4,800 ft.
- Expected clearance: 1,200 ft.
- Without correcting for the pressure change, actual clearance could be only **600 ft**.

4. Additional Warning Signs

- Flying from **warm air into colder air** without correction can result in true altitude being lower than indicated.
- In the Northern Hemisphere, sustained drift to the right (starboard) may indicate movement towards low pressure. Check your altimeter setting.

5. Abnormally High Pressure: Above 31.00" Hg

- Cold, dry air masses can produce pressure above 31.00" Hg.
- Most standard altimeters cannot be set above 31.00" Hg.
- In these conditions, **true altitude can be higher than indicated altitude**.
- Obtain the actual setting from ATC and set **31.00" Hg as advised**, maintaining it throughout the flight.
- Exercise extra caution during flight planning and flight, particularly during VFR night operations; both IFR and VFR pilots must be vigilant.

6. Adjust Aerodrome Requirements

For every **0.10" Hg above 31.00" Hg**, increase:

- **Ceiling requirement:** 100 ft.
- **Visibility requirement:** ¼ statute mile (s.m.).

Remember: **¼ s.m. = 1,320 ft ≈ 402 m.**

🧠 Final Memory Shortcut

Remember	Key rule
RESET	Use local pressure; typically every 100 NM below FL180.
CALCULATE	0.10" Hg ≈ 100 ft.
PROTECT	High to Low — Look Out Below!
HIGH PRESSURE	Above 31.00" Hg, follow ATC guidance and apply extra caution.
AERODROME	Per 0.10" Hg above 31.00": +100 ft ceiling, +¼ s.m. visibility.

Altimetry and Altimeter Setting Procedures

The altimetry and altimeter setting procedures help in ensuring **adequate vertical separation from terrain and other aircrafts using barometric pressure**. They take into account atmospheric pressure changes, terrain characteristics, air traffic flow and ATS services, local/regional procedures.

Quick revision of a few earlier stated concepts:

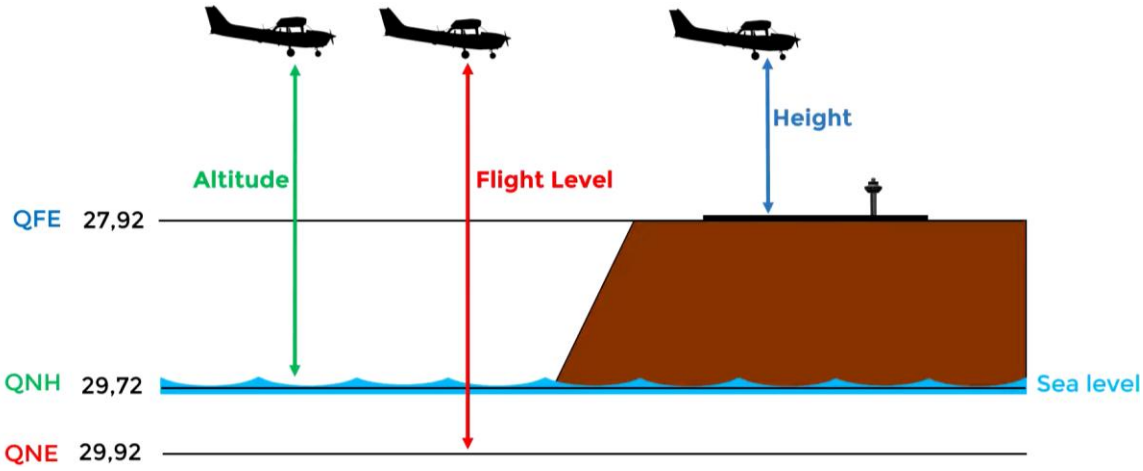
Altitude: Vertical distance between mean sea level and a point in the air.

Height: Vertical distance between a point on the ground and a point in the air.

Elevation: Vertical distance between mean sea level and a point on the ground.

Flight level: Vertical distance between the isobar 29.92" Hg (standard) and a point in the air.

- A standard altitude reference (29.92" Hg or 1013.25 hPa) used to maintain vertical separation between flight/s at higher altitudes. These higher altitudes vary by regions or countries.
- Flight levels are also known as FL.
- They are represented by three digit numbers expressed in 100 foot increments e.g. FL360 = 36,000 ft.
- Like two other barometric references (QNH or QFE), flight levels are represented by QNE. In short, QNE refers to the standard pressure level of 29.92" Hg or 1013.25 hPa.



There are three main barometric references used in the altimeter setting procedures:

QNH – Pressure level that identifies the mean sea level. Therefore, will indicate **Altitudes**.

QFE – Pressure level that identifies the airport/aerodrome level. Therefore, will indicate **Heights**.

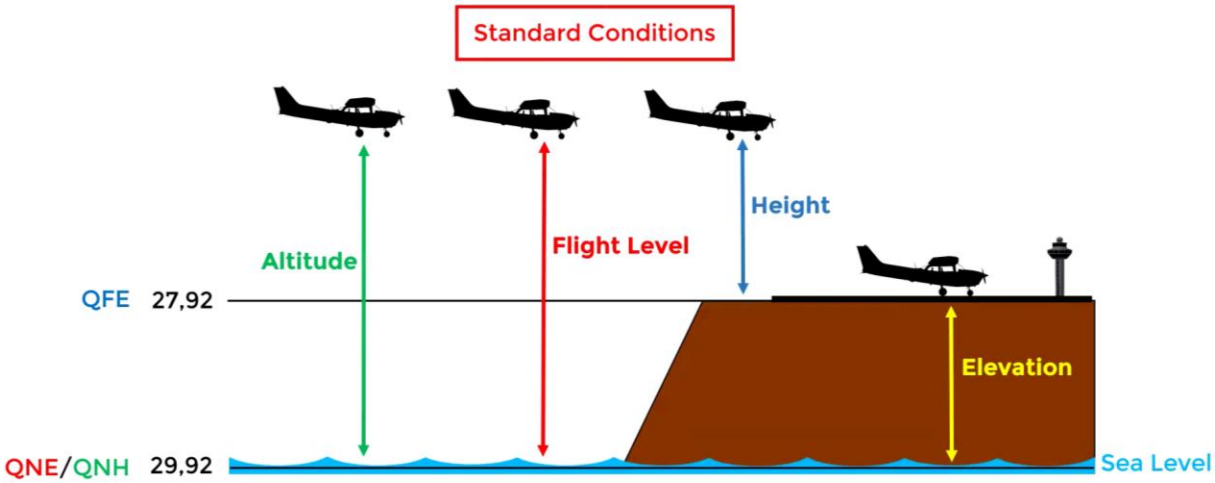
QNE - Standard pressure level of 29.92" Hg or 1013.25 hPa. Therefore, will indicate **Flight Levels**.

A few Cases:

Standard Conditions: In standard conditions we know the mean sea level is 29.92" Hg or 1013.25 hPa. This would mean:

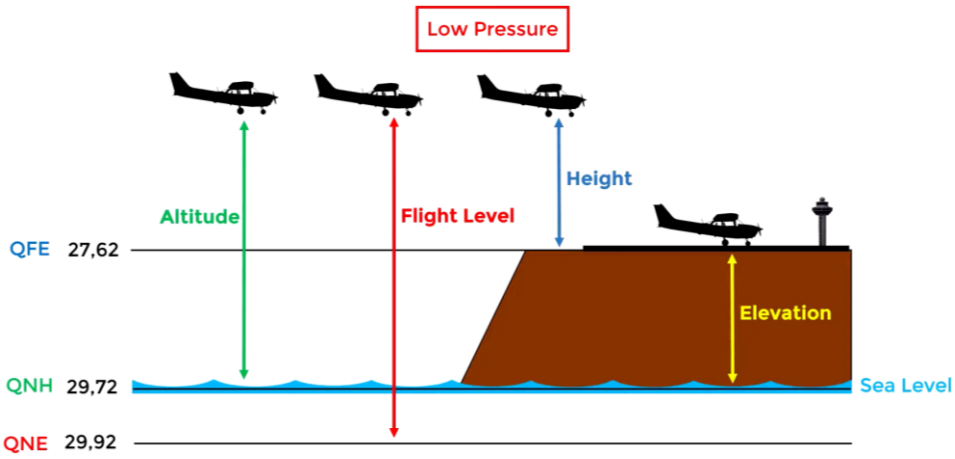
- QNH and QNE will be equal and therefore the altitude and flight level will be the same.

- Using this setting while on the ground the altimeter will read elevation
- If the QFE is 27.92” Hg, and the pilot sets this as barometric reference the altimeter will read the height of the aircraft above that airport.



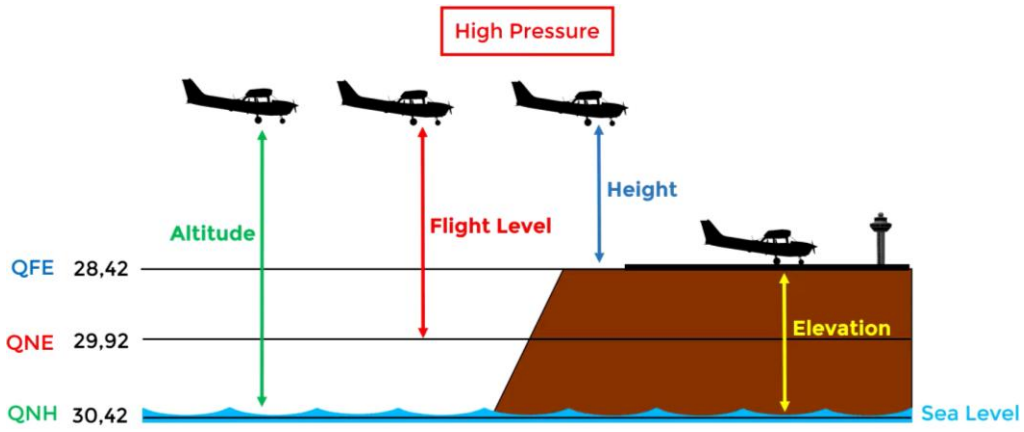
Low Pressure: A lower than standard pressure condition means that the sea level pressure is lower than 29.92” Hg. Let's suppose it is 29.72 in this example:

- 29.72” Hg would become the QNH as we adjust the barometric reference the altimeter will read the altitude of the aircraft in flight.
- The altimeter will read airport elevation if the aircraft is on the ground.
- In this situation the QNE or standard pressure level will be below mean sea level. If we adjust it as barometric reference the altimeter will indicate the flight level. As we can see is higher than the altitude.
- Finally, let's assume that the QFE at this airport is 27.62” Hg. If it is set as barometric reference the altimeter will indicate the height above that airport



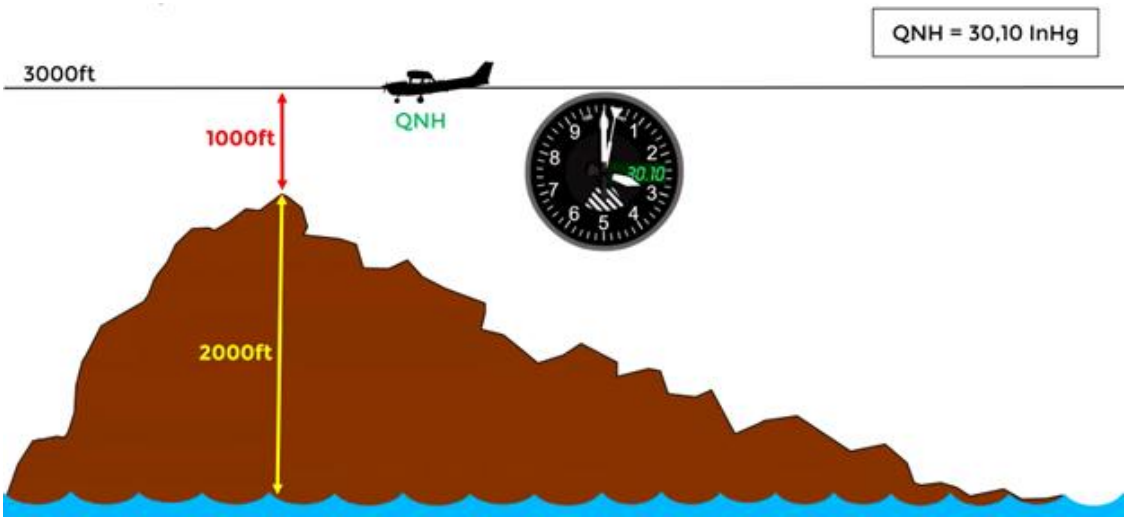
High Pressure: In a higher than standard pressure condition, the sea level pressure will be higher than 29.92” Hg. Let's assume it is 30.42” Hg:

- 30.42” Hg would become the QNH, it in the altimeter it will read the altitude or elevation above sea level
- QNE or standard pressure level will be above sea level. if it is adjusted in the altimeter it will read the flight level which as we can see is lower than the altitude.
- Finally, if the QFE is for example 28.42” Hg it will allow the altimeter to read height above the airport.

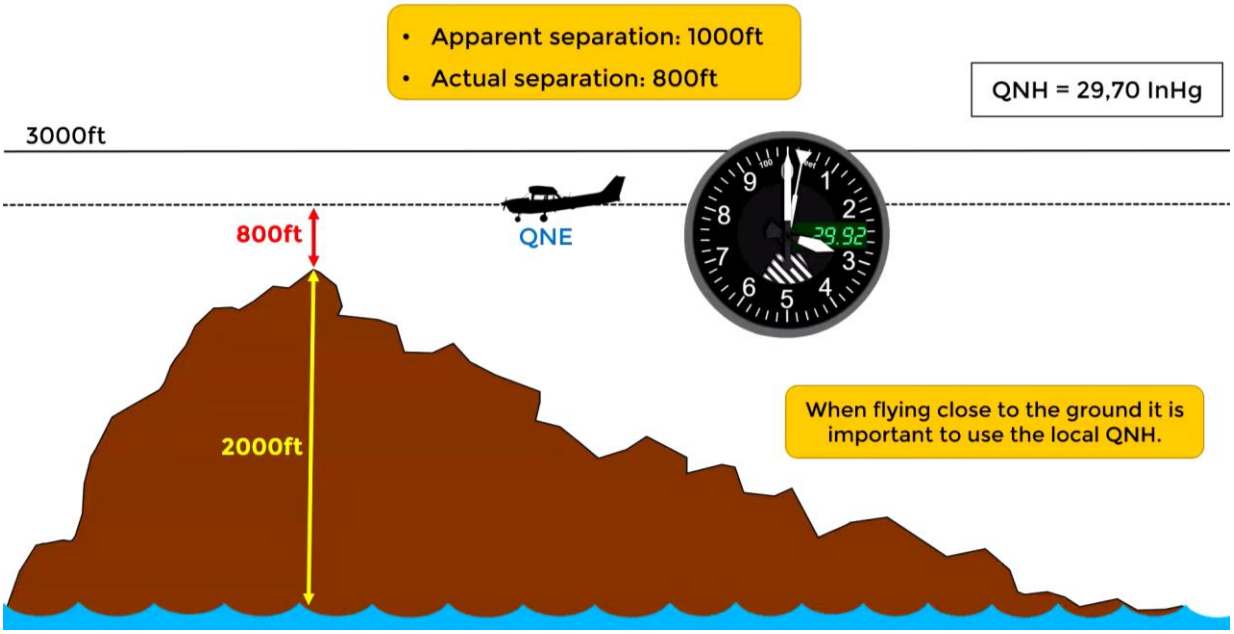


Separation with the terrain and obstacles: To ensure adequate separation with terrain and obstacles it is necessary to use the QNH. Since, it allows the pilot to know the real altitude above sea level.

- QFE of a certain airport could also be used for this purpose. However, it is little used worldwide since it leads to confusion and mistakes.
- Most charts and procedures show the elevation of terrain and obstacles in relation to sea level for example, here the top of this mountain has an elevation of 2000 feet above sea level. Therefore, a pilot who is planning to fly through this area with a separation of 1000 feet would have to fly at an altitude of 3000 feet using the local QNH. In this case it is 30.10” Hg.

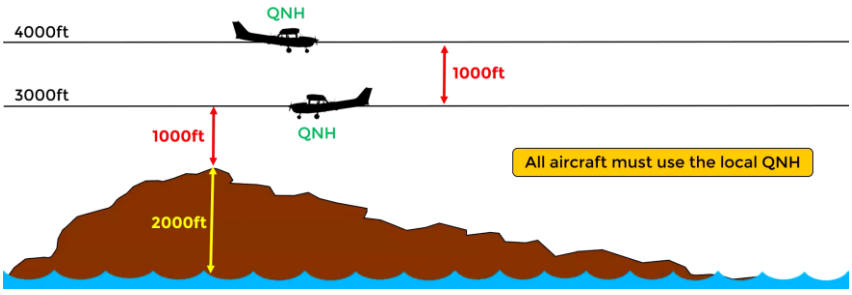


- If otherwise this aircraft uses an altimeter setting other than the QNH the apparent separation with terrain will be different from the actual separation and can be dangerous. In the below case the QNH is 29.70 " Hg and the altimeter reads 29.92" Hg i.e. the QNE.



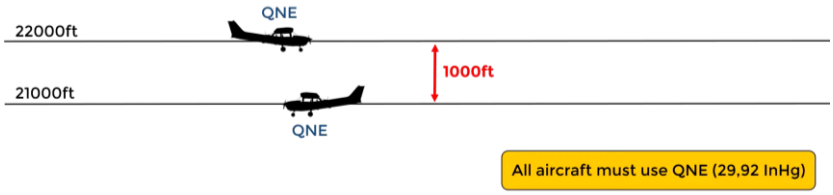
Flying at low altitudes

- To have an adequate separation with terrain the altimeter must be set with the local QNH
- To ensure an adequate separation with other aircraft all of them must use the same altimeter setting

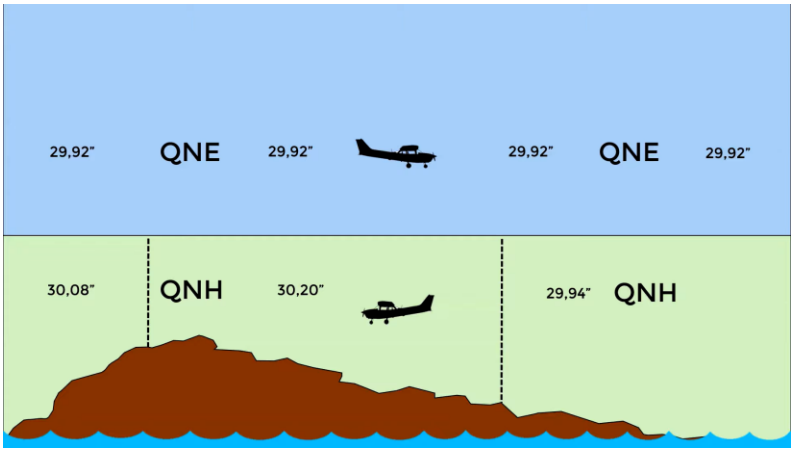


Flying at high altitude

- There are no terrain or obstacles
- There are only nearby aircraft flying
- Only action required is to ensure same altimeter setting is used
- The only setting that remains constant regardless of the conditions is the QNE since it is always 29.92" Hg or 1013 hPa.



Now having seen these two scenarios, where at low altitudes all aircraft must use QNH. However, the problem with this is that the QNH does not have a fixed value it changes depending on the conditions of the different areas. Therefore, in this case the pilot has to update the QNH setting on a regular basis depending on the information provided by the ATC. On the other hand, aircraft operating at high altitudes will use QNE which is always the same 29.92" Hg or 1013 hPa.



Where the change from QNH to QNE and vice versa should be made

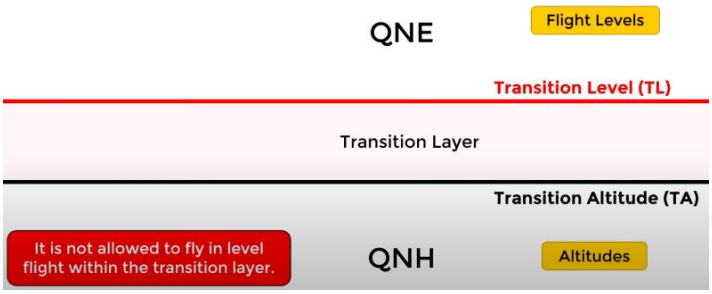
A few concepts:

Transition Altitude (**TA**) - The altitude below which all aircraft must use the local QNH and therefore, express their vertical position in terms of altitudes.

Transition Level (**TL or TRL**) - The level above which all aircraft must use QNE and therefore express their vertical position in terms of flight levels.

Transition Layer (**TL**) - Air space between the transition altitude and the transition level.

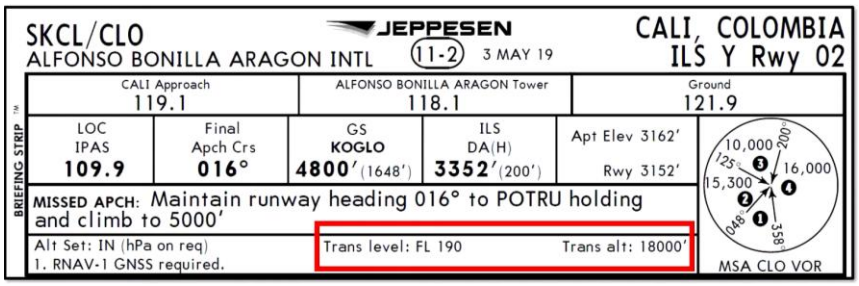
- It is not allowed to fly leveled within this layer
- The aircrafts operating here must be climbing or descending
- Aircrafts climbing through the transition layer must change from QNH to QNE
    - When crossing the transition altitude within the transition layer these aircrafts must express their vertical position in terms of flight levels since they are using now QNE
- Aircraft descending through the transition layer must change from QNE to QNH
    - When crossing the transition level within the transition layer these aircrafts must express their vertical position in terms of altitudes since they are using now QNH



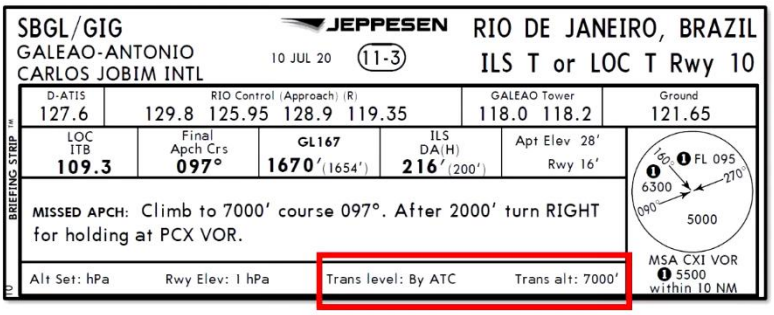
How to identify the transition altitude and transition level for a certain area

Depending on terrain elevations and as provided by the civil aviation authority a transition altitude and level will be published for a certain area or airport. These values can be found in the charts of each airport or in the aeronautical information publication of each country.

In this chart for CALI, Colombia that the transition altitude is eighteen thousand feet and the transition level is flight level one niner zero. Therefore, in this case below eighteen thousand feet all aircraft must use local QNH. While above flight level one niner zero all aircraft must use QNE here the transition layer is between eighteen thousand feet and flight level one niner zero.

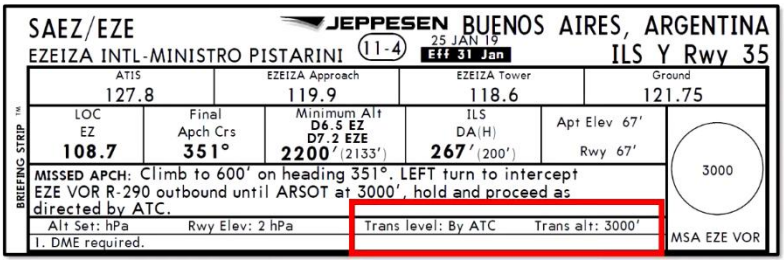


In some cases, the transition altitude may have a fixed value, but the transition level is established by the ATC. in these cases, the air traffic control calculates the appropriate transition level to be used based on the current QNH and the published transition altitude this transition level is reported through the ATIS (Automatic Terminal Information Service, is a continuous broadcast of recorded aeronautical information at busier airports) or a communications VHF.



ATC calculates the TL based on the current QNH and the published transition altitude.

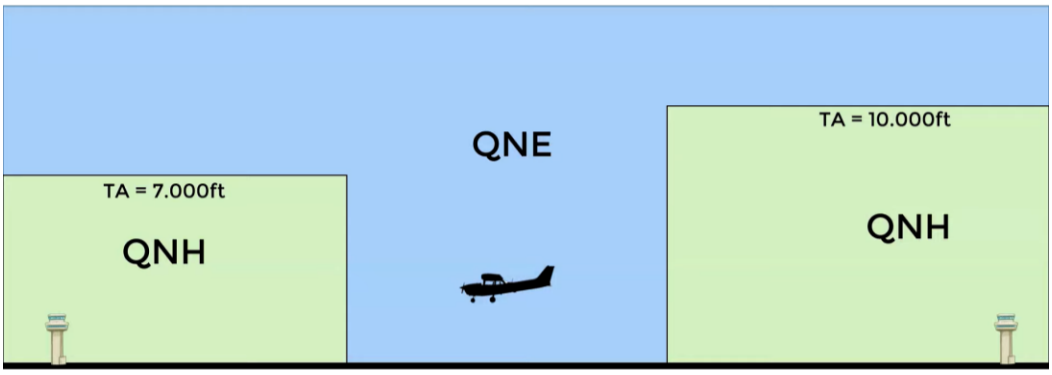
- ATIS
- VHF Frequency



Areas without Transition Altitude

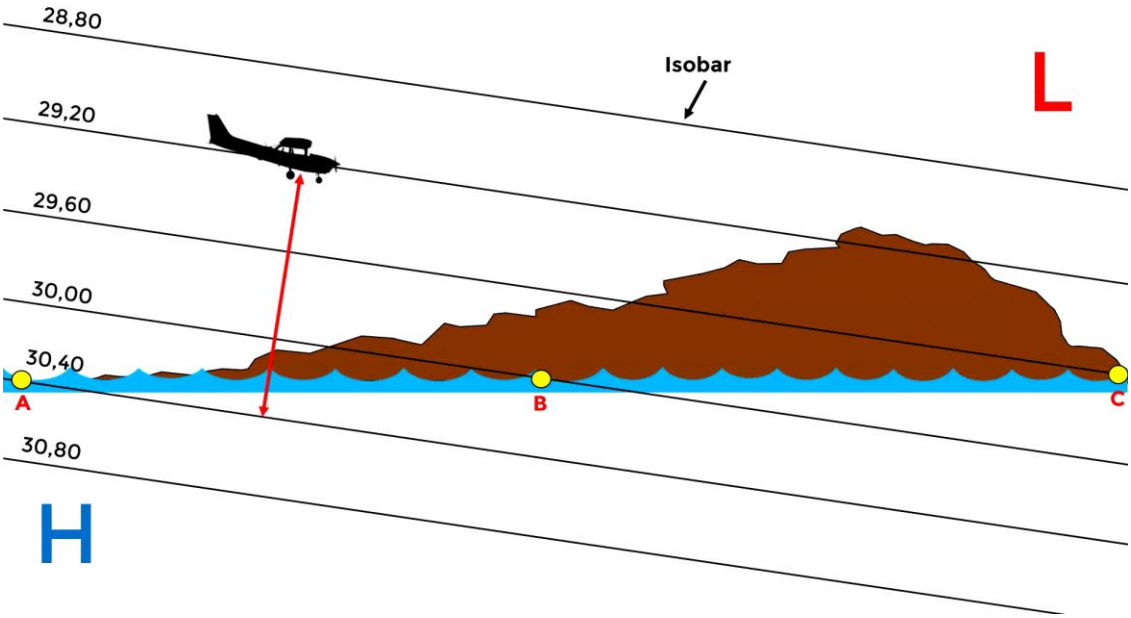
In areas where a transition altitude or level is not published or when there is no QNH information all aircraft must use the QNE and express the vertical position in terms of flight levels.

In this example this aircraft departed using the local QNH from an area where the published transition altitude is 10,000 feet but then it enters an area where there is not a published transition altitude or level. In such case the pilot must set the QNE and use flight levels until entering an area with a published transition altitude and where QNH information is provided usually this occurs in uncontrolled airspace in remote areas where there is no QNH information. Such as for example jungles, deserts, oceans, etc. In such scenarios the pilot must exercise extra caution since there's no way to assure an adequate terrain and obstacle clearance while using QNE.

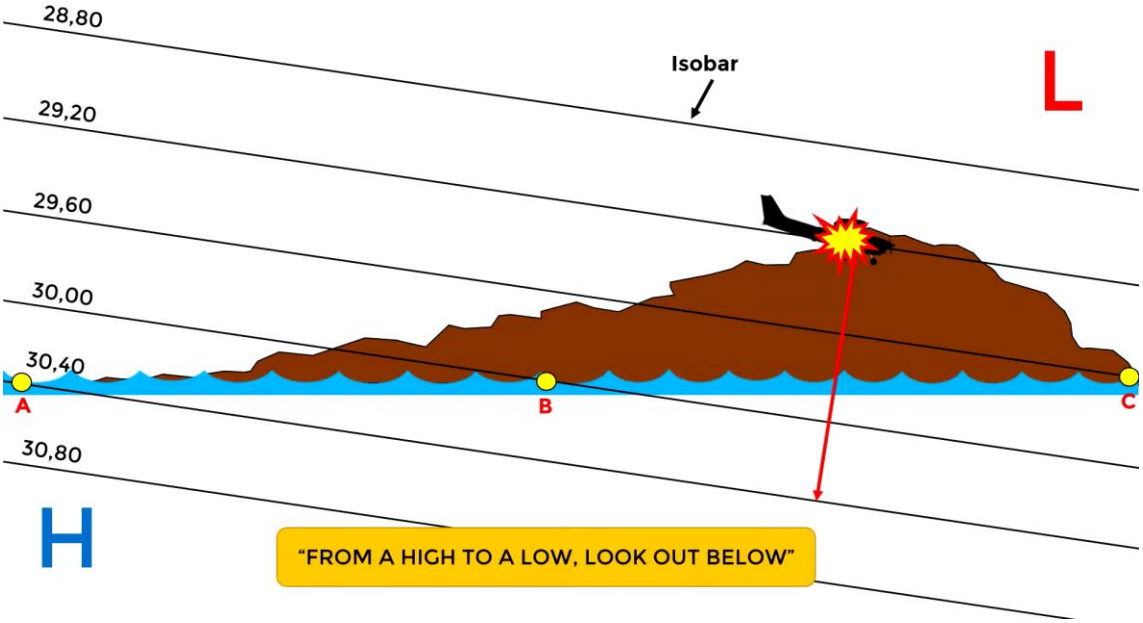


Flying from high pressure areas to Low Pressure areas.

When flying from high pressure areas to low pressure areas, the isobars will be inclined as below:

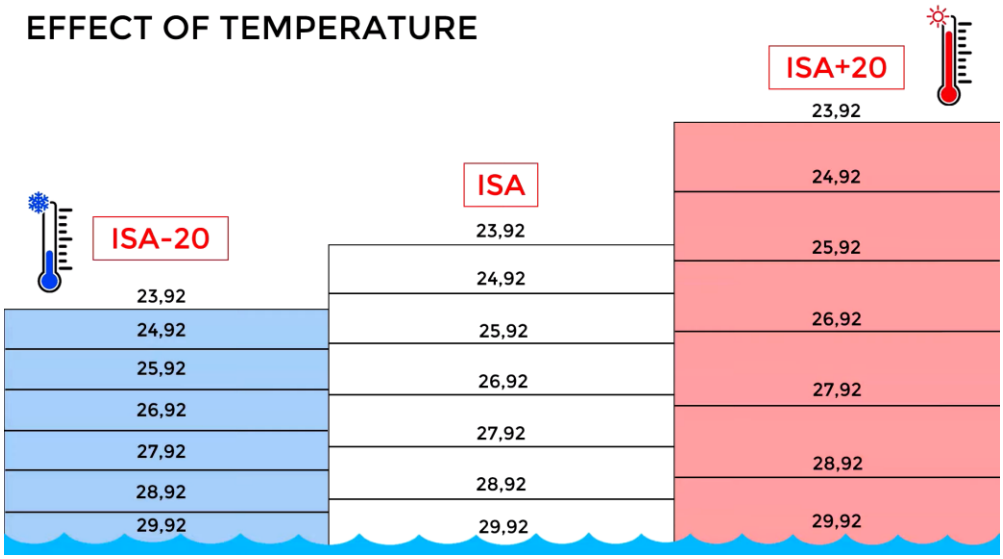


On the left side we can see that in the point A the QNH is 30.40" Hg, then point B dropped to 30.00" Hg, and for point C, QNH further dropped to 29.60" Hg. Under these conditions an aircraft flying from point A to point C would depart with a QNH setting of 30.40" Hg. However, if the pilot does not update the QNH in a long time while flying to the low-pressure area, the aircraft will be gradually descending. This happens because the aircraft is no longer using the correct QNH which can lead to reduced terrain separation or even a crash this situation is colloquially expressed in the phrase "*from high to low look out below*".

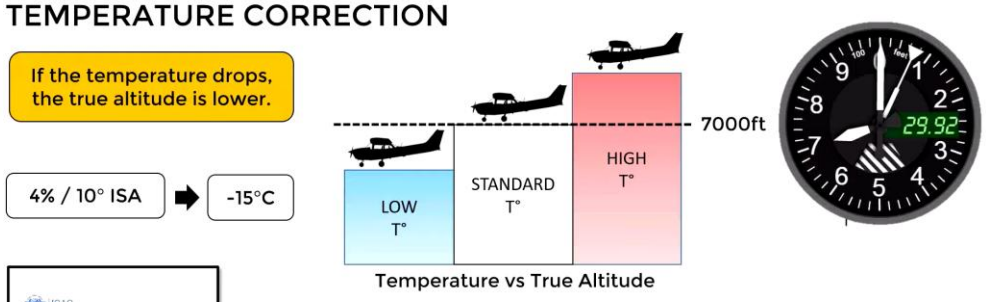


Effect of temperature

If the temperature is colder than standard, for example, in ISA- 20 conditions the isobar will be closer together. If the temperature is warmer than standard for example an ISA+20 the isobars will be further apart. This means that air temperature affects the rate at which the pressure reduces with altitude and therefore it will affect the reading of barometric altimeters.



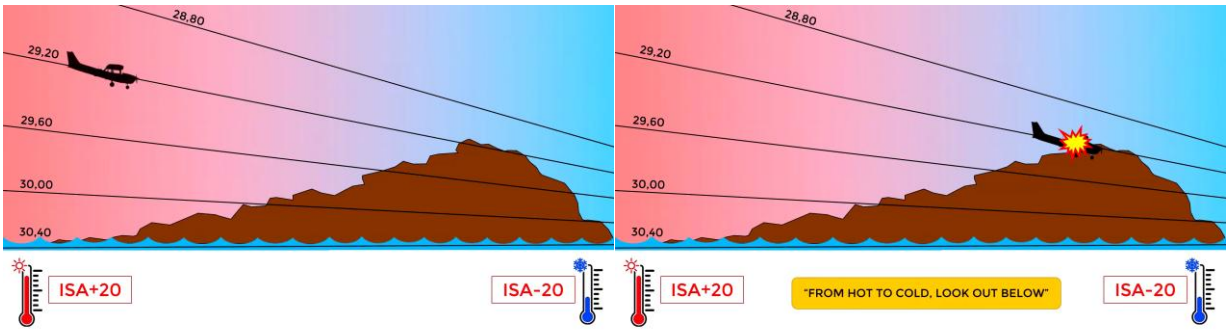
This implies is that if we are flying for example at 7,000 feet under standard conditions and the temperature reduces, the true altitude will be lower than seven thousand feet. While if the temperature increases the true altitude will be higher than 7000 feet this happens precisely because of the change in the spacing of the isobars with the temperature. According to the ICAO document 8168 the true altitude changes by 4% / 10 degrees of ISA deviation, this rule of thumb is useful down to -5° C. With colder temperatures it is necessary to consult the following table published in the document.





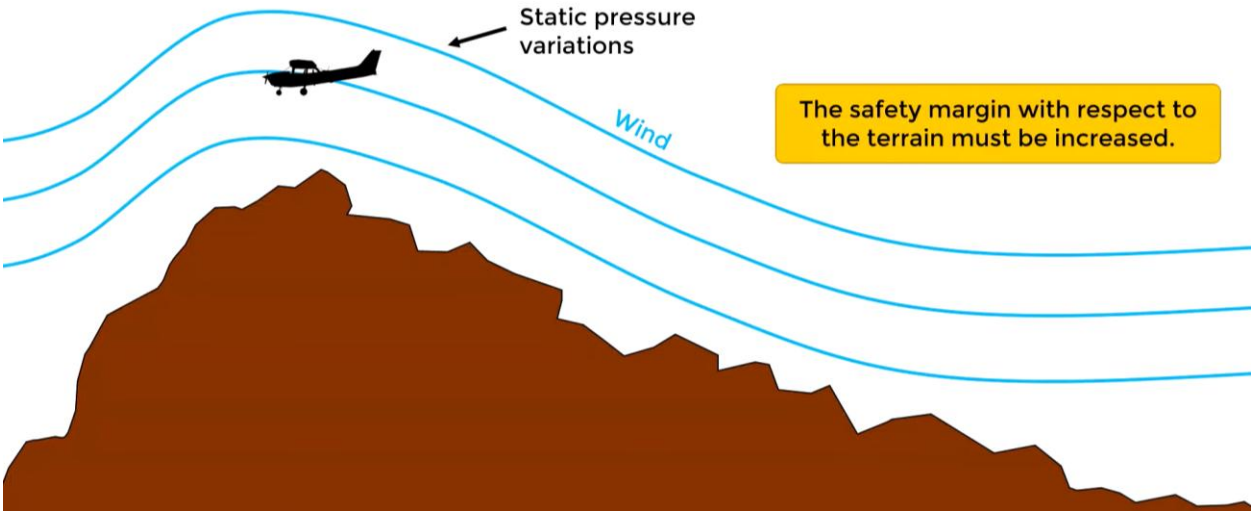
Aerodrome temperature (°C)	Height above the elevation of the altimeter setting source (feet)													
	200	300	400	500	600	700	800	900	1000	1500	2000	3000	4000	5000
0	20	20	30	30	40	40	50	50	60	90	120	170	230	280
–10	20	30	40	50	60	70	80	90	100	150	200	290	390	490
–20	30	50	60	70	90	100	120	130	140	210	280	420	570	710
–30	40	60	80	100	120	140	150	170	190	280	380	570	760	950
–40	50	80	100	120	150	170	190	220	240	360	480	720	970	1210
–50	60	90	120	150	180	210	240	270	300	450	590	890	1190	1500

Temperature leads to a situation quite similar to pressure. An aircraft is flying from a warmer area to a colder area the isobars will be arranged in this way as we can see in the warmer area the isobars are farther apart, while in the colder area they are closer together this means that even if the QNH remains constant an aircraft flying to a colder area will gradually descend. This can lead to a reduced terrain and obstacle clearance or even a crash this is colloquially expressed in the phrase "*from hot to cold look out below*".



Corrections in Mountainous Areas

Apart from these measurement errors of the altimeter with pressure and temperature changes there's another one to take into account it occurs in mountainous areas when the wind blows at high speeds, this produces local static pressure fluctuations due to the Bernoulli effect. This means that an aircraft flying in these areas may present erroneous altitude indications. It is necessary to increase the altitude safety margin with respect to the terrain.

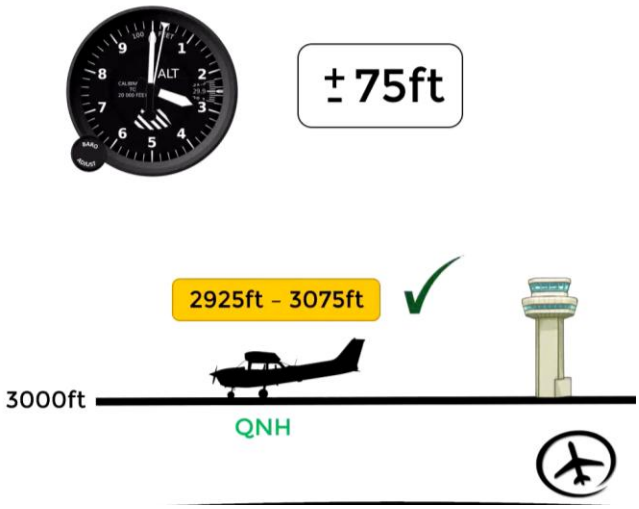


The altimeter error tolerance

The barometric altimeters may have an acceptable margin of error which should be checked before each flight. The exact values are published on the ICAO document 8168. Most manufacturers and

operators consider that an adequate margin is plus or minus 75 feet. this means that if for example an aircraft is on the ground at an airport with an elevation of 3000 feet, when adjusting the current QNH the altimeter should read between 2925 feet and 3075 feet to be considered suitable for air operations otherwise the altimeter must be checked and recalibrated by a certified aeronautical workshop.

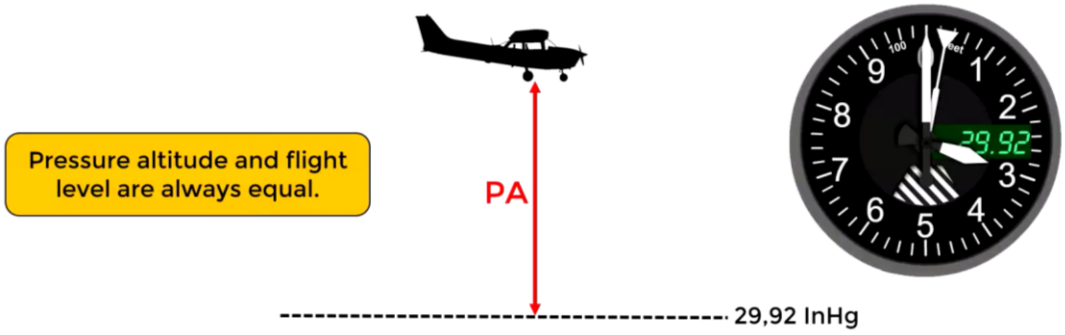




Altitude Types

Pressure Altitude (PA):

- It is the altitude in relation to the **Standard Datum Plane** (Isobar 29.92 lnHg / 1013 hPa) i.e. corresponds to the pressure at sea level under standard conditions.
- It is the altitude indicated by an altimeter adjusted with the QNE (29.92 lnHg / 1013 hPa). Pressure altitude is exactly the same as the definition of Flight Level. Therefore, the pressure altitude will always be equal to the corresponding FL.



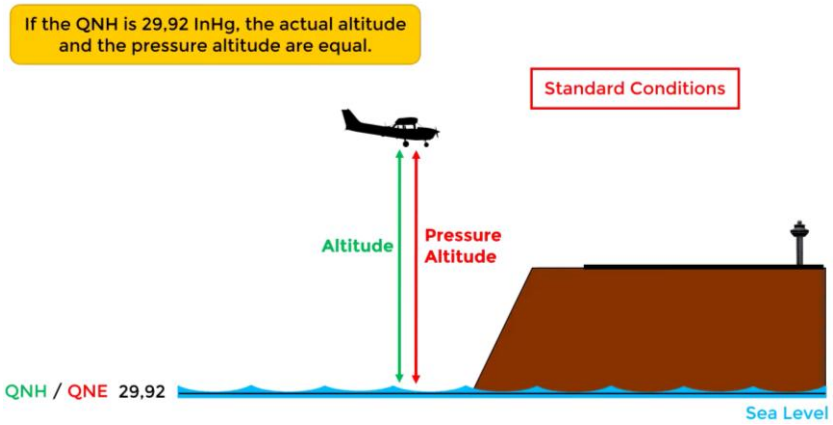
This standardized pressure level is used for various calculations and performance assessments in aviation, especially when dealing with non-standard atmospheric conditions.

Pressure altitude is equal to the indicated altitude when the altimeter setting is set to, 29.92” Hg.

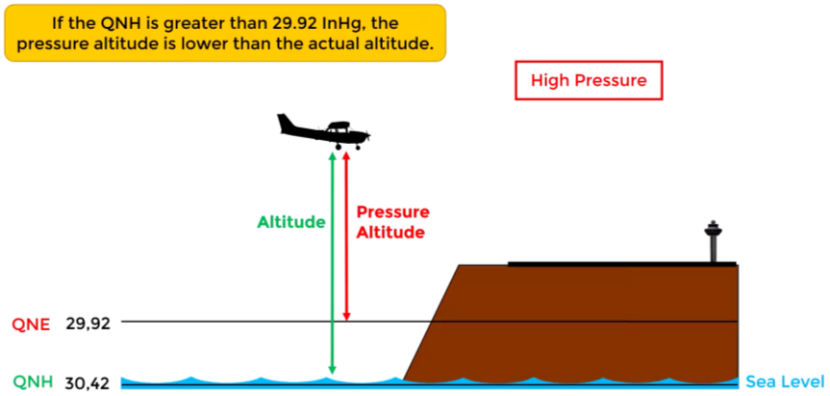
It is used for performance calculations and flight level computation.

Variation in Pressure Altitudes

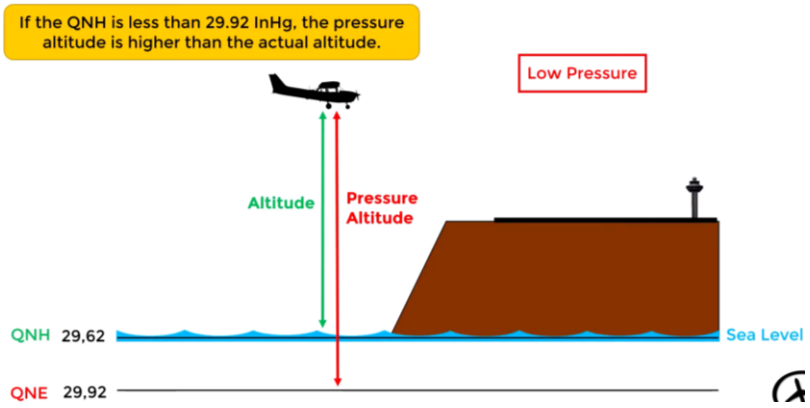
In Standard Conditions, the sea level pressure is 29.92 In Hg. Therefore, the QNH and QNE are equal. We can say that the pressure altitude is equal to the actual altitude.



A higher than Standard Pressure conditions the QNH will be higher than the QNE which means that the 29.92 isobar would be above the sea level. Therefore, in this case the pressure altitude will be lower than the actual altitude.

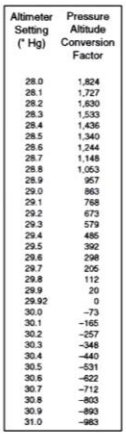


For lower than standard pressure conditions the QNH will be lower than the QNE which means that the 2992 isobar would be below the sea level. Therefore, the pressure altitude will be higher than the actual altitude.



Calculating Pressure Altitude

A table with correction values depending on the QNH, or directly use the altimeter of the aircraft, or use an approximate formula to determine the pressure altitude.



Altimeter Setting (" Hg)	Pressure Altitude Conversion Factor
28.0	1,824
28.1	1,727
28.2	1,630
28.3	1,533
28.4	1,436
28.5	1,340
28.6	1,244
28.7	1,148
28.8	1,053
28.9	957
29.0	863
29.1	768
29.2	673
29.3	579
29.4	485
29.5	392
29.6	298
29.7	205
29.8	112
29.9	20
29.92	0
30.0	–73
30.1	–165
30.2	–257
30.3	–348
30.4	–440
30.5	–531
30.6	–622
30.7	–712
30.8	–803
30.9	–893
31.0	–983

Table



Altimeter

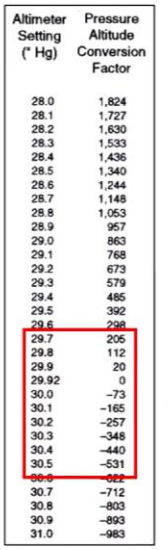
PA = ALT + 30 (1013 - QNH)

PA = ALT + 1000 (29,92 - QNH)

Approximate formula

Pressure Altitude Calculation using the Table:

Suppose we have QNH of 30.20 InHg, and an Indicated Altitude of 12,000 feet. If we look at the table, we can see the different correction values to be applied depending on the QNH. In this case for 30.20 InHg, the correction value is -257. Therefore, all we have to do is take 12,000 feet and subtract 257 feet obtaining a pressure altitude of 11,743 feet.



Altimeter Setting (" Hg)	Pressure Altitude Conversion Factor
28.0	1,824
28.1	1,727
28.2	1,630
28.3	1,533
28.4	1,436
28.5	1,340
28.6	1,244
28.7	1,148
28.8	1,053
28.9	957
29.0	863
29.1	768
29.2	673
29.3	579
29.4	485
29.5	392
29.6	298
29.7	205
29.8	112
29.9	20
29.92	0
30.0	–73
30.1	–165
30.2	–257
30.3	–348
30.4	–440
30.5	–531
30.6	–622
30.7	–712
30.8	–803
30.9	–893
31.0	–983

QNH: 30.20"
Altitude: 12000ft

Altimeter Setting	Conversion Factor
29.7	205
29.8	112
29.9	20
29.92	0
30.0	–73
30.1	–165
30.2	–257
30.3	–348
30.4	–440
30.5	–531

12000 - 257 = 11743

PA = 11743ft

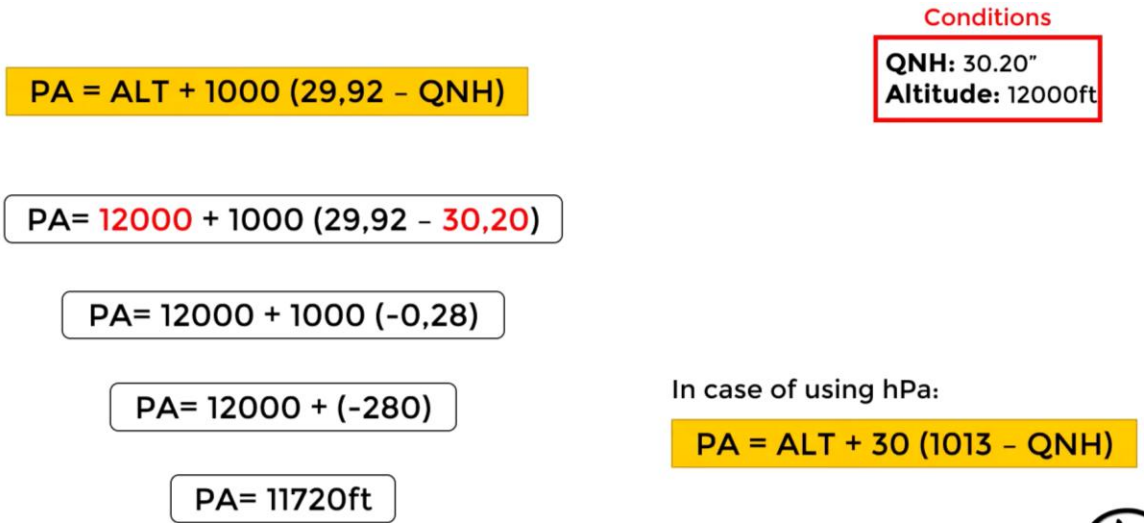
Using Altimeter

Using the same conditions (QNH of 30.20" Hg) as in the previous example. Altimeter will be adjusted with the local QNH of 30.20" Hg, indicating an altitude of 12000 feet. To determine the

pressure altitude all we have to do is adjust 29.92 in the altimeter and read the altitude indication in this case 11,740 feet.



Using Formula



Density Altitude (DA)

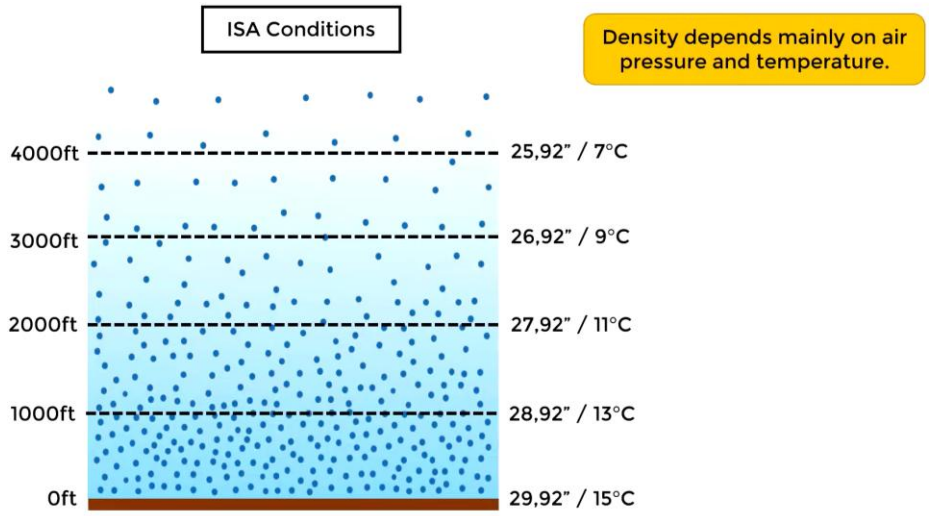
An altitude in the ISA atmosphere where a given density is found.

It is a critical concept, air density is a determining factor in aircraft performance since the thrust produced by the engines, the lift produced by the wings, among other things depend on air density. Also, the definition of density altitude the isa atmosphere is mentioned since this is the main frame of reference to be used.

Some basics:

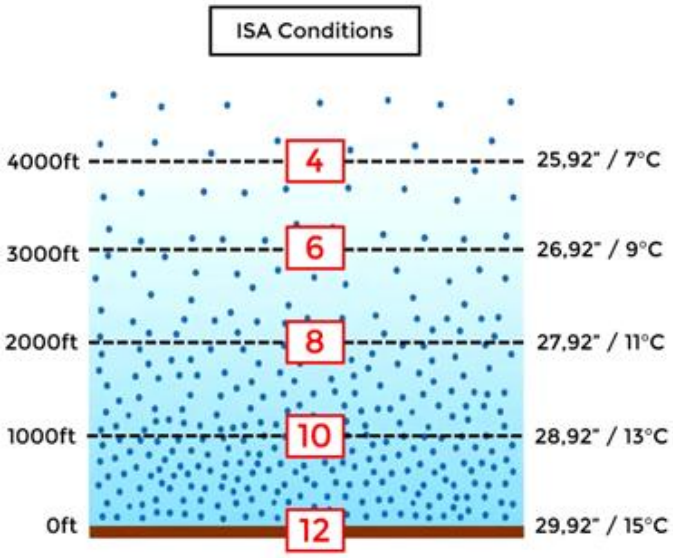
- At low altitudes, air molecules are closer, therefore, higher air density.
- At high altitudes, air molecules are apart, therefore, lower air density.
- At sea level we have a pressure of 29.92" Hg and 15° C as temperature.
- Pressure decreases at the rate of 1" / 1000 ft and temperature at ~2° C / 1000 ft.

- Air density depends mainly on pressure and temperature although it also depends on humidity (it has not been accounted for this study).



Based on the above basics, the pressure and temperature at a certain altitude are always the same this means that the density at a certain altitude will also be always the same. To illustrate this let's assign a certain density value (fictitious) to each altitude in the ISA atmosphere. Again, to reiterate, these will be fictitious density values which do not obey to any unit of measurement since they will only be assigned for the purposes of the example.

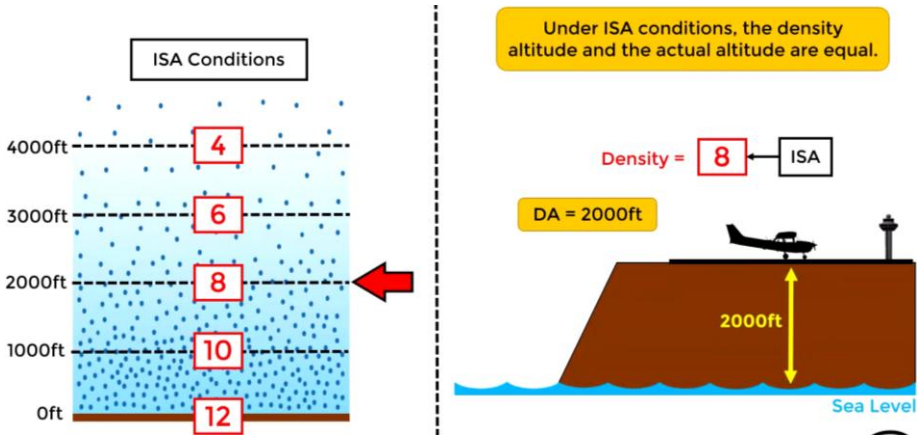
For 29.92" Hg and 15 °C, we have assumed a density value of 12 (fictitious value), at 1000 feet the combination of 28.92" Hg and 13° C results in a density of 10 (fictitious value), at 2000 feet the combination of 27.92" Hg and 11° C results in a density of 8 (fictitious value), 3000 ft. a density of 6 (fictitious value), and at 4000 ft. a density of 4 (fictitious value).



Density altitude during standard conditions

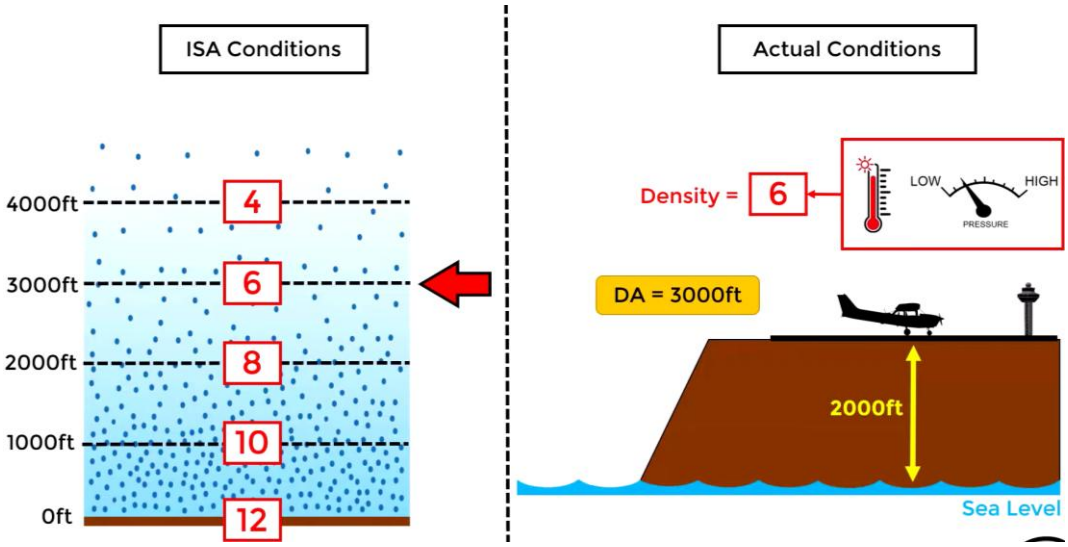
In standard conditions the density altitude will be equal to the actual altitude.

- We have standard ISA conditions
- We are at 2000 ft.
- At two thousand feet we find an air density of eight. Therefore, we say that the density altitude is 2000 ft.



Density altitude during higher than standard temperature and lower than standard pressure

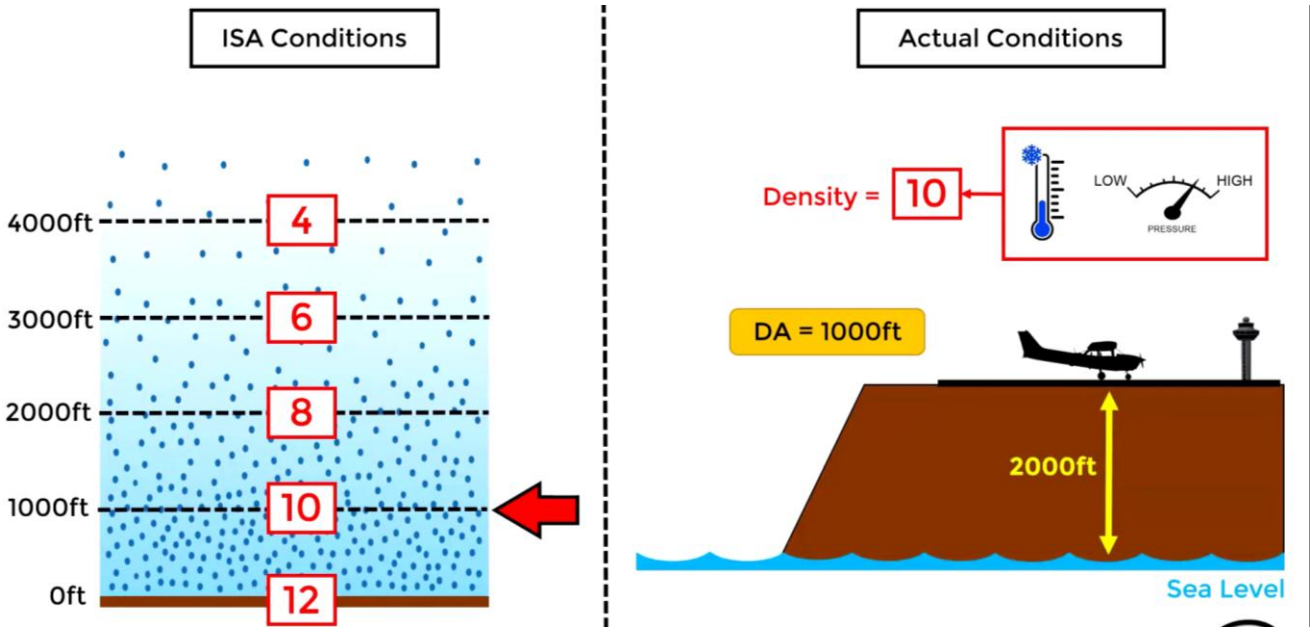
We now have a higher than standard temperature and a lower than standard pressure. The combination of these two factors will cause the air density to be reduced. Let's say for example that this combination of factors results in a density of 6. If we then look for this density value in the ISA model, we can see that we find it at 3000 feet. Therefore, we say that under these conditions the airport has a density altitude of 3000 feet although the actual elevation is 2000 feet. This implies that the aircraft performance will behave as if the aircraft were flying at 3000 feet even though the actual elevation is 2000 feet. Simply due to the fact that the conditions caused the air density to be lower.



Density altitude during lower than standard temperature and higher than standard pressure

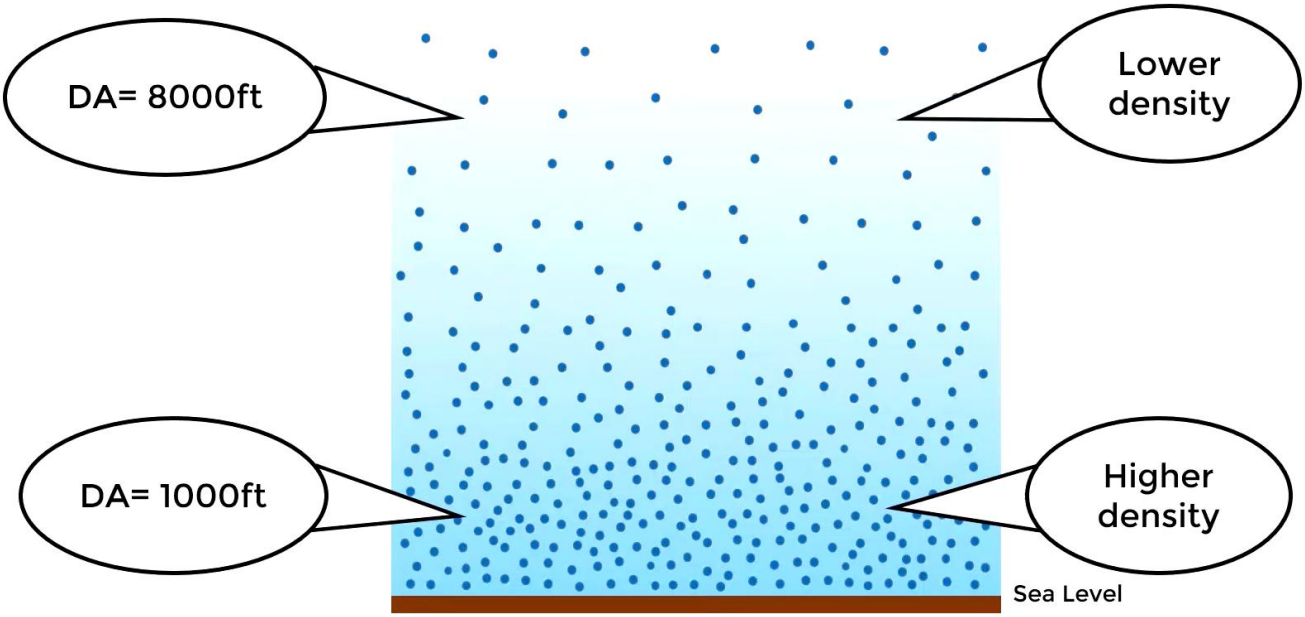
- Now the opposite happens when we have **higher than standard pressure** and **lower than standard temperature** conditions. In this case the combination of these factors will cause the air density to increase.

- suppose for example that the combination of these factors results in a density of ten.
- If we then look for this density value in the ISA model, we will find it at 1000 ft. Therefore, we say that this airport under these conditions has a density altitude of 1000 feet.
- Although its actual elevation is 2000 feet, this means that the performance of an aircraft will behave as if it were flying at 1000 feet due to the increased air density.



In conclusion:

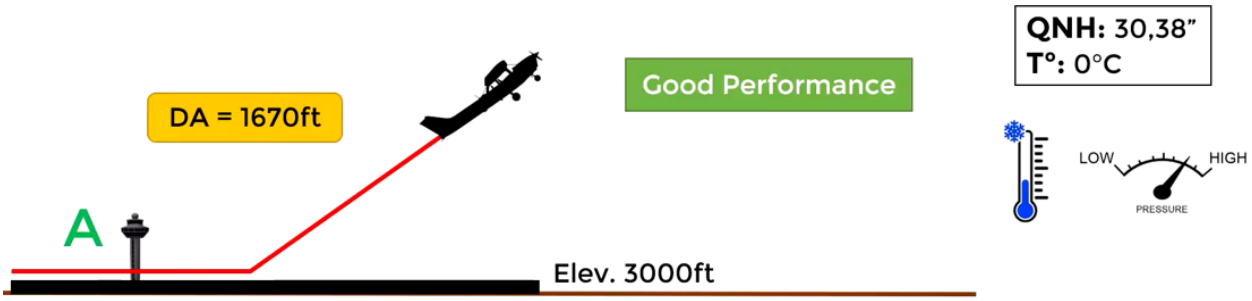
- If the density altitude is low, it means that the air density is high.
- If the density altitude is high, it means that the air density is low.



High Pressure and Low Temperature Example

Airport "A" with an elevation of 3000 feet reports a QNH of 30.38" Hg and a temperature of 0° C, this indicates a relatively high pressure and low temperature. In this situation:

- The air density to be high. Thus, it means the density altitude will be low.
- In this case, DA is at 1670 feet.
- In other words, aircrafts operating at this airport will behave as if they were flying at 1670 feet. Which means that they will present a good performance due to high air density.



Low Pressure and Hight Temperature Example

Airport "B" which also has an elevation of 3000 feet reports a QNH of 29.80" Hg and a temperature of 30° C, this indicates a relatively low pressure and high temperature. In this situation:

- A low air density, Thus, density altitude will be high.
- In this case, the DA is at 5600 feet.
- In other words, aircraft operating at this airport will behave as if they were flying at 5600 feet which means that they will present a poor performance due to the reduced air density.



How to Calculate Density Altitude

The following are the key methods for calculating Density Altitude:

- Flight Computer
- Density altitude charts
- Formula

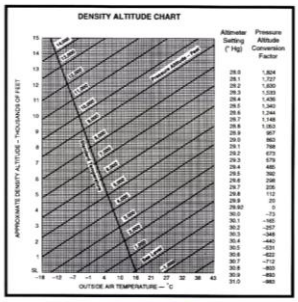
Some, calculate and ATIS broadcast the density altitude based on the current conditions that way the pilot doesn't need to calculate it manually.

- There are several methods for calculating the density altitude:

Some ATIS calculate and broadcast DA information.



Flight Computer



Table

DA = PA + 120 (OAT- ISA)

Approximate formula

Fight Computer

A device used for performing calculations essential for flight planning and navigation. It's essentially a circular slide rule, often called an E6-B or CRP-1, etc. While electronic flight computers exist, the manual E6-B remains popular due to its reliability, ease of use, and lack of reliance on electrical power.

Below is a simulator for E6-B:

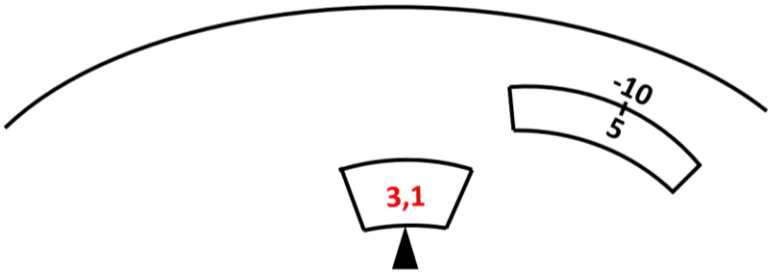
https://mediafiles.aero.und.edu/aero.und.edu/aviation/trainers/e6b/

We will use the inner scales specifically the little scale in the center in the upper right scale. To calculate the DA we must have the Outside Air Temperature (OAT), and the Pressure Altitude (PA)

Step 1: Align the PA with the corresponding OAT in the upper right scale

Step 2: Read the resulting density altitude in the center scale.

Determine the density altitude with a PA of 5000ft and an OAT = -10°C.

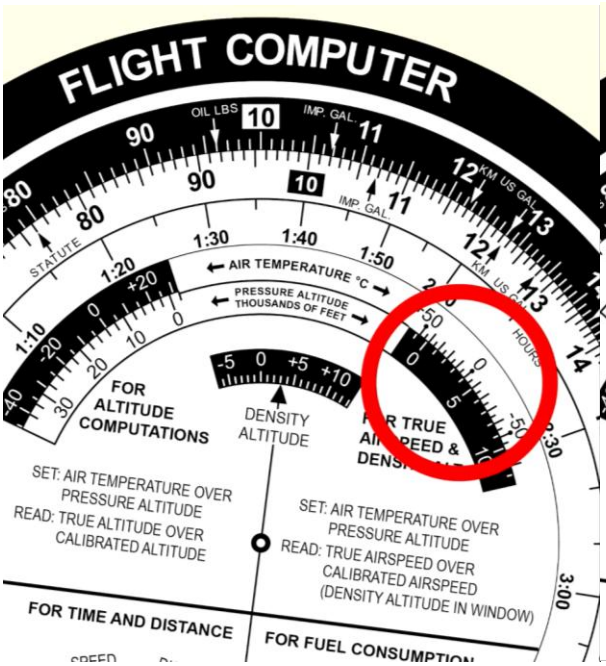


DA = 3100ft

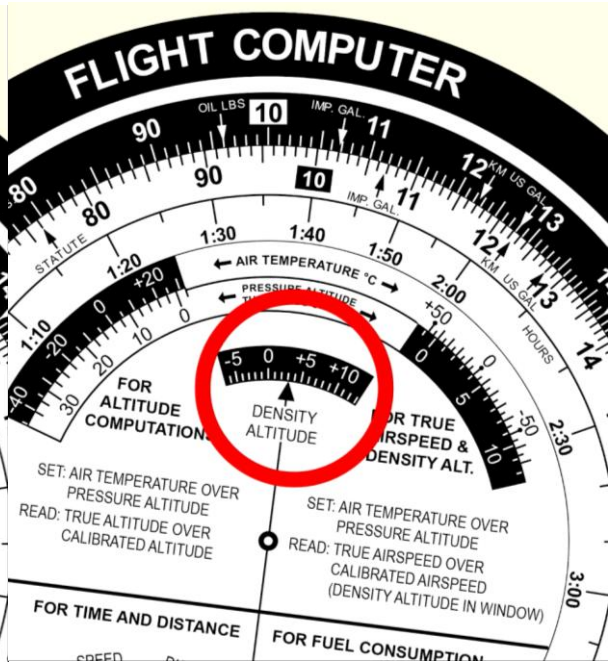
Step 1: align 5000 feet with minus 10 degrees in the upper right scale

Step 2: once done we read 3.1 in the center scale which corresponds to a DA of 3100 feet

Step 1:



Step 2:



APPROXIMATE FORMULA

- It is required to know in advance the pressure altitude, the current temperature and the ISA temperature at that altitude.

$$DA = PA + 120\,(OAT - ISA)$$

DA = Density Altitude

PA = Pressure Altitude

OAT = Actual outside air temperature

ISA = Standard temperature at that altitude

- How to determine ISA temperature for a given pressure altitude?

$$ISA = 15 - \frac{(2 \times ALT)}{1000}$$

EXAMPLE:

PA: 5000ft
OAT: -10°C

$$ISA = 15 - \frac{(2 \times 5000)}{1000} \Rightarrow ISA = 15 - (10) \Rightarrow ISA = 5°C$$

DA= PA + 120 (OAT - ISA)

DA= 5000 + 120 (-10 - 5)

DA= 5000 + 120 (-15)

DA= 5000 + (-1800)

DA= 3200ft

Calculating DA using graph

Using the same previous example, where PA = 5000 feet and OAT = -10° C.

Step 1: Identify the line that represents 5000 feet of PA. Which is this one highlighted in yellow.

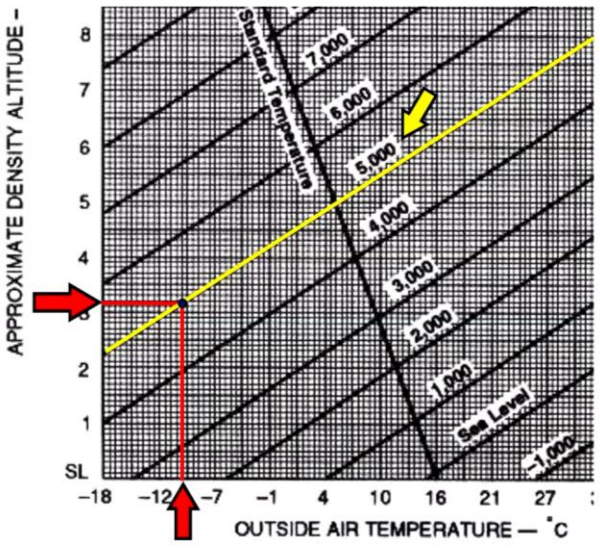
Step 2: In the X-axis we look for the current air temperature which is -10° C.

Step 3: Draw a vertical line until intercepting the 5000 feet line

Step 4: Finally, draw a horizontal line to the left and read the resulting DA. Which in this case is 3150 feet approximately.

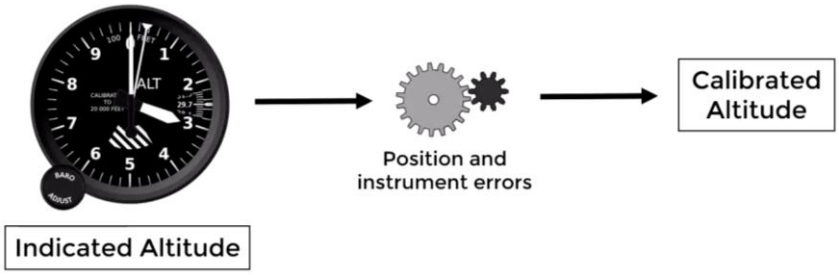
TABLE

PA: 5000ft
T°: -10°C



DA= 3150ft

Calibrate Altitude - The indicated altitude corrected for position and instrument errors. However, usually these errors tend to be small. In practice it is normally assumed that the indicated altitude and calibrated altitude are roughly equal.



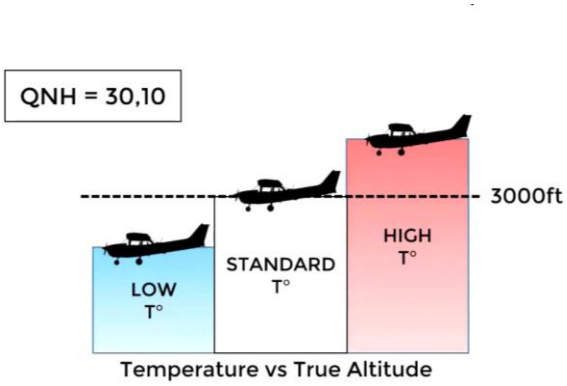
In practice it is assumed that the calibrated and indicated altitude are roughly equal.

To obtain the calibrated altitude, the manufacturer publishes an altitude calibration table in the aircraft manual which specifies the corrections to be applied to obtain the calibrated altitude. In this table altitude airspeed and aircraft configuration are considered to determine the correction values.

True Altitude: Actual altitude in relation to mean sea level. It is obtained by <u>correcting the indicated and calibrated altitude for non-standard temperature and pressure</u>.

Temperature corrections: The altitude reading is affected by the temperature of the air. For example, if an aircraft is flying under standard temperature conditions with the current QNH of 30.10" Hg adjusted on the altimeter it will indicate the true altitude.

- If the temperature is lower than standard, even using the correct QNH the true altitude will be lower than the indicated altitude.
- if the temperature is higher than standard, the true altitude will be higher than the indicated altitude.

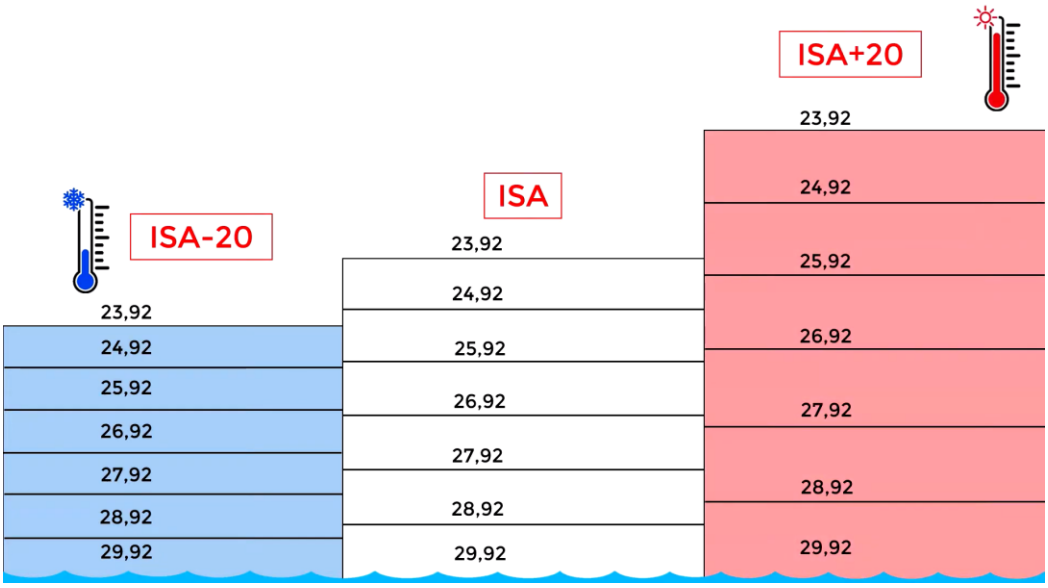


The altitude reading is affected by air temperature.



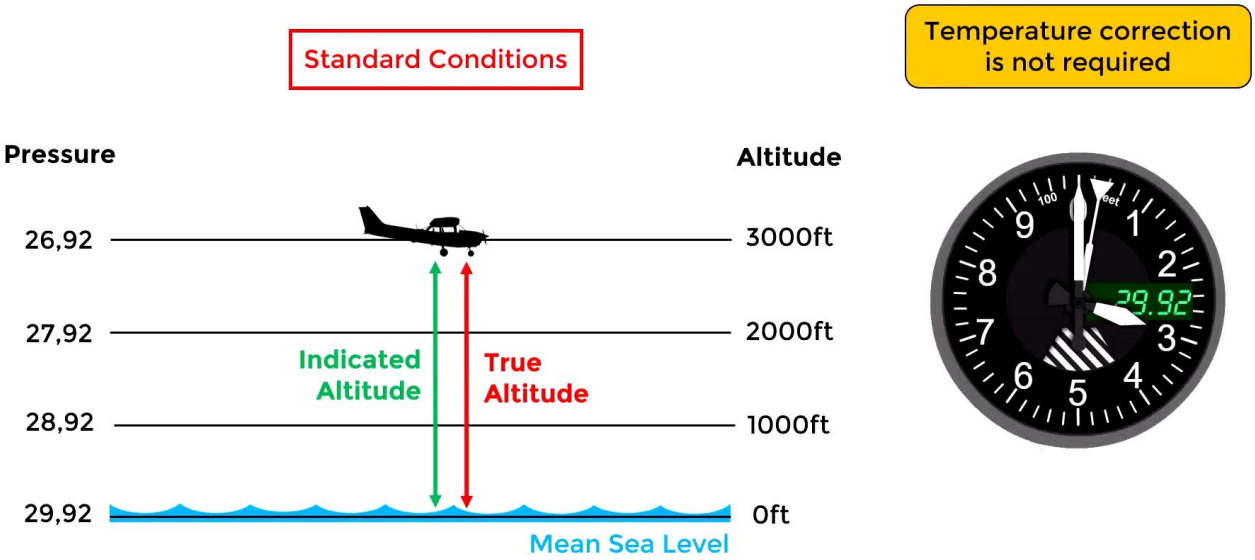
This temperature error occurs due to the pressure distribution with altitude at different temperatures. In the below graph, we have the typical distribution of isobars under standard temperature conditions. When the temperature is lower than the standard, the pressure changes more rapidly with altitude and therefore, the isobars will be less spaced. On the other hand, when the temperature is higher than standard, the pressure changes slowly with altitude and therefore, the isobars will be further

apart and as we can see the pressure at sea level is the same on all three cases which means that the QNH is the same the only thing that changes is the pressure distribution with altitude. As we can see the pressure at sea level is the same on all three cases which means that the QNH is the same the only thing that changes is the pressure distribution with altitude.



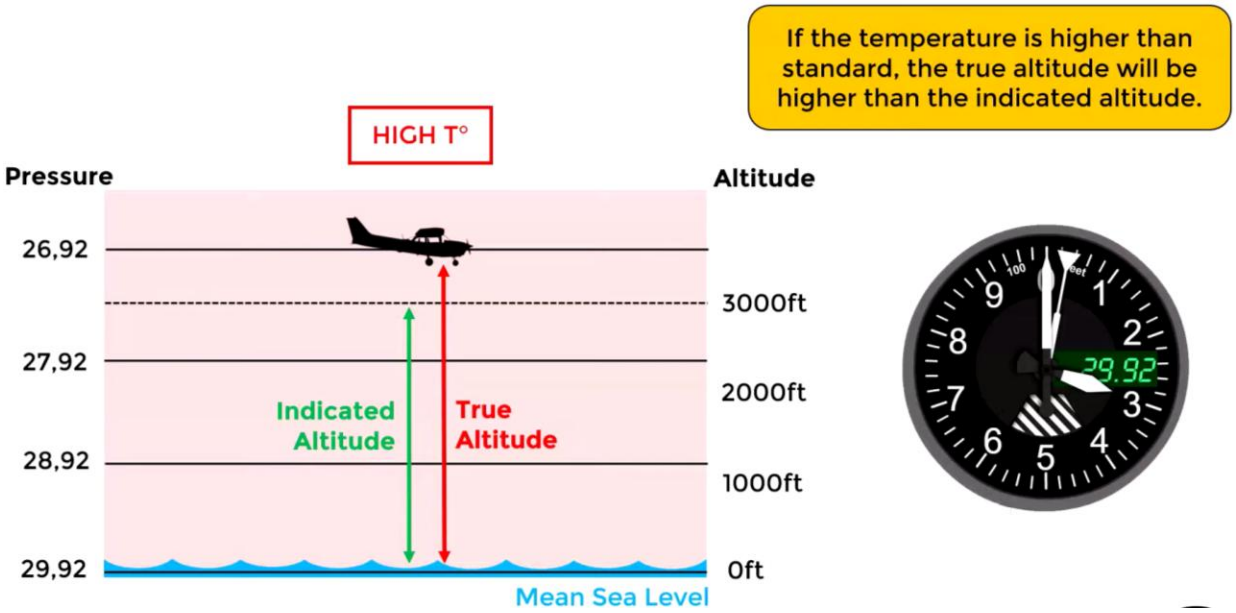
Standard Conditions

Under standard temperature conditions the indicated altitude and true altitude will be the same. Therefore, a temperature correction is not required.



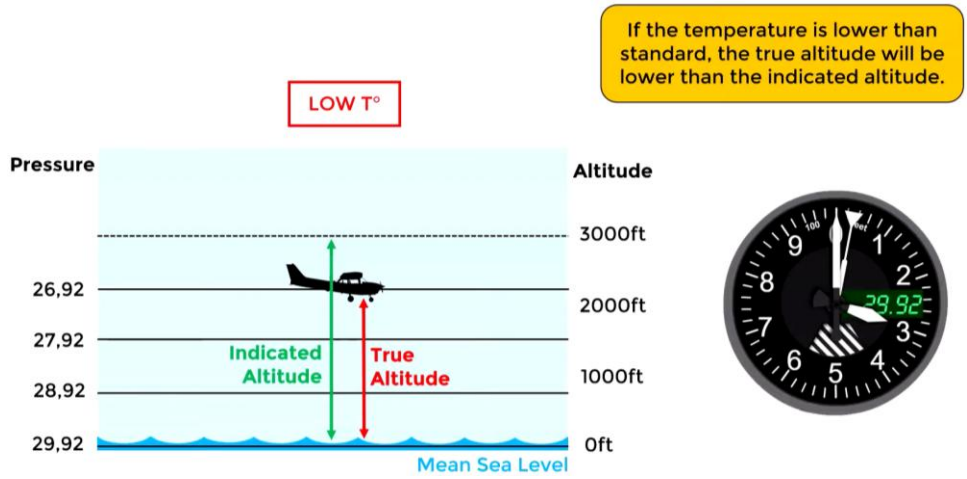
High Temperature Conditions

In high temperature conditions the isobars will be further apart. As we can see in this case the true altitude is higher than the indicated altitude.



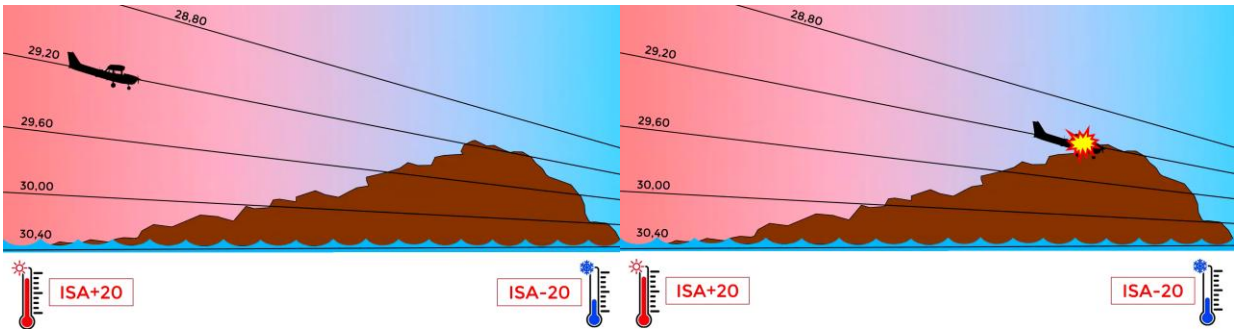
Low Temperature Condition

If the temperature is low in this case the isobars will be closer together. Therefore, the true altitude will be lower than the indicated altitude.

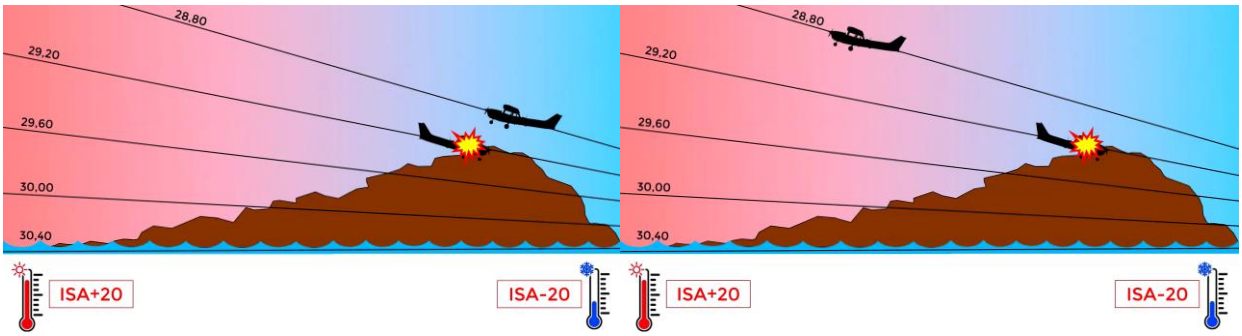


In the below images we can see change in the spacing of the isobars depending on the temperature. It is important to note that the pressure at sea level remains constant. Therefore, the QNH is the same in both areas (hot and cold).

The aircraft flies from a high temperature area to a low temperature area. Even though it uses the correct QNH it will gradually descend which could constitute a risk of collision with terrain.



Conversely if an aircraft flies from a low temperature area to a high temperature area it will gradually climb despite its altimeter will be indicating a constant altitude.



Methods of calculating True Altitude

There are several methods to calculate the true altitude:



Flight Computer

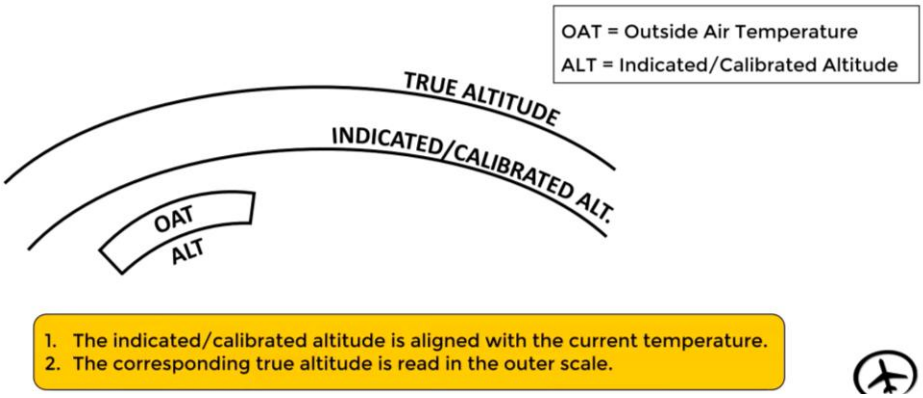
$$TA = ALT + \frac{4\ (ALT)\ \Delta ISA}{1000}$$

Approximate formula

Flight computer – We must know the indicated or calibrated altitude and the current temperature at that altitude.

Step 1: Align the altitude with the corresponding temperature.

Step 2: Read the corresponding true altitude in the outer scale.

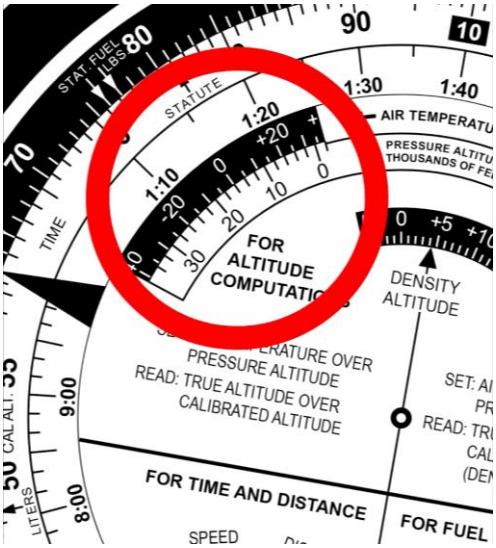


Example

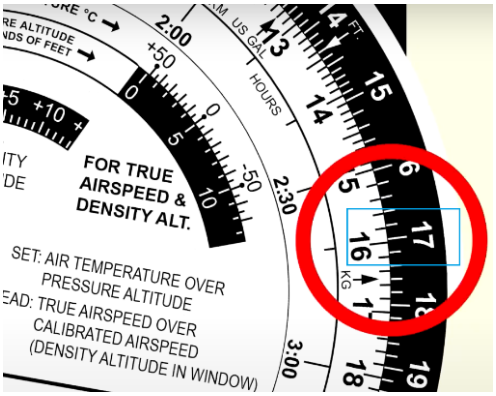
OTA = 0° C

ALT = 16,000 feet

Step 1: align the altitude of sixteen thousand feet with 0° C on the scale.



Step 2: Look for sixteen thousand feet in the middle scale and read the corresponding true altitude on the outer scale.



In this case, the true altitude is ~17,000 feet, this means that even though the altimeter is indicating 16,000 feet with the current QNH. The aircraft is actually flying at seventeen thousand feet due to the relatively high temperature.

Determine the true altitude if flying at an indicated altitude of 16,000ft, with a temperature of 0°C. ➡ **17.000ft**



Calculating True Altitude Using Formula

- It is required to know in advance the ISA deviation (ΔISA).

$$TA = ALT + \frac{4\ (ALT)\ \Delta ISA}{1000}$$

TA = True Altitude

ΔISA = ISA Deviation

ALT = Indicated/Calibrated Altitude

- How to determine ISA deviation?

1. Find ISA temperatura at that altitude

$$ISA = 15 - \frac{(2 \times ALT)}{1000}$$

2. Determine ISA deviation

$$\Delta ISA = (OAT - ISA)$$

Determine the true altitude if flying at an indicated altitude of 16,000ft, with a temperature of 0°C.

$$ISA = 15 - \frac{(2 \times 16000)}{1000}$$

$$ISA = -17°C$$

$$\Delta ISA = (0°C - (-17°C))$$

$$\Delta ISA = +17°C$$

$$TA = 16000 + \frac{4\ (16000)\ (+17°C)}{1000} \rightarrow TA = 16000 + 1088 \rightarrow TA = 17088ft$$

Practical Application of True Altitude

Under normal conditions only pressure correction is applied, this means that the only thing that is done is adjust the QNH on the altimeter. Temperature correction is not usually applied so the altitude used in most operations is the indicated altitude rather than the true altitude.



INDICATED ALTITUDE

The current QNH is adjusted to correct the reading for pressure.

Usually, temperature correction is not applied

Temperatures well below standard require an addition of a safety factor to the minimum flight altitudes to ensure adequate separation from terrain and obstacles. The correction factors to be used are published in the ICAO document 8168.



Table 2-4-1 b). Values to be added by the pilot to minimum promulgated heights/altitudes (ft)

Aerodrome temperature (°C)	*Height above the elevation of the altimeter setting source (feet)*													
	200	300	400	500	600	700	800	900	1 000	1 500	2 000	3 000	4 000	5 000
0	20	20	30	30	40	40	50	50	60	90	120	170	230	280
–10	20	30	40	50	60	70	80	90	100	150	200	290	390	490
–20	30	50	60	70	90	100	120	130	140	210	280	420	570	710
–30	40	60	80	100	120	140	150	170	190	280	380	570	760	950
–40	50	80	100	120	150	170	190	220	240	360	480	720	970	1 210
–50	60	90	120	150	180	210	240	270	300	450	590	890	1 190	1 500

This table is designed for an aerodrome at sea level.

For example, the minimum descent altitude or MDA of a particular instrument approach procedure is 400 feet and the temperature is - 40 ° C. Using the table we can see that the correction to be added to the MDA is 100 feet obtaining a corrected minimum descent altitude of 500 feet.

Most airlines establish specific policies as to when temperature correction should be applied.

Minimum Descent Altitude (MDA) = 400ft
Aerodrome temperatura = -40°C

400ft + 100ft = 500ft

Corrected MDA= 500ft

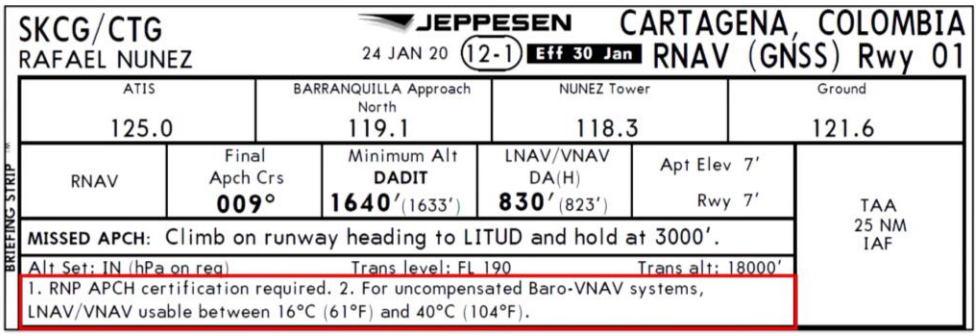
Table 2-4-1 b). Values to be added by the pilot to minimum promulgated heights/altitudes (ft)

Aerodrome temperature (°C)	*Height above the elevation of the altimeter setting source (feet)*													
	200	300	400	500	600	700	800	900	1 000	1 500	2 000	3 000	4 000	5 000
0	20	20	30	30	40	40	50	50	60	90	120	170	230	280
–10	20	30	40	50	60	70	80	90	100	150	200	290	390	490
–20	30	50	60	70	90	100	120	130	140	210	280	420	570	710
–30	40	60	80	100	120	140	150	170	190	280	380	570	760	950
–40	50	80	100	120	150	170	190	220	240	360	480	720	970	1 210
–50	60	90	120	150	180	210	240	270	300	450	590	890	1 190	1 500

In some cases like in the approach procedures with vertical guidance that use uncompensated BARO-VNAV systems and operational temperature range is specified in the chart.

For example, here the second note establishes that this procedure may be executed without temperature correction provided the temperature is between 16 and 40 ° C. If the current temperature is outside this range the crew must apply the corresponding temperature corrections to all the altitudes of the procedure.

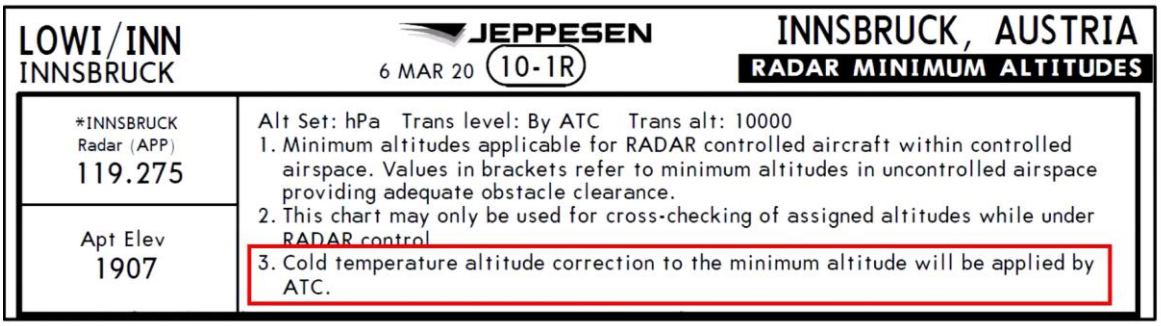
For uncompensated BARO-VNAV systems, an approved temperature range is specified.



If the temperature is outside this range, altitudes must be manually compensated for temperature in order to perform the procedure.

The civil aviation authority of each province/state/country usually takes into account the effect of temperature when establishing minimum flight altitudes so that the pilot does not have to manually calculate the correction. Here we can see an example, this is a radar minimum altitudes chart for Innsbruck, Austria. As we can see it says that ATC will apply the cold temperature corrections when

necessary. Therefore, the pilots doesn't have to worry about it normally this is done by making a survey of the average temperatures in the area and based on this the most conservative safety margins are applied.



LOWI/INN
INNSBRUCK

JEPPESEN
6 MAR 20 10-1R

INNSBRUCK, AUSTRIA
RADAR MINIMUM ALTITUDES

*INNSBRUCK
Radar (APP)
119.275

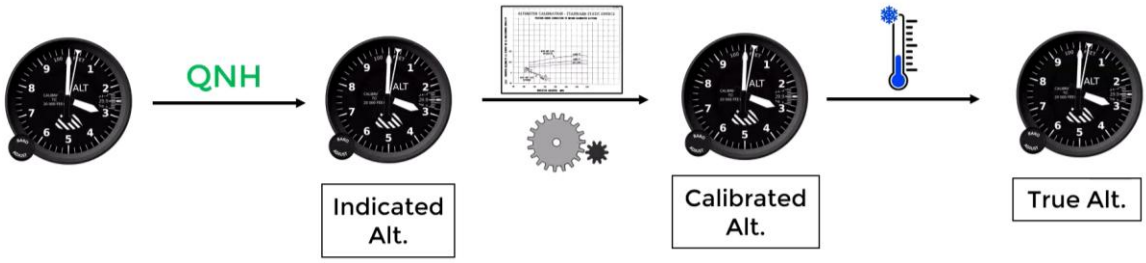
Apt Elev
1907

Alt Set: hPa Trans level: By ATC Trans alt: 10000
1. Minimum altitudes applicable for RADAR controlled aircraft within controlled airspace. Values in brackets refer to minimum altitudes in uncontrolled airspace providing adequate obstacle clearance.
2. This chart may only be used for cross-checking of assigned altitudes while under RADAR control.
3. Cold temperature altitude correction to the minimum altitude will be applied by ATC.

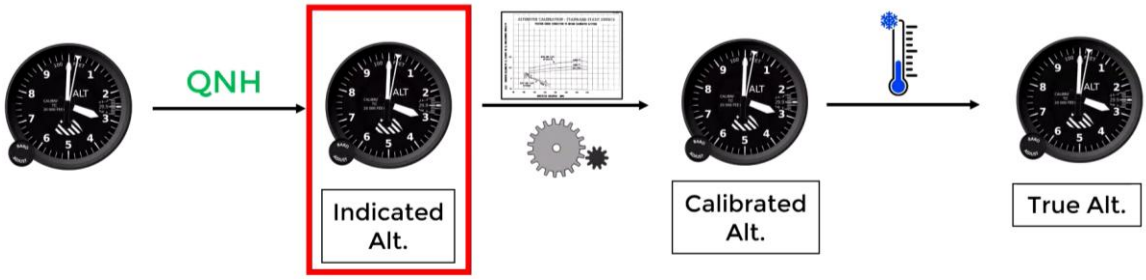
A study is made of the average temperatures in the area and, based on this, the most conservative safety margins are applied.

Summary

If we adjust the current QNH on the altimeter we will obtain the indicated altitude, if we then apply the instrument and position error corrections we obtain the calibrated altitude, and if we then apply the temperature correction we obtain the true altitude.



In practice the indicated altitude is the most commonly used so the pilot only needs to worry about updating the QNH on a regular basis while flying.



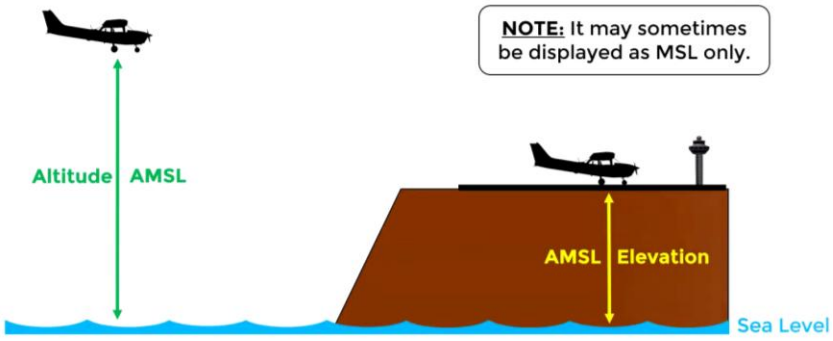
ABBREVIATIONS

Vertical position of an aircraft in flight can be expressed in different terms, such as:

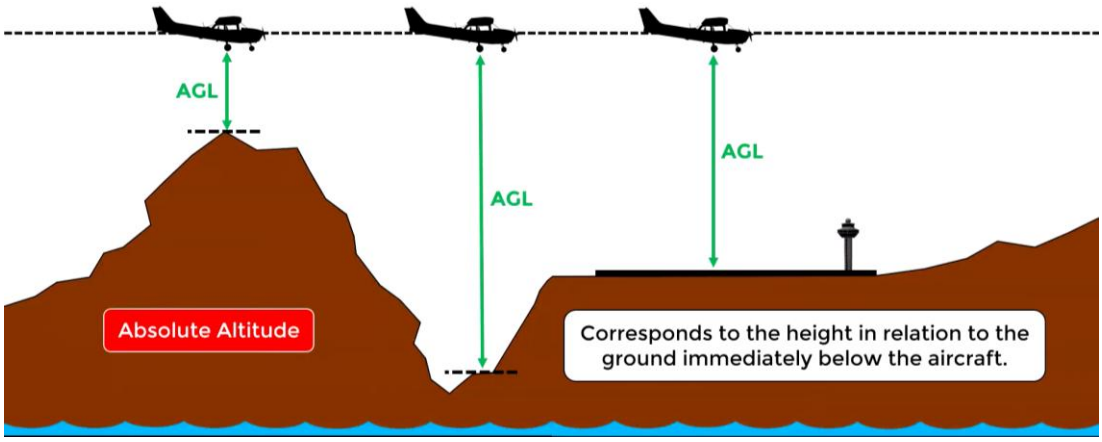
- Altitude
- Height
- Elevation
- Flight Level

Various abbreviations are used to refer to these terms, let's review them:

- **AMSL – Above Mean Sea Level**
  - This term is used to express either altitudes or elevations, since, both of them use the mean sea level as reference.
  - Whenever the reference used is the mean sea level it will be used.
  - This term may be displayed as MSL only



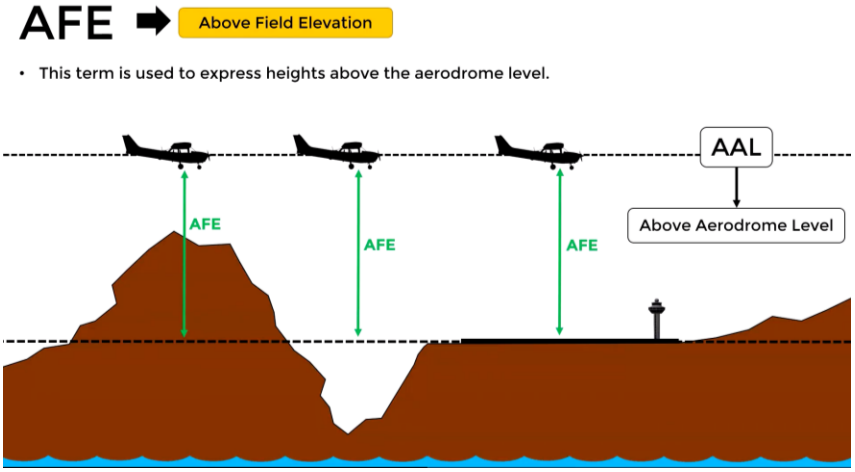
- **AGL – Above Ground Level**
  - Expressed as height above a point on the ground.
  - Refers to the height in relation to the ground immediately below the aircraft not the height in relation to the aerodrome level.



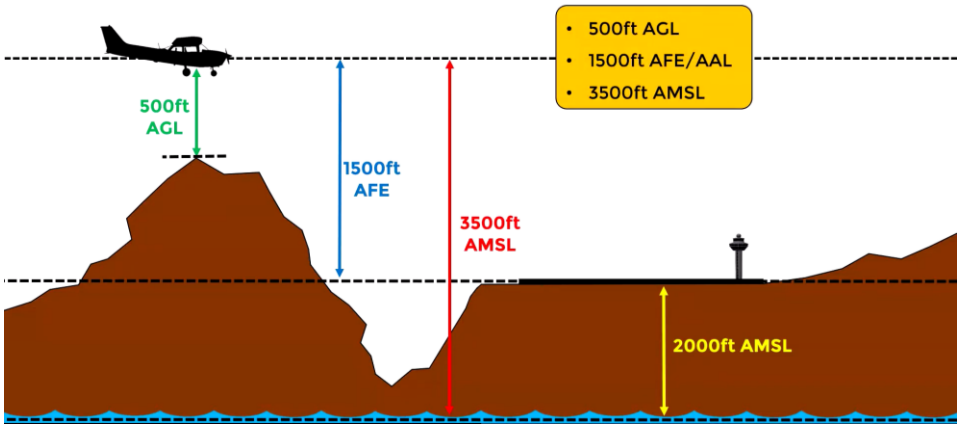
  - If the aircraft maintains level flight and it passes over this valley its AGL height will actually be greater. Since, the height is now being measured in relation to the valley and if the aircraft keeps flying and passes over this airport its AGL height will change once again.
  - Sometimes referred to as **Absolute Altitude.**

- AGL height can be measured by radio altimeter. The radio altimeter or radar altimeter is the only instrument that can provide an accurate AGL reading in real time. It emits radio waves vertically to the ground where they impact and bounce back to the aircraft so based on the time it takes for a wave to go and return is how the current AGL height is determined and shown in the instrument.

- **AFE - Above Field Elevation**
  - Specify heights in relation to a certain aerodrome level.
  - Regardless of the terrain conditions below the aircraft, if it keeps level flight the AFE height will remain constant.
  - Sometimes instead of using AFE the abbreviation AAL i.e. Above Aerodrome Level is used.



- Let's see an example combined with AMSL, AGL, and AFE



Recommended videos for these topics:

Flight Instruments
https://www.youtube.com/playlist?list=PLJ-0SVdtegU-SS5Kp12Yw70iJbx4ApHhD
Why Your Altimeter Lies in Hot Weather (and why it matters on an IFR approach)
https://www.youtube.com/watch?v=aY9ueoxCcdc

Altimeter Regions

Canadian airspace is divided, for purposes of altimetry, into an Altimeter Setting Region and a Standard Pressure Region. The geographic limits of these two regions are illustrated in Fig.4.19.

The Altimeter Setting Region

Prior to take-off from an airport located in the Altimeter Setting Region, a pilot must set the aircraft altimeter to the current altimeter setting of that airport or, if that altimeter setting is not available, to the elevation of the airport.

While cruising within the Altimeter Setting Region, pilots must adjust their altimeters to the reported current altimeter setting of the nearest station along their route of flight, or where such stations are separated by more than 150 nautical miles, to the nearest station to the route of flight.

When approaching the airport of intended landing, the altimeter must be set to the current altimeter setting of that airport. The Altimeter Setting Region includes the airspace up to 18,000 feet only.

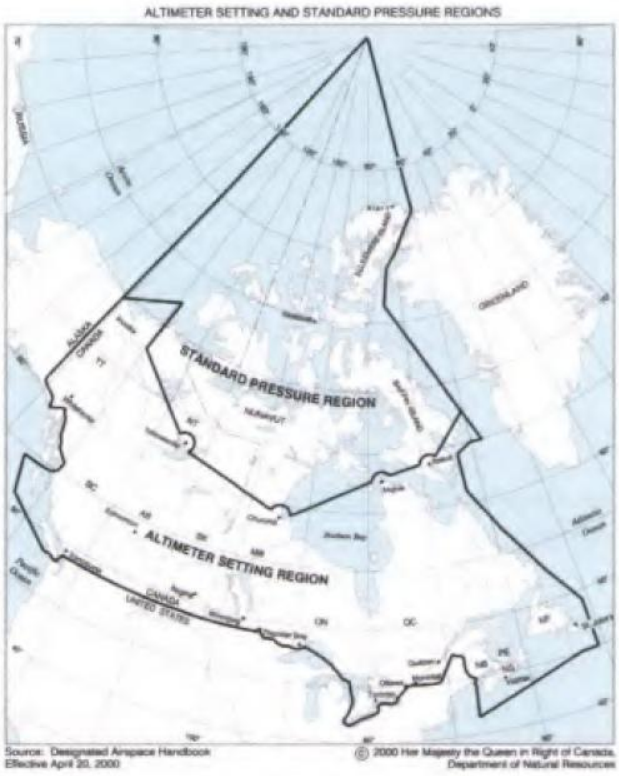


Fig.4.19 Altimeter Regions

The Standard Pressure Region

The Standard Pressure Region includes that area of Northern Canada depicted and so designated in Fig.4.19 and also all airspace above 18,000 feet anywhere in Canada.

In the Standard Pressure Region, the altimeter is set to standard pressure (29.92" Hg). For take off and climb, from an airport in this region, the altimeter should be set to the current altimeter setting (or the airport elevation) and reset to the standard pressure setting upon reaching the cruising

altitude. For descent and landing, the current altimeter setting of the airport of intended landing should be set on the altimeter.

The Altimeter Setting Region must be considered in the context of possible altimeter pressure errors. When the altimeter setting is set on the barometric scale, the altimeter will register the indicated height above sea level. When you land at an airport from which the altimeter setting was obtained by radio, your altimeter will record the altitude of the field above sea level.

The altitude at which an aeroplane flies when using indicated height above mean sea level (MSL) as its reference is known as cruising altitude. The area in which the indicated height above sea level is used is known as the altimeter setting region.

When taking off within the altimeter setting region, or approaching to land, the altimeter should be set to the current altimeter setting of the field. If the altimeter setting is not available, prior to take-off, you can set the altimeter to read the elevation of the field above sea level. During flight, the altimeter should be progressively reset to the altimeter setting of the nearest station along the route of flight.

The Standard Pressure Region must be considered in the context of possible altimeter pressure errors. Over transoceanic routes and certain continental areas, altimeter settings are not available. Flights over these areas are flown at pressure altitude (defined below). Because the altimeter is less reliable at high altitudes, flights above 18,000 feet msl are also flown at pressure altitude.

When taking off within the standard pressure region, the altimeter should be set to the current altimeter setting of the field; if this is not obtainable, to the elevation of the field above sea level. Immediately before reaching cruising altitude, the altimeter should be re-set to 29.92" Hg. When approaching to land at an airport within the Standard Pressure Region, the altimeter should be re-set to the current altimeter setting of the field just prior to commencing the descent from cruising altitude or just prior to descending below the flight level at which a holding procedure has been conducted.

When flying out of the Standard Pressure Region into the Altimeter Setting Region, or vice versa, the altimeter should be adjusted to the setting appropriate to the area in which the aeroplane is being flown.

Field level pressure is the actual barometric pressure (not corrected to sea level) at any particular airport. If a pilot obtains the field level pressure by radio from an airport they are approaching and sets it on their barometric scale, their altimeter will register 0 feet when they land.

Summary of Altimeter Regions

Altimeter Setting Region (ASR)

- **Covers:** All airspace **up to 18,000 ft**.
- **Altimeter setting (QNH):**
    - **Before takeoff:** Set to airport's current altimeter setting (or airport elevation if unavailable).

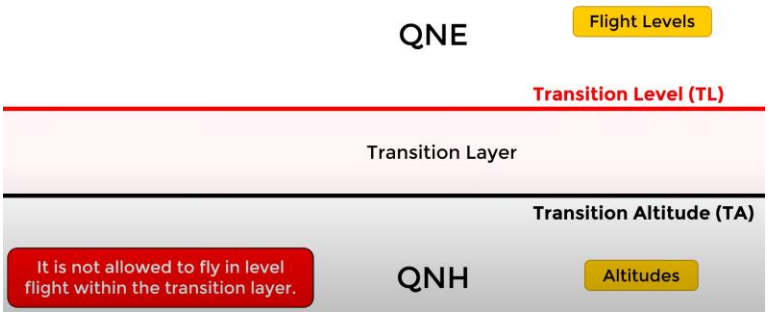
    - **While cruising:** Update to **nearest station's** altimeter setting (use closest station if >150 NM apart).
    - **Before landing:** Set to destination airport's altimeter setting.
- **Purpose:** Altimeter setting (QNH) reads **altitude ASL**.

Standard Pressure Region (SPR)

- **Covers:**
    - **All airspace above 18,000 ft (FL 180)** across Canada.
    - **Northern Canada** as depicted on charts (refer to the map above).
- **Altimeter setting:**
    - **Cruising:** Set to **standard pressure 29.92" Hg**.
    - **Takeoff/climb:** Use airport's altimeter setting, then switch to **29.92"** when reaching cruising altitude.
    - **Descent/landing:** Switch back to airport's altimeter setting before arrival.
- Must reset altimeter when entering/exiting SPR unless ATC says otherwise.

Transition Level — FL 180

- **Not used for cruising in Canada.**
- **Climb:** Passing **upward** through FL 180 → **Change** current altimeter setting **mean sea level (QNH) → 29.92"**.
- **Descent:** Passing **downward** through FL 180 → **Change 29.92" → QNH**.



Topic	Remember
Standard Pressure Region	Northern Canada + all airspace above 18,000 ft
Standard Pressure Setting	29.92" Hg
Altimeter Setting Region	Uses local airport altimeter setting
Above 18,000 ft	Pressure altitude (29.92)
Below 18,000 ft	Altitude above MSL using local setting
Takeoff in SPR	Local setting → then 29.92 before cruise
Landing in SPR	Reset to destination setting before descent
Takeoff in ASR	Use local setting or airport elevation
Enroute in ASR	Update to nearest station's setting

Moving between regions	Change to appropriate setting
Field Level Pressure	Altimeter reads 0 ft at touchdown
Altimeter Setting	Altimeter reads airport elevation when landed

ALTIMETER TEMPERATURE ERROR

When an aircraft is in flight, and due to Due to continual heating and cooling, it can be assumed that the altitude indication of an altimeter is always in error as a result of temperature variations. The amount of error depends on the degree to which the average temperature of the column of air between the aircraft and the ground differs from the average temperature of the standard atmosphere for the same column of air.

If the actual temperature of the air column in which the aeroplane is flying is colder tan standard air, the true altitude of the aeroplane above sea level will be lower than the indicated altitude. If the actual temperature is warmer than standard air, the true altitude will be higher.

Since all altimeters in the same area are equally affected by temperature error, air traffic regulations require you to fly only at indicated altitude.

Abnormally cold temperature factors can result in altimeter temperature errors. It is important for a pilot to be able to calculate his/her true altitude above sea level. This is particularly essential in mountainous areas when the temperatures are very low. Pressure altimeters are calibrated to indicate true altitude under conditions of international standard atmosphere (ISA). Any deviation from ISA will result in an erroneous reading on the altimeter. In very cold temperatures, the true altitude will be much lower than the indicated altitude and could be critical in obstacle clearance. The very cold, dense air will cause an altimeter error of as much as 20%. The altimeter might read 8,000 feet, for example, when the true altitude of the aircraft is only 6,400 feet. In IFR operation, an aircraft maintaining the minimum obstacle clearance altitude (MOCA) while flying through the mountains may be well below a safe altitude if the pilot is trusting the altimeter indication and has not calculated the true altitude. Pilots experienced in winter operations in the mountains must always be conscientious about calculating true altitude before commencing any instrument approach to landing in order to ensure the safety of the flight.

The values derived from an altitude correction chart such as the one shown in Fig.2.50 should be added to the published procedure altitudes, minimum sector altitudes and DME arcs to ensure adequate obstacle clearance. The altimeter setting of the destination airport should be that used in the calculations.

Height above the elevation of the altimeter source (feet)														
A/D Temp °C	200	300	400	500	600	700	800	900	1000	1500	2000	3000	4000	5000
0°	0	20	20	20	20	40	40	40	40	60	80	140	180	220
-10°	20	20	40	40	40	60	80	80	80	120	160	260	340	420
-20°	20	40	40	60	80	80	100	120	120	180	240	380	500	620
-30°	40	40	60	80	100	120	140	140	160	240	320	500	660	820
-40°	40	60	80	100	120	140	160	180	200	300	400	620	820	1020
-50°	40	80	100	120	140	180	200	220	240	360	480	740	980	1220

Note: Values should be added to published altitudes

Example: Aerodrome Elevation 2262 Aerodrome Temperature -50°C

	Altitude	HAA	Correction	Indicated Alt.
Procedure Turn	4000 ft.	1738 ft.	+420 ft.	4420 ft.
FAF	3300 ft.	1038 ft.	+240 ft.	3540 ft.
MDA Straight-in	2840 ft.	578 ft.	+140 ft.	2980 ft.
Circling MDA	2840 ft.	578 ft.	+140 ft.	2980 ft.

Fig.2.50 Altitude Correction Chart

To calculate true altitude, it is important to know that all computers currently in use are fitted with a sliding scale for correcting temperature error and converting indicated altitude to true altitude. However, the correction is based on the pressure altitude rather than the indicated altitude. To find the former at any time, simply set the barometric scale to 29.92 and the altimeter will record the pressure altitude.

THE AIRSPEED INDICATOR (ASI)

The Airspeed Indicator AKA **ASI** tells:

- The speed at which the plane is travelling through the air (not over the ground).
- The dial is calibrated in KTS and miles per hour.



Fig.2.51 Airspeed Indicator

Static Pressure

- static pressure is the same as the atmospheric pressure
- it is exerted equally in all directions regardless of the object is moving or is still

Dynamic Pressure

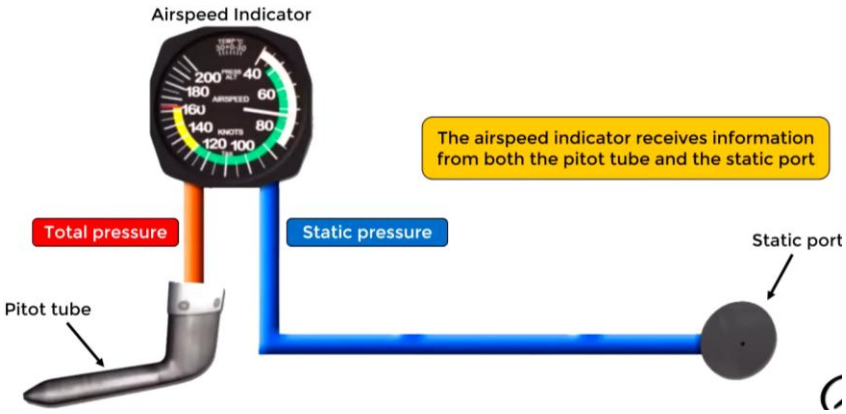
- Pressure produced by the air when an object moves through it.

- Is proportional to the speed and air density. Therefore, the faster we move the higher the dynamic pressure and vice versa.

Total Pressure

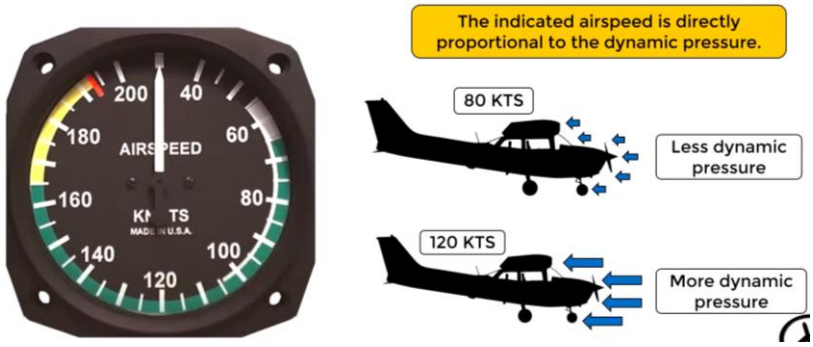
- Total Pressure = Dynamic + Static Pressure
- When measuring the pressure of a jet of air we will obtain the total pressure.

The airspeed indicator is connected to both the pitot and static pressure sources.



Reading of speed (through the air) or Dynamic Pressure = Pressure in the pitot pressure system (Total Pressure) - the pressure in the static system (Static Pressure).

The air speed is directly proportional to the dynamic pressure. If for example an aircraft flies at 80 KTS it will experience less dynamic pressure than if it were flying at 120 KTS.



Functioning of the instrument

If we look inside the instrument, we will see the following:

- The static port is directly connected to the inside of the case.
- The pitot tube is connected to a diaphragm/Aneroid Capsule (corrugated metal expansion box called the aneroid capsule) which in turn is connected to gears, moving the needles.

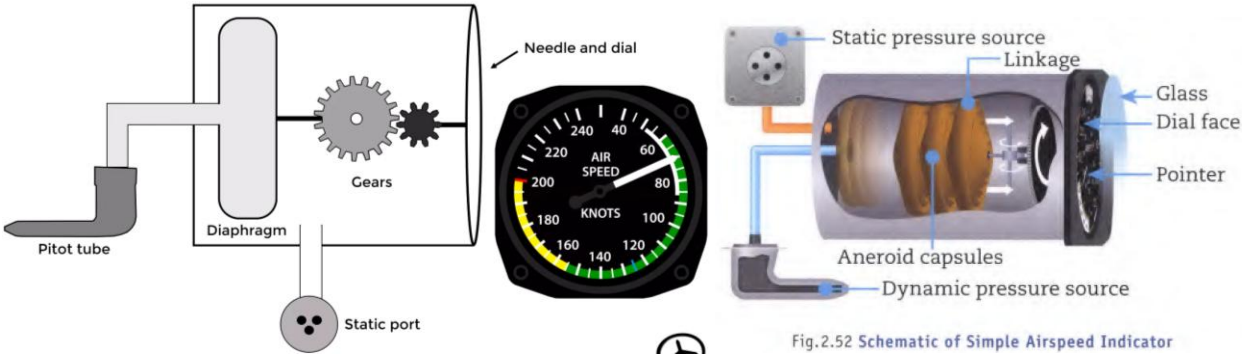
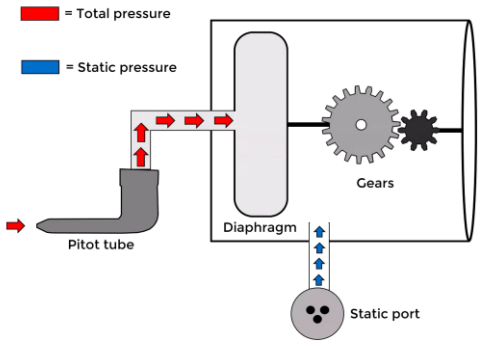
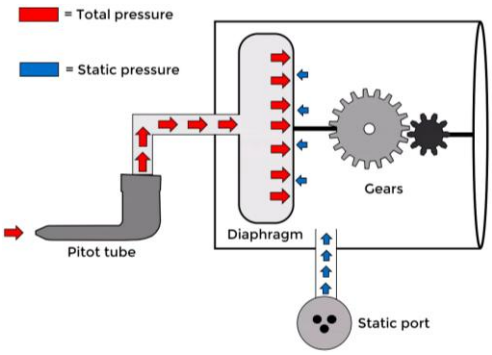


Fig.2.52 Schematic of Simple Airspeed Indicator

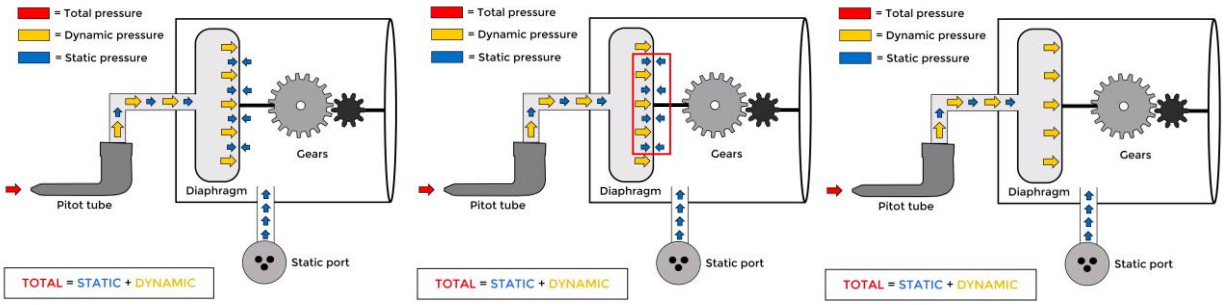
- The static port allows the whole case to be filled with static pressure
- The pitot tube causes the diaphragm to fill with total pressure



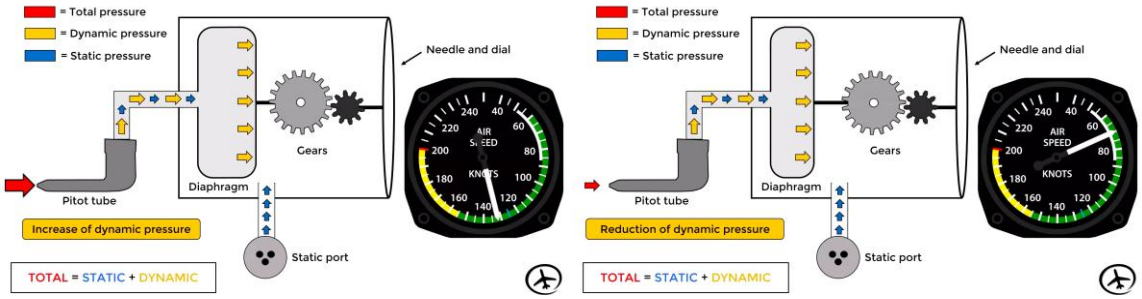
- The total pressure inside the diaphragm tries to expand it while the static pressure in the case tries to contract



- As we know, Total Pressure = Static pressure + dynamic pressure. We can express it as a combination of these two pressures:
    - Part of the pressure inside the diaphragm is dynamic.
    - Part is static
- The static pressure inside the diaphragm is the same as outside of it. They can be cancelled out now there's only dynamic pressure.



- Thus, the expansion or contraction of the diaphragm will depend only on dynamic pressure.
    - If dynamic pressure increases, the diaphragm will expand making the gears move the needle to indicate a higher speed.
    - If dynamic pressure reduces this will cause the diaphragm to contract making the gears move the needle once again but now to indicate a lower speed.



The ASI instrument is not totally perfect which means that the airspeed indication has some errors. That's the reason why we have different speed terms and definitions in aviation each of these speeds is corrected by certain errors. So, let's see each of them in more detail.

Indicated Airspeed (IAS) - The speed indicated on the instrument is not corrected for any type of error.

- It is the difference between the pressure in the pitot pressure system (total pressure) and the pressure in the static system (static pressure).
- Expressed in KTS or Miles/hour.

Errors associated with Indicated Airspeed

- **Instrument error** - Error due to manufacturing imperfections and wear of the gears inside the instrument.
- **Position Error** – Error produced due to changes of flaps and gear configuration, eddies that are formed as the air passes over the wings changes of angle of attack and some other

maneuvers. This error affects directly the pressure measured by both the pivot tube and the static port.

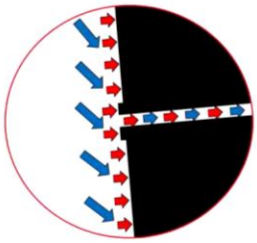
Small manufacturing imperfections and gear wear of the instrument.

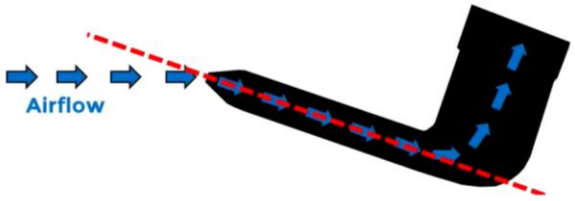




POSITION ERROR

Pitot-static system errors in pressure measurement due to maneuvers or changes in aircraft configuration.





Calibrated Airspeed (CAS) - Refers to the IAS corrected for the instrument and position errors. If we take the airspeed indicated on the instrument and we correct it for instrument and position errors, we will obtain the CAS.



Determination of Calibrated Airspeed

The manufacturers design special tables of airspeed calibration which are based on certain configurations and speeds, and they are published in the aircraft manual. In this example, this is an airspeed calibration table for a Cessna 172.

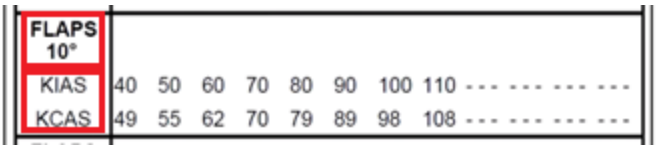
AIRSPEED CALIBRATION
NORMAL STATIC SOURCE

CONDITION:
Power required for level flight or maximum rated RPM dive.

FLAPS UP												
KIAS	50	60	70	80	90	100	110	120	130	140	150	160
KCAS	56	62	70	79	89	98	107	117	126	135	145	154
FLAPS 10°												
KIAS	40	50	60	70	80	90	100	110	---	---	---	---
KCAS	49	55	62	70	79	89	98	108	---	---	---	---
FLAPS 30°												
KIAS	40	50	60	70	80	85	---	---	---	---	---	---
KCAS	47	53	61	70	80	84	---	---	---	---	---	---

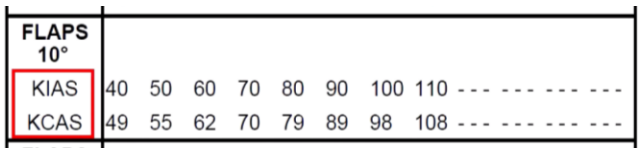
For example, if we fly with an indicated airspeed of 100 KTS with the flaps extended to 10 degrees. What will be our calibrated airspeed.

In this case we have to refer to the row of flaps 10 degrees and we can see that we have both the indicated and calibrated air speeds.



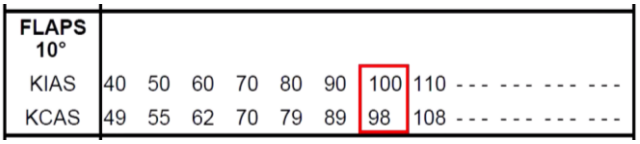
FLAPS 10°												
KIAS	40	50	60	70	80	90	100	110	---	---	---	---
KCAS	49	55	62	70	79	89	98	108	---	---	---	---

Note, when speeds are expressed in KTS it is common to include a K before the speed term.



FLAPS 10°												
KIAS	40	50	60	70	80	90	100	110	---	---	---	---
KCAS	49	55	62	70	79	89	98	108	---	---	---	---

Now in the table we just have to look for 100 KTS of indicated airspeed and then read the corresponding calibrated air speed of 98 KTS.



FLAPS 10°												
KIAS	40	50	60	70	80	90	100	110	---	---	---	---
KCAS	49	55	62	70	79	89	98	108	---	---	---	---

As we can see the difference between the indicated and calibrated air speeds is quite small. Therefore, in practice, it is common to assume that the indicated and calibrated air speed are roughly equal.

Compressibility Error

In the design of the pitted static system the air is assumed to behave as an ideal gas. One of the characteristics of an ideal gas is that it cannot be compressed. However, the reality is that when flying

at high speeds the air actually compresses. Thus, generating certain errors in the measurement of air speed.

Normally the compressibility effects start to become significant above 250 KTS. Therefore, this error is almost nil in small and slow aircraft, but if we fly at high speeds we must then add a compensation for the air compressibility factor.

Equivalent airspeed (EAS) is the calibrated air speed corrected for compressibility errors. So, in other words if we take the calibrated air speed and we apply the compressibility errors correction we get the equivalent airspeed.



This compressibility error correction is only carried out when flying at high speeds and the truth is that calculating this correction manually can be a bit complex that's why normally this correction is automatically calculated in real time by the avionics computers of the aircraft so the pilot doesn't have to worry about this correction.

$$EAS = \frac{CAS}{\left[1 + \frac{1}{8}(1-\delta)M^2 + \frac{3}{640}(1 - 10\delta + 9\delta^2)M^4\right]}$$

M is Mach number

P is static pressure

P_0 is standard sea level pressure (1013.25 hPa)

$$\delta = \frac{P}{P_0}$$

Normally this correction is calculated by the onboard avionics computers.

Density Error

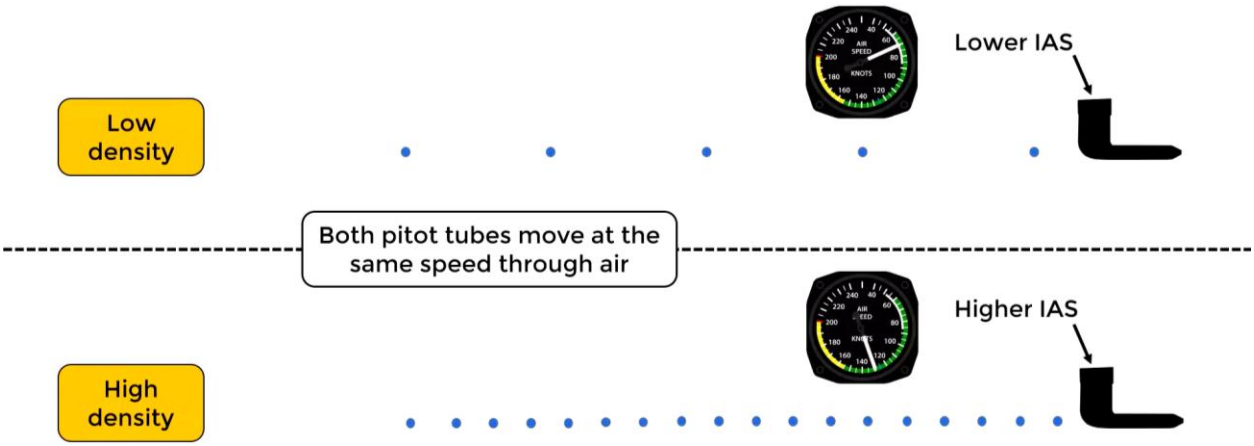
The dynamic pressure depends on two factors, the speed of the aircraft and the density of the air.

For example, here we have two pitted tubes from different aircraft. In this case the tube at the top will move in air with lower density which means that the air particles will be farther apart while the tube at the bottom will move in air with higher density which means that the air particles will be closer together.



In this case we can observe that the tube at the top as it moves in less dense air it captures less air particles which means that it experiences a lower dynamic pressure. Therefore, the air speed indicator will show a lower indicated air speed.

The tube at the bottom moving in denser air captures a larger amount of air particles which means that it experiences a higher dynamic pressure. Therefore, the airspeed indicator will show a higher indicated airspeed.

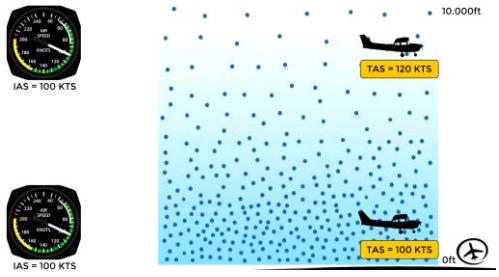


Both tubes are moving at the same speed through the air. The simple fact of moving in different densities result in different indicated air speeds. Therefore, the air speed reading needs to be corrected by density error which gives as a result the **true airspeed**.

True Airspeed (TAS) - The air speed corrected by density error.

- The density of air depends mainly on temperature and static pressure.
- The instrument is calibrated to indicate the true airspeed only at sea level under standard conditions. Thus, the airspeed indicator assumes that the density is always that of sea level and with standard temperature.
- At sea level the indicated air speed and the true air speed will be roughly equal. However, as we climb and the density starts reducing then this relationship will change.

For example, an aircraft flies at a very low altitude close to sea level and its airspeed indicator shows 100 KTS. Since near the sea level the indicated and true air speeds are roughly equal the true air speed will also be 100 KTS. However, if the aircraft flies higher in lower density air and its indicator also shows 100 KTS that would mean that the aircraft must be flying at a higher true airspeed in this case 120 KTS. This way the true airspeed continues to increase as we climb with a constant indicated airspeed.



Calculation of True Airspeed

There are two methods of calculating True Airspeed:

- Formula
- Flight Computer

Formula is based on the rule of thumb which says that the true air speed increases by two percent for each 1000 feet of altitude increase.

→1000 ft = 2% increase in airspeed.

- Calculate the TAS with an IAS of 110 KTS under the following conditions:

PA= 10.000ft
OAT= 5°C

Using the approximate formula

$$\frac{1.000ft}{10.000ft} = \frac{2\%}{?\,\%} \rightarrow \frac{(10.000ft) \times (2\%)}{1.000ft} = 20\%$$

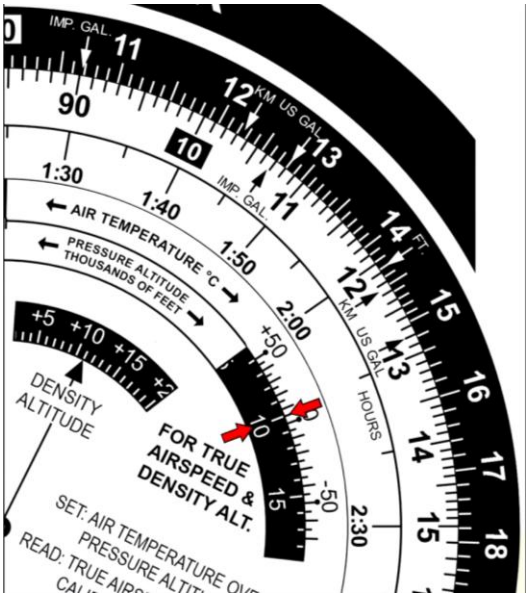
110 x (0,20)= 22 → 110 + 22 = 132

110 x (1,20)= 132

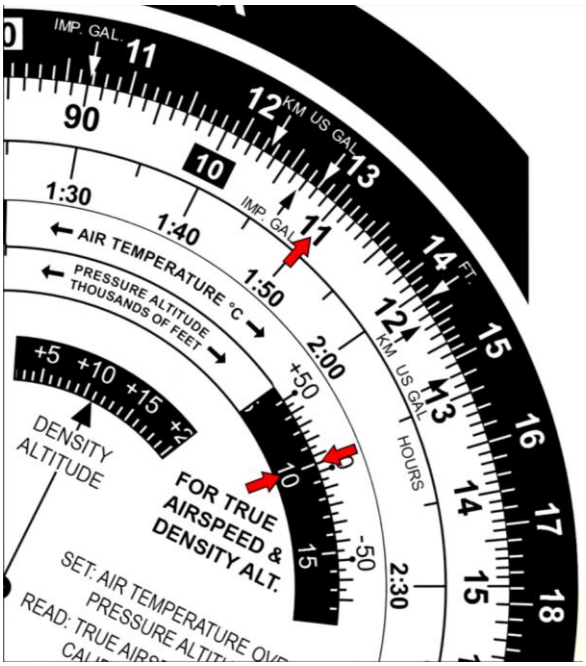
Using Flight Computer

Using the same example where we have an PA = 10,000 ft. IAS = 110 KTS and OAT = 5° C.

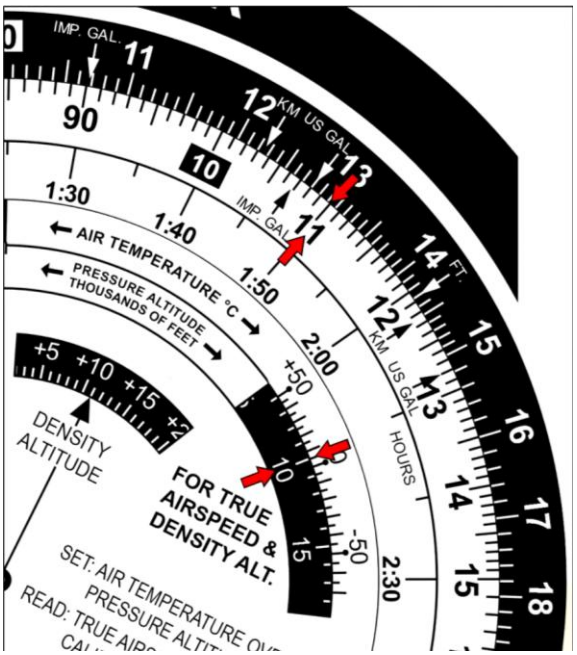
Step 1: Align the air temperature of 5° C with the pressure altitude of 10,000 feet



Step 2: Look for 110 in the middle scale



Step 3: Finally, we observe that it corresponds to approximately 130 in the outer scale.



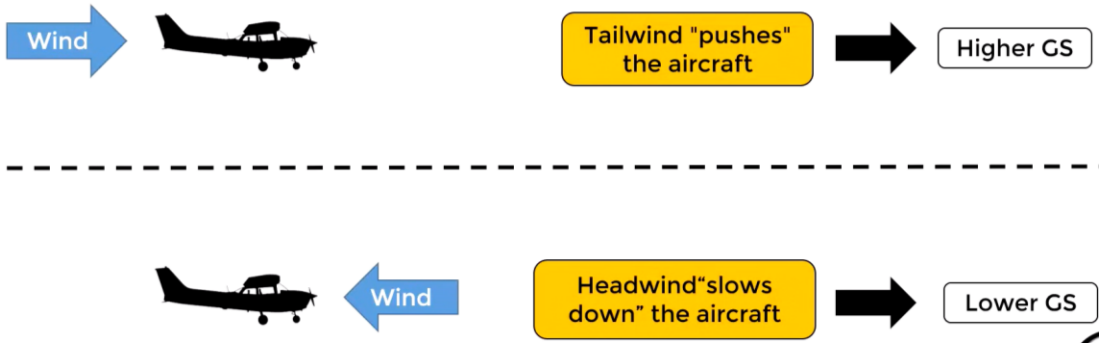
This means that the resulting true air speed is 130 KTS.

Ground Speed (GS)

- The actual speed of the aircraft in relation to the ground.
- The true airspeed (TAS) corrected by wind.

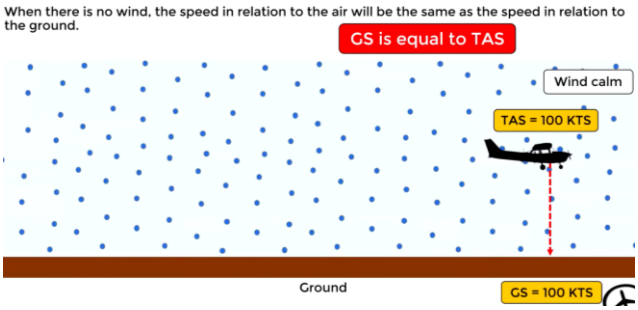
Let's see how the wind affects the speed of an aircraft:

- When an aircraft experiences a tailwind, it will literally be pushed by the wind in other words its speed in relation to the ground will increase.
- When the aircraft experiences a headwind, it will be slowed down by the wind in other words its speed in relation to the ground will decrease.



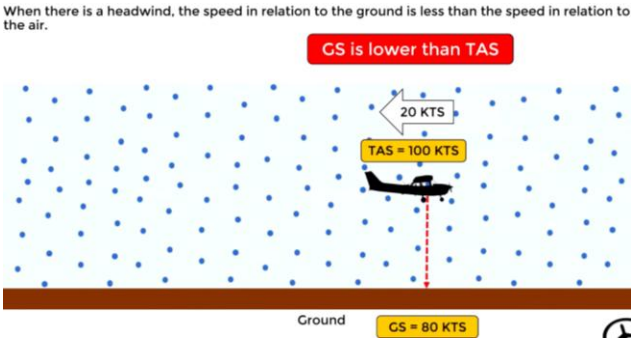
To determine the ground speed we must apply the wind correction

Case 1: No wind, in this case GS = TAS.



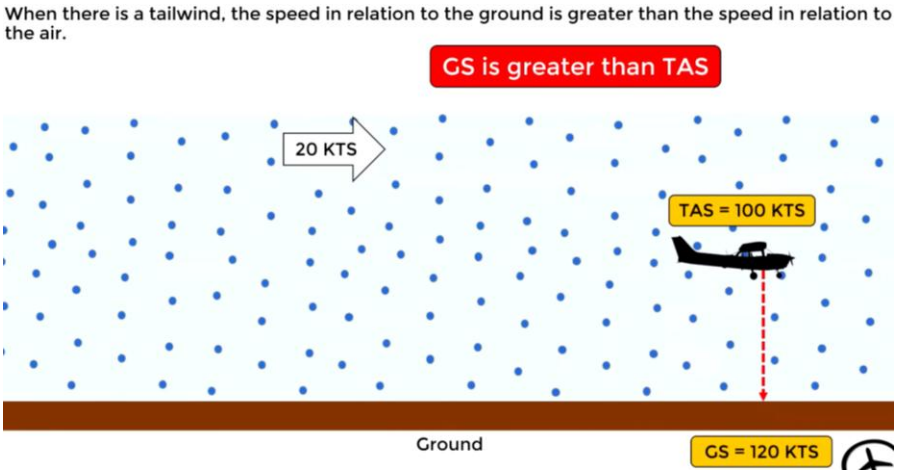
Case 2: Headwinds, in this case GS < TAS

The aircraft is flying with a headwind of 20 KTS this means that the air mass is moving against the aircraft at 20 KTS. If the aircraft is moving at 100 KTS in relation to the air mass, we will observe that its speed in relation to the ground will be 20 KTS lower. So, the ground speed will be 80 KTS.



Case 3: Tailwinds, in this case GS > TAS

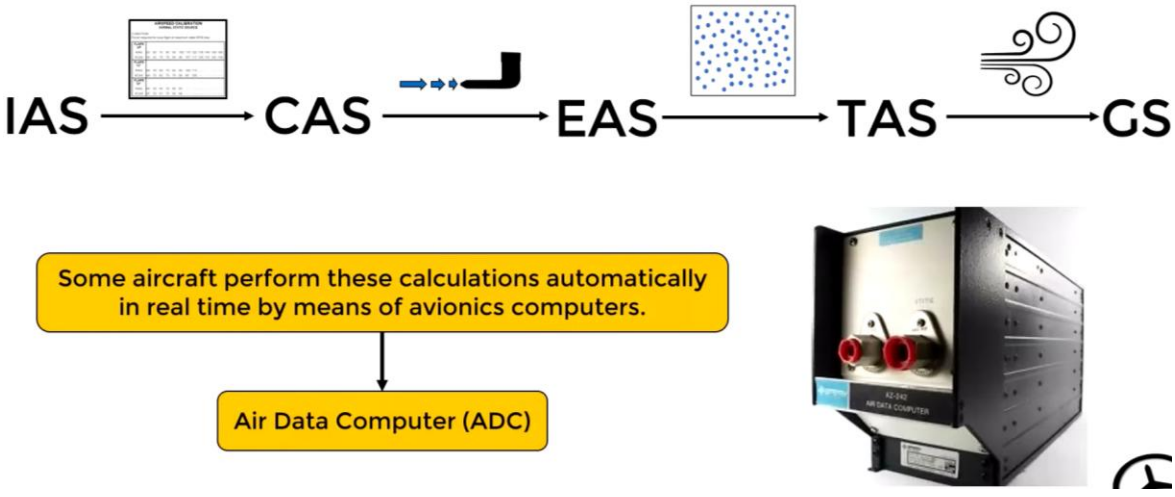
An aircraft that is flying with a tailwind of 20 KTS. As the aircraft travels through the air mass at 100 KTS, while at the same time the air mass is moving in the same direction at 20 KTS this means that the aircraft is moving at 120 KTS in relation to the ground.



Summary

- The Indicated Airspeed (IAS) corrected for instrument and position errors we will get the Calibrated Airspeed (CAS).
- If then we apply compressibility corrections, we will get the Equivalent Airspeed (EAS).
- if then we apply density correction we will get the True Airspeed (TAS).
- Finally, if we apply wind correction, we obtain the Ground Speed (GS).

Some aircraft perform these calculations automatically in real time by means of an air data computer. However, if the aircraft is not equipped with this avionics computer the pilot will have to do these calculations manually.



Practical applications of these speeds

Indicated airspeed depends directly on the dynamic pressure. Therefore, it is useful to stablish:

- Structural Speed limitations
- Take-off and landing speeds
- Stall speeds
- Performance-related speeds (Vx and Vy)

True airspeed is the actual speed of the aircraft in relation to the air. Therefore, it is used in:

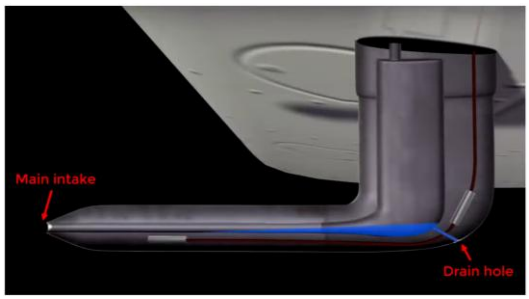
- ATC flight plan
- MACH speed calculation
- Endurance calculations
- Wind correction angle (WCA) calculation
- General flight planning aspects

Ground speed is the actual speed of the aircraft in relation to the ground. It is used for calculating:

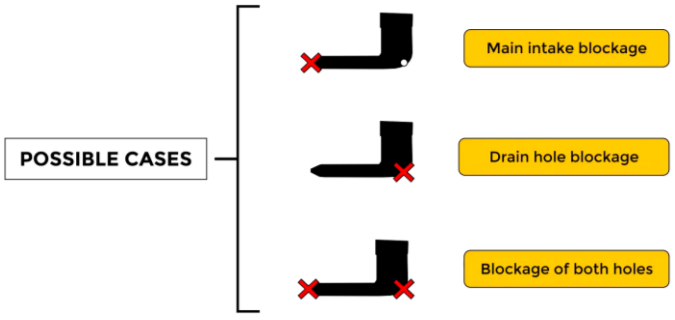
- Flight time
- Estimated time
- Fuel consumption
- Range
- Climb and descent
- General flight planning

Pitot Tube and Blockages

The pitot tube has two holes a main intake in the front and a drain hole at the rear.

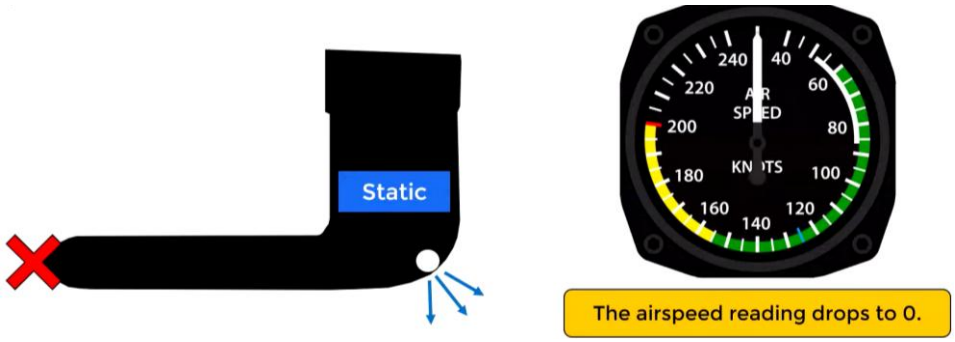


There are three possible cases. Main intake blockage, drain hole blockage, blockage of both holes.



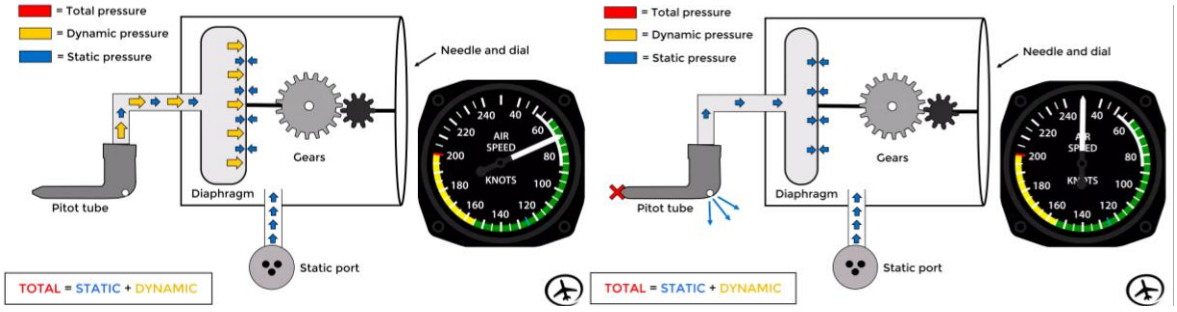
Main intake blockage

- Dynamic pressure can no longer enter the tube.
- Residual dynamic pressure in the tube will escape through the drain hole.
- Leaving only the static pressure.
- This makes the airspeed reading drop to zero.



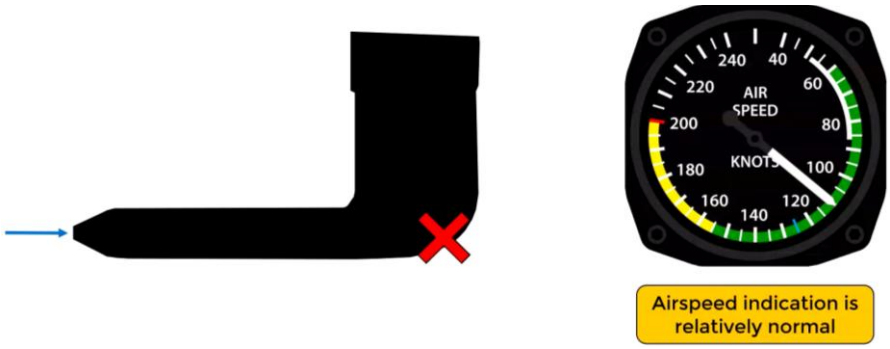
Why does the airspeed drop to 0

The dynamic pressure that was inside the diaphragm will escape through the drain hole of the pitot tube leaving only static pressure inside then the diaphragm will contract moving the needle to indicate zero.



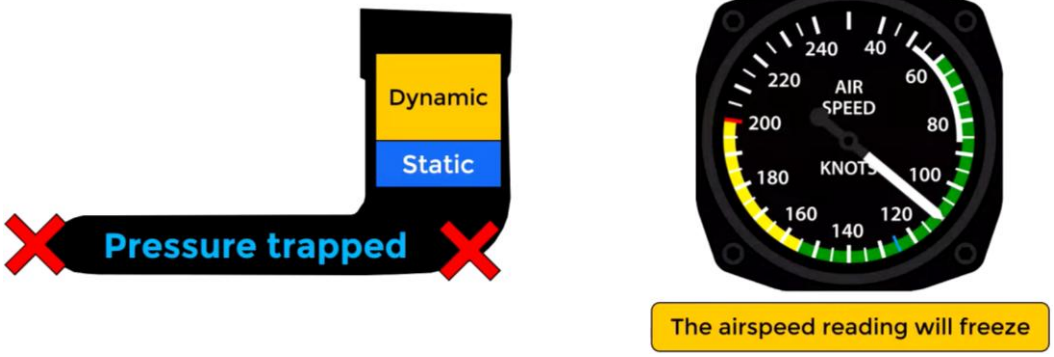
Drain hole blockage

- The total pressure continues to enter the tube through the main intake.
- The airspeed is initially little effected.
- Water can accumulate inside the tube which may lead to large measurement errors over time.



Blockage of both holes

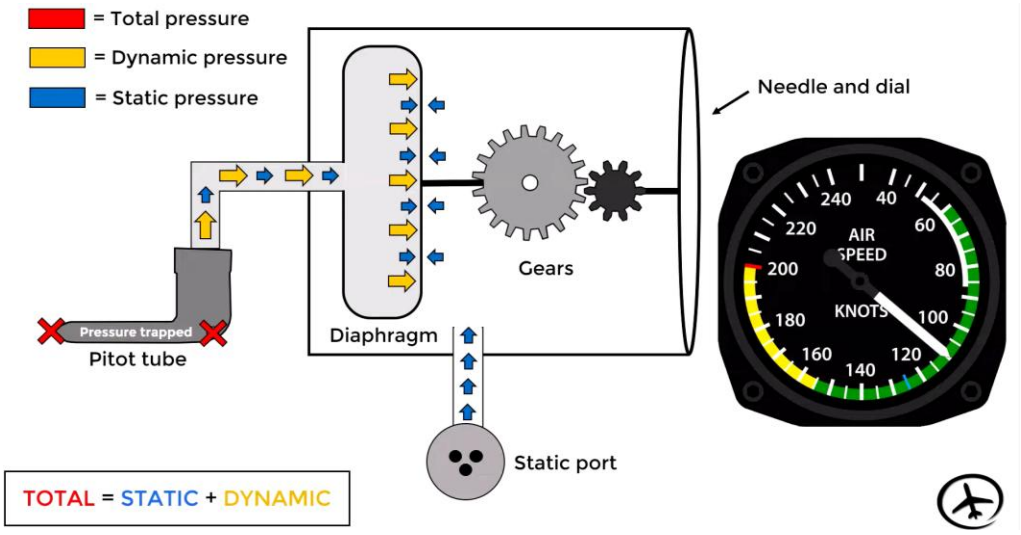
- The total pressure can no longer enter the tube.
- The pressure inside the tube cannot escape.
- The pressure trapped inside the system will give a false air speed reading.
- If the aircraft maintains altitude the airspeed indicator will freeze indicating the last speed measured before the blockage.



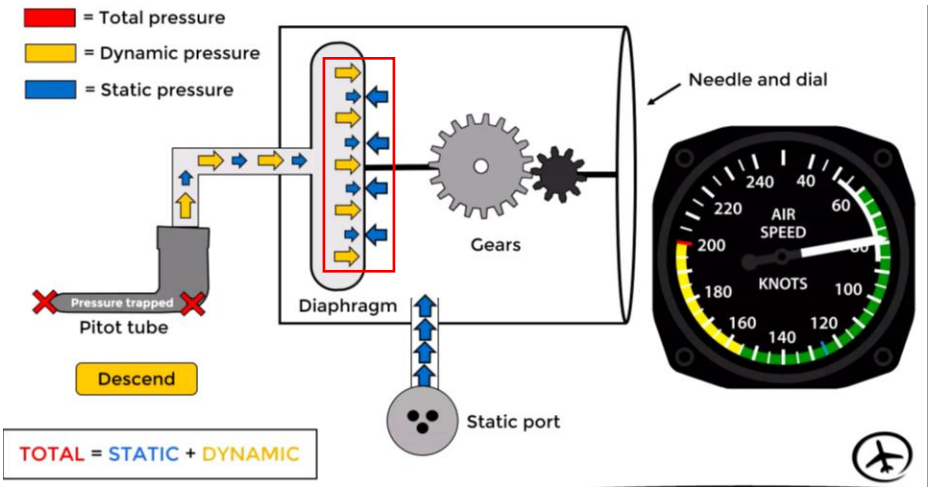
- If the aircraft climbs the airspeed reading will increase, and if the aircraft descends the indication will decrease. The instrument will act as if it were an altimeter.

Why does the airspeed indicator turn to an altimeter

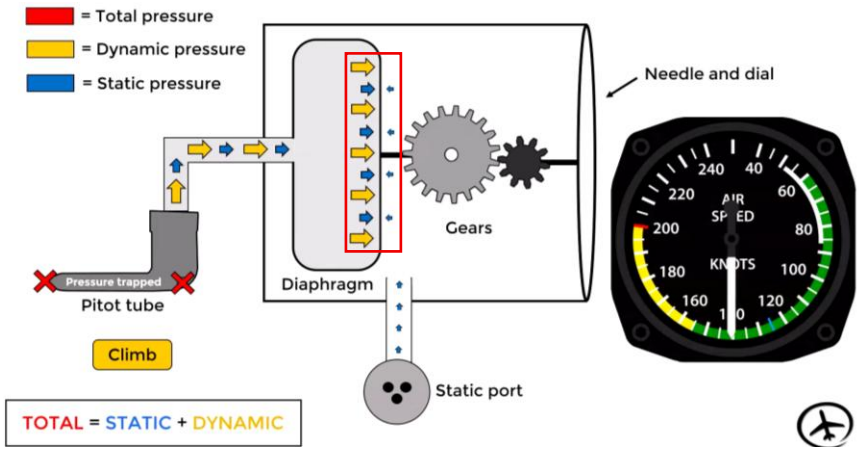
- In the event of blockage of both pitot tube holes, the pressure inside the pitted lines will be trapped.
- The dynamic and static pressure inside the diaphragm will remain constant.
- the airspeed indication freezes if the aircraft maintains altitude.



- In this situation if the aircraft descends, the static pressure provided by the static port will increase. As we know the atmospheric pressure increases as we descend.
- The static pressure inside and outside the diaphragm will be different.
- The higher static pressure outside the diaphragm will cause the diaphragm to contract, causing the instrument to indicate a lower speed



- If the aircraft climbs, the static pressure outside the diaphragm will reduce causing it to expand and therefore, making the instrument indicate a higher airspeed.



How to solve the blockage problem

In most cases the blockages occur due to ice. By activating the pitot hear would melt the accumulated ice. In spring, fall and winter, pitot heat should be used. When the plane is on ground, the pitot system should be covered as it avoids water, dust, mosquitos, snow, etc.





The pitot heat should be turned on when icing conditions are suspected.

As a rule of thumb, 10°C might be selected as the critical temperature. Turn the pitot heat on just prior to take-off and turn it off after landing since extended use of pi tot heat on the ground may cause the heater element to burn out.

Static port blockage

- As long as the aircraft maintains a constant altitude the air speed reading will not be affected.
- If the aircraft climbs or descends the airspeed indicator will act contrary to an altimeter i.e. we have exactly the opposite reaction to a pitot tube blockage.
- If the aircraft climbs the airspeed reading will decrease and if the aircraft descends the reading will increase.



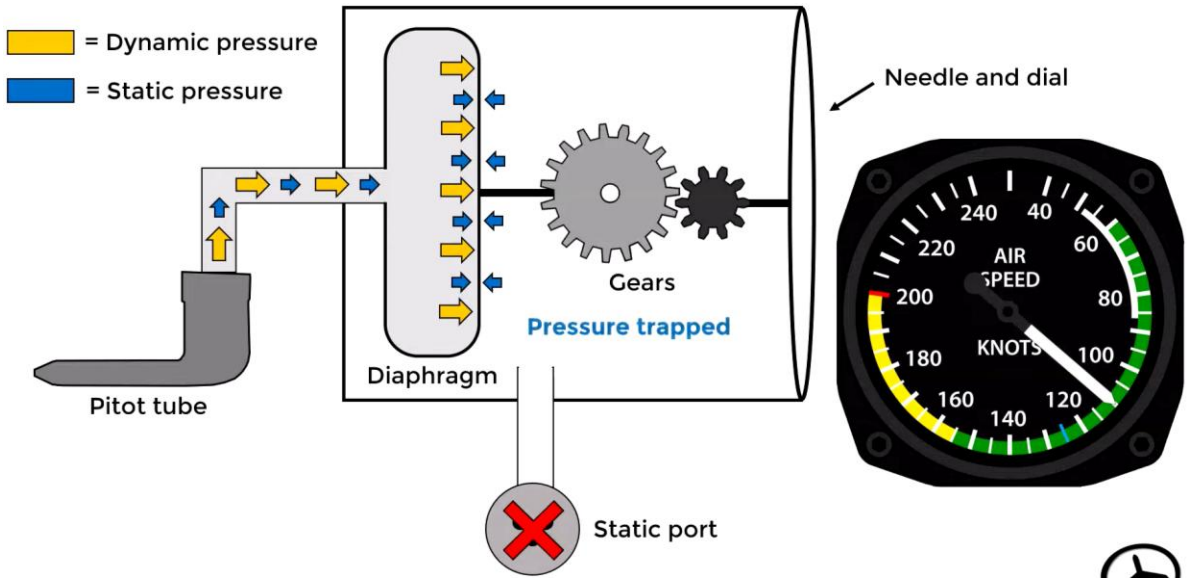
If the aircraft climbs, the airspeed reading will decrease.



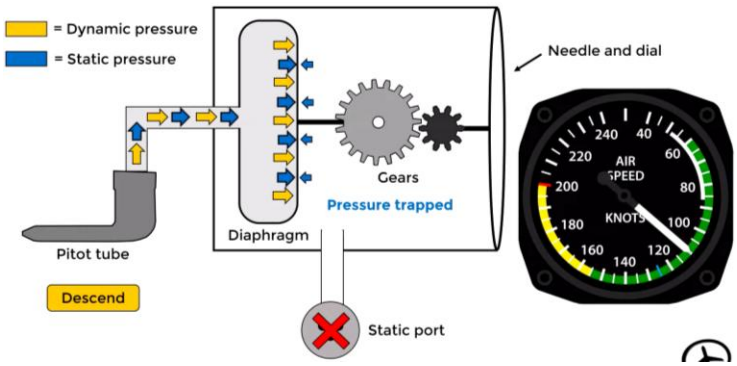
If the aircraft descends, the airspeed reading will increase

Why does the airspeed indicator turn to a reverse altimeter

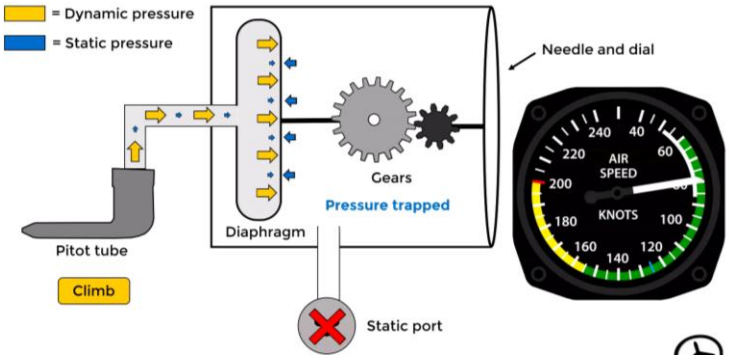
- The static pressure at the time of blockage will be trapped inside the instrument case. Therefore, the pressure outside the diaphragm will remain constant.
- As long as the aircraft maintains altitude, the diaphragm will expand and contract normally depending on dynamic pressure changes.



- If the aircraft descends the static pressure entering through the pitted tube will increase, while the static pressure trapped inside the instrument case remains constant. Therefore, the diaphragm will expand making the instrument indicate a higher speed.



- If the aircraft climbs here the static pressure entering through the pitted tube will decrease gradually. While the static pressure trapped inside the case remains constant. Thus, causes the diaphragm to contract and therefore, make the instrument indicate a lower speed.



Solving Static Port Blockage

In most aircrafts, an alternate static source allows static pressure to enter the system. When suspecting a static port blockage, this alternate static source must be activated. However, there is something to keep in mind it is that the static pressure provided by this alternate source is slightly different from the real one. It is necessary to consult the correction tables for the airspeed indicator published in the aircraft manual regarding the use of the alternate static source.

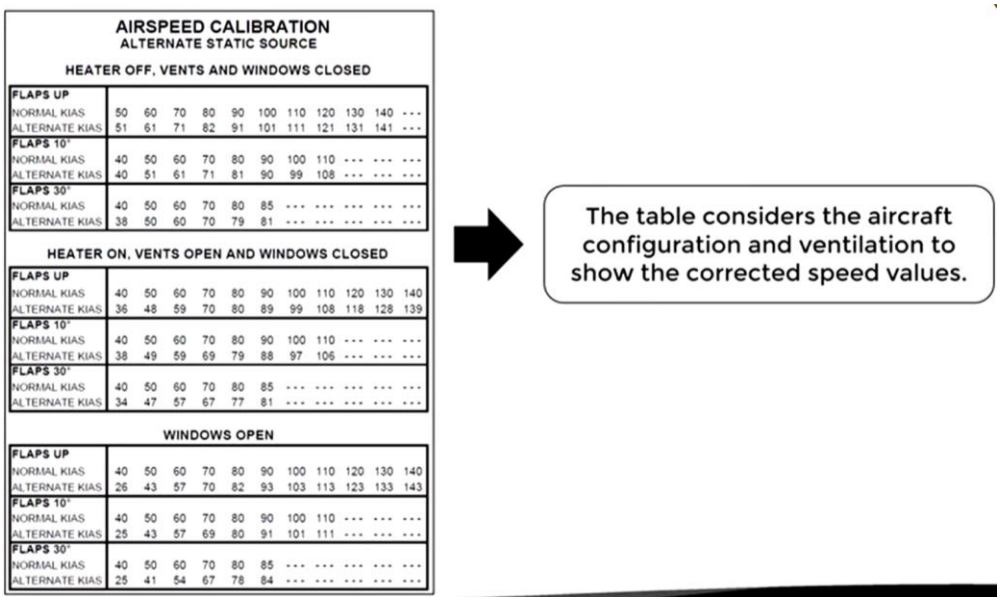


When a static port blockage is suspected, the alternate static source must be activated.

↓

The correction tables in the manual for the airspeed indicator should be consulted.

This correction table considers the aircraft configuration and ventilation in order to show the corrected speed values, and it is important to make sure that the table being used is for the alternate static source instead of the normal one.



AIRSPEED CALIBRATION
ALTERNATE STATIC SOURCE

HEATER OFF, VENTS AND WINDOWS CLOSED

FLAPS UP											
NORMAL KIAS	50	60	70	80	90	100	110	120	130	140	---
ALTERNATE KIAS	51	61	71	82	91	101	111	121	131	141	---
FLAPS 10°											
NORMAL KIAS	40	50	60	70	80	90	100	110	---	---	---
ALTERNATE KIAS	40	51	61	71	81	90	99	108	---	---	---
FLAPS 30°											
NORMAL KIAS	40	50	60	70	80	85	---	---	---	---	---
ALTERNATE KIAS	38	50	60	70	79	81	---	---	---	---	---

HEATER ON, VENTS OPEN AND WINDOWS CLOSED

FLAPS UP											
NORMAL KIAS	40	50	60	70	80	90	100	110	120	130	140
ALTERNATE KIAS	36	48	59	70	80	89	99	108	118	128	139
FLAPS 10°											
NORMAL KIAS	40	50	60	70	80	90	100	110	---	---	---
ALTERNATE KIAS	38	49	59	69	79	88	97	106	---	---	---
FLAPS 30°											
NORMAL KIAS	40	50	60	70	80	85	---	---	---	---	---
ALTERNATE KIAS	34	47	57	67	77	81	---	---	---	---	---

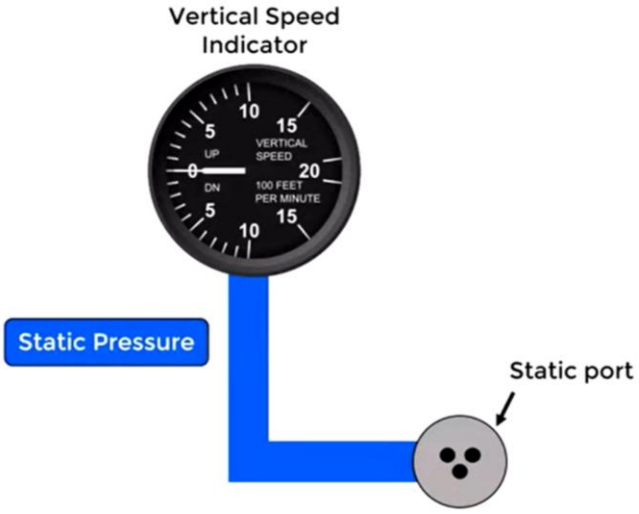
WINDOWS OPEN

FLAPS UP											
NORMAL KIAS	40	50	60	70	80	90	100	110	120	130	140
ALTERNATE KIAS	26	43	57	70	82	93	103	113	123	133	143
FLAPS 10°											
NORMAL KIAS	40	50	60	70	80	90	100	110	---	---	---
ALTERNATE KIAS	25	43	57	69	80	91	101	111	---	---	---
FLAPS 30°											
NORMAL KIAS	40	50	60	70	80	85	---	---	---	---	---
ALTERNATE KIAS	25	41	54	67	78	84	---	---	---	---	---

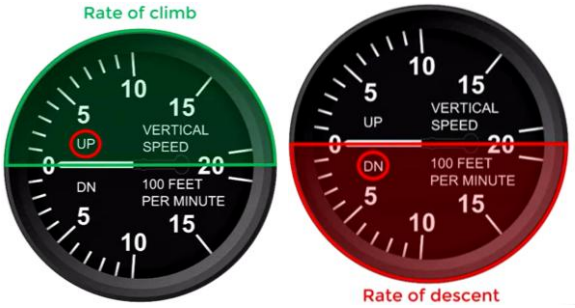
THE VERTICAL SPEED INDICATOR (VSI)

The vertical speed (vertical velocity, or rate of climb) indicator shows the rate, in feet per minute (may vary depending upon the design of the instrument), at which the aeroplane is ascending or descending.

- VSI is connected only to the static port i.e. it receives static pressure information only.



- VSI is divided into two parts:
  - Upper part – Rate of climb
  - Lower part – Rate of descent



- In the image below, the needle is pointing zero. Therefore, it indicates that the aircraft is in level flight at a constant altitude.



- In the below image the needle moves up:
  - The first image showcases a rate of climb of 300 feet per minute
  - The second image shows a rate of climb of 1000 feet per minute



- In the below image the needle moves down:
  - The first image showcases a rate of descent of 500 feet per minute
  - The second image shows a rate of descent of 800 feet per minute



How does VSI works – Basic principle

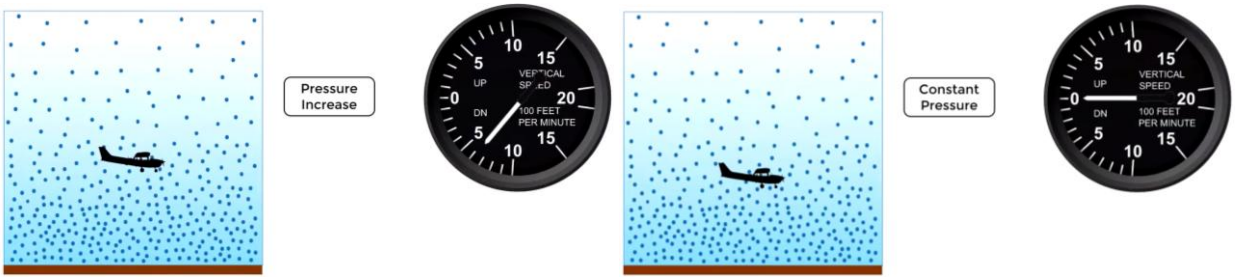
The principle on which it operates is the change in barometric (static) pressure which occurs with any change in height.

The static pressure decreases with altitude. At lower altitudes there is a higher pressure while at higher altitudes there is a lower pressure

- If there are not static pressure changes it means that the aircraft is in level flight and therefore the VSI will indicate zero feet per minute.
- If the aircraft starts climbing, the pressure will gradually reduce and therefore, the instrument will interpret it as an altitude increase. Once the aircraft stops climbing and maintains level flight again the pressure will remain constant and the instrument will return to indicate zero feet per minute.



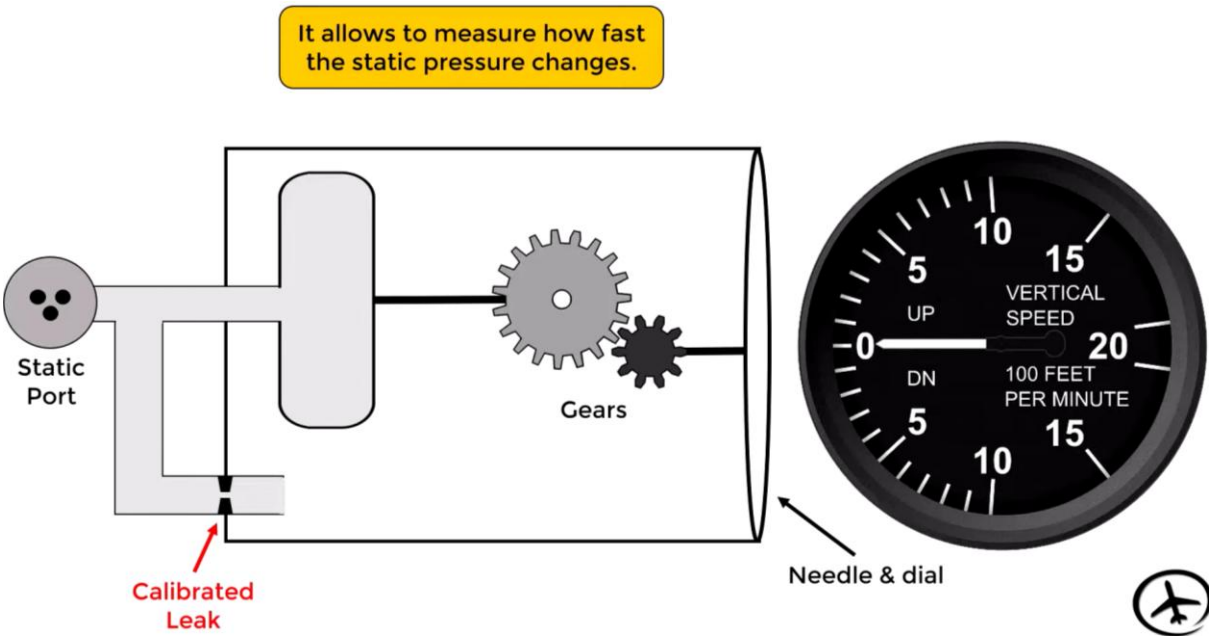
- If the aircraft starts descending, the static pressure will gradually increase so the instrument will interpret this as an altitude decrease indicating the corresponding descent rate. Again, when the aircraft levels off the pressure will remain constant and the VSI will return to zero feet per minute.



In general terms then the VSI interprets changes in pressure as changes in altitude and measuring how fast the pressure is changing helps it can determine the rate of climb or descent.

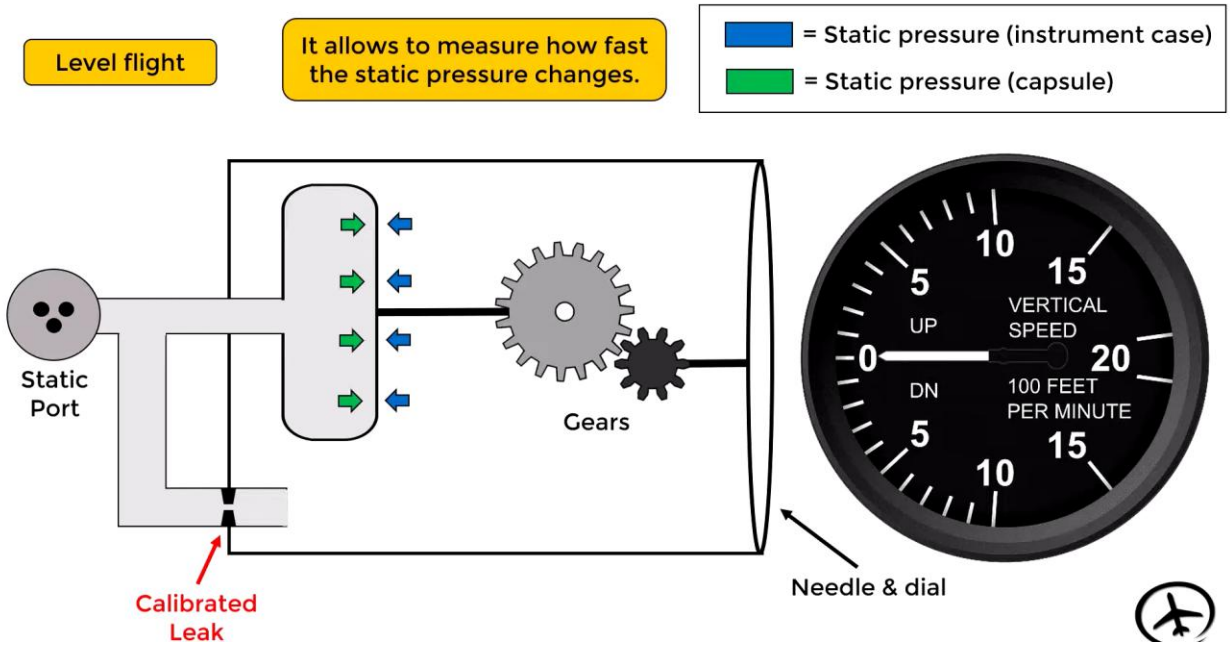
How does VSI works – detailed view

The static port is directly connected to a diaphragm inside the instrument case, which in turn is connected to gears that move the needle in the dial. A second static line is connected to the instrument case. However, it incorporates a calibrated leak which causes the pressure changes in the instrument case to be more gradual this way it is possible to measure how fast the static pressure is changing.

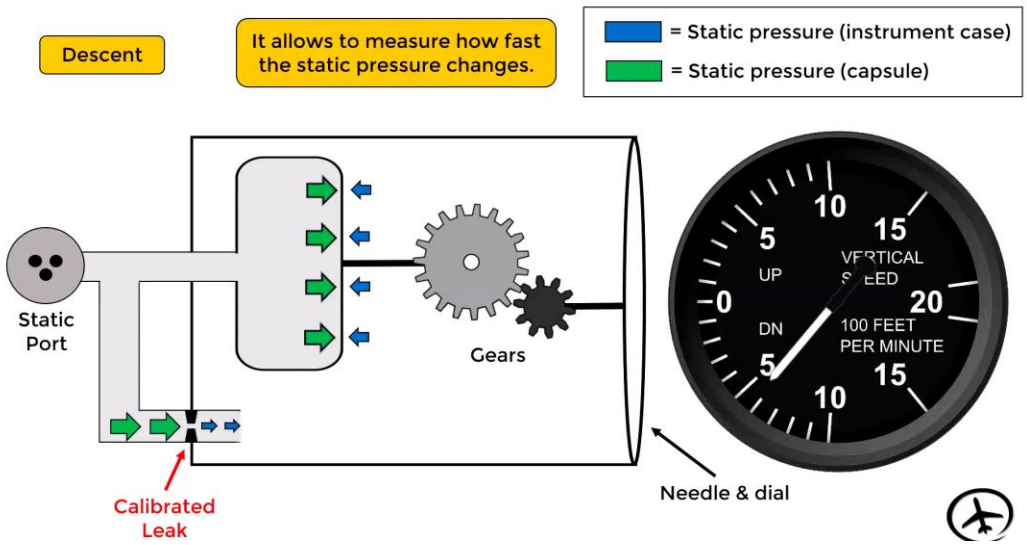


Level Flight

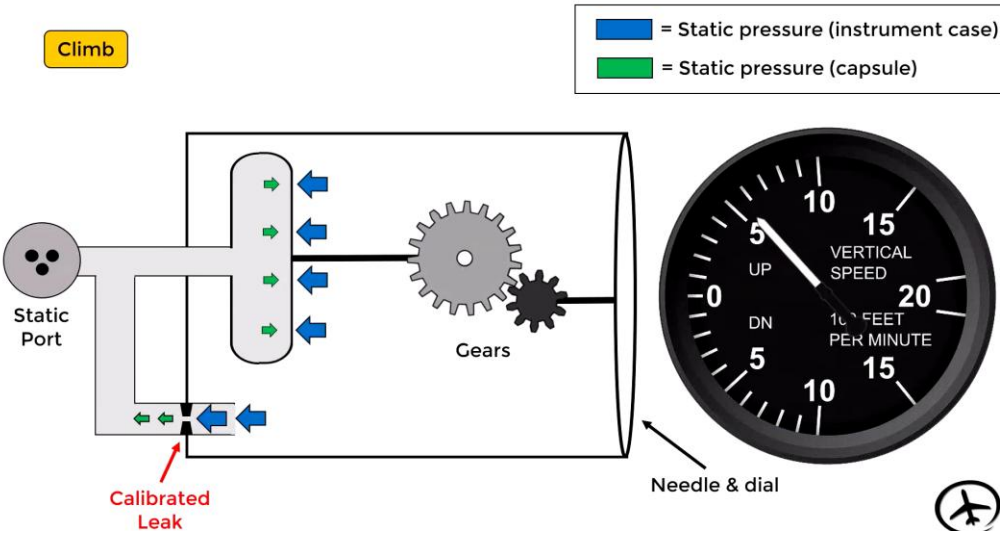
In this case the pressure inside the instrument case and the pressure inside the capsule are equal. Therefore, the needle will 0.0 feet per minute.



Descent - In this case, the static pressure will increase in the capsule almost immediately. However, inside the case the pressure doesn't increase that fast due to the calibrated leak which restricts the flow to the case. Accordingly, a temporary pressure difference is created between the capsule and the case causing the capsule to expand and move the gears to indicate a rate of descent.



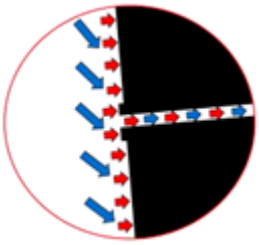
Climb - In this case, the static pressure will decrease almost immediately in the capsule. However, due to the calibrated leak it doesn't decrease that fast in the instrument case. This creates a pressure differential which causes the capsule to contract moving the needle to indicate a rate of climb.



Instrument Errors - VSI suffers from instrument error which is caused due to mechanical imperfections and wear of the gears. It also experiences position errors due to inaccuracies of the pitot static system while measuring the static pressure caused by maneuvers and changes in aircraft configuration.







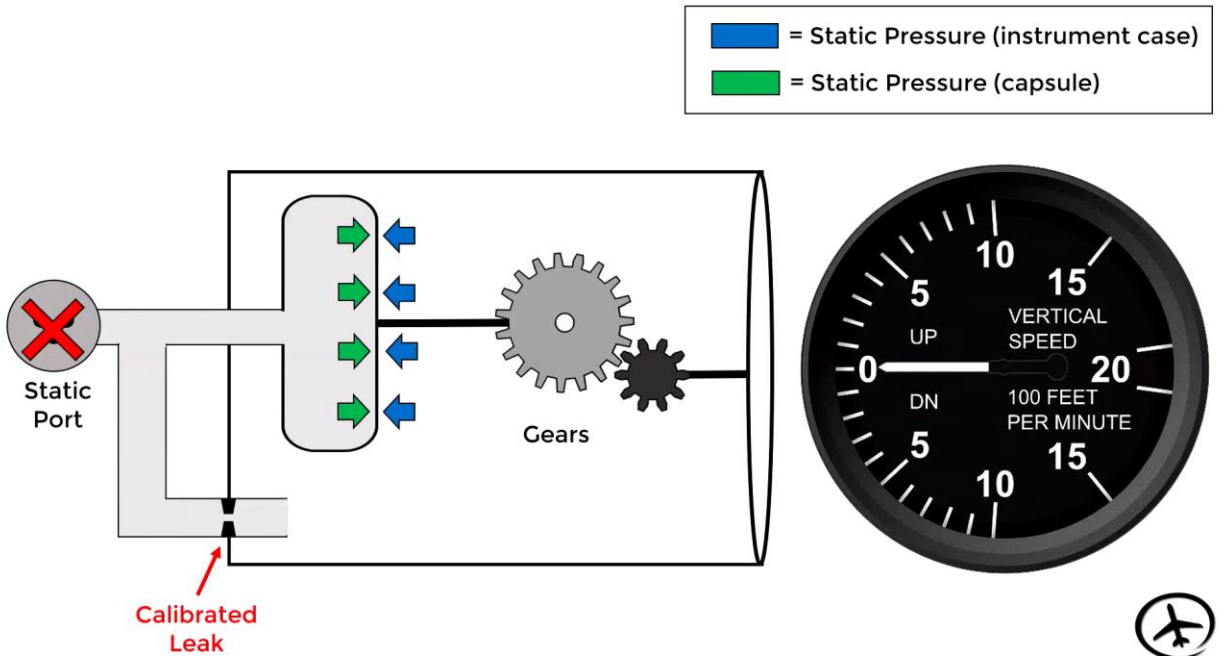
Lag error

- Delay in the instrument indication due to rapid changes in climber descent rate. This happens because the capsule doesn't have enough time to contract or expand rapidly.
- The delay varies between 3 and 6 seconds.
    - It happens when the aircraft is in a constant descent, and it suddenly levels off. In this case, it will take a few seconds for the VSI to indicate a level flight.
    - This can also happen when the aircraft is in level flight, and it suddenly starts a climb. Again, it will take a few seconds for the VSI to indicate the corresponding rate of climb.
- If pitch changes are made slowly, the lag is minimal, and the instrument will give a fairly accurate representation of the vertical speed at any given moment. If the pitch changes are large and rapid, the amount of lag in the instrument will be sizeable.

Instantaneous vertical speed indicators are available that do not have lag and are therefore very accurate. They operate on a more complicated arrangement of pistons and cylinders that give an almost instantaneous indication of change in vertical speed.

Static Port Blockage

In case of static port blockage, VSI reading will gradually return to zero feet per minute and it will remain still indicating level flight. This happens because the pressure in the capsule and inside the case is the same and since there's no pressure differential the instrument will interpret it as a level flight.



To solve this situation most aircraft incorporate an alternate static source which must be activated in case a static port blockage is suspected.

2.2.2 Gyro Instruments

Gyroscopic Instruments in 3D

https://www.youtube.com/watch?v=giiRLMesFFA

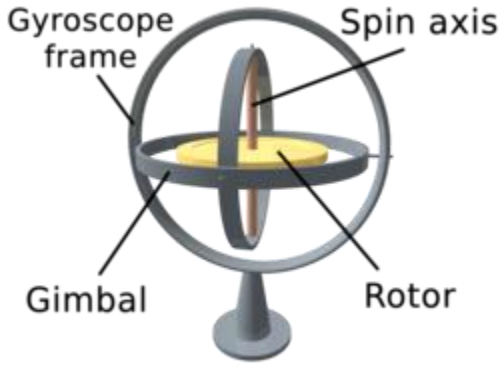
Gyroscopic Instruments

https://www.youtube.com/watch?v=hVsx4XWafXg&t=80s

Several flight instruments utilize gyroscopes for their operation. The most common instruments containing gyroscopes are the turn coordinator, heading indicator, and the attitude indicator. To understand how these instruments operate requires knowledge of the instrument power systems, gyroscopic principles, and the operating principles of each instrument.

Gyroscopic Principles

The gyroscope is a rotor, or spinning wheel, rotating at high speed in a universal mounting, called a gimbal, so its axle (a central shaft that a wheel or gear rotates) can be pointed in any direction.



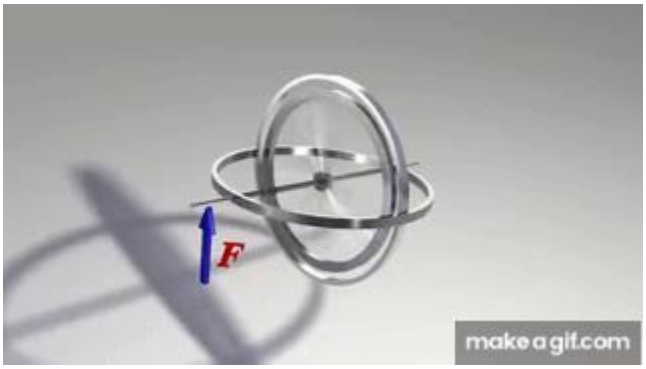
All of the practical applications of the gyroscope are based upon two fundamental characteristics: gyroscopic inertia (or rigidity in space) and precession.

Rigidity in Space - Refers to the principle that a gyroscope remains in a fixed position in the plane in which it is spinning.

By mounting this wheel, or gyroscope, on a set of gimbal rings, the gyro is able to rotate freely in any direction. Thus, if the gimbal rings are tilted, rotate, or otherwise moved, the gyro remains in the plane in which it was originally spinning.



Precession - The tilting or turning of a gyro in response to a deflective force. The reaction to this force does not occur at the point at which it was applied; rather, it occurs at a point that is 90° later in the direction of rotation.



This principle allows the gyro to determine a rate of turn by sensing the amount of pressure created by a change in direction.

The rate at which the gyro precesses is proportional to the deflective force and inversely proportional to the speed of the rotor.

- This means a stronger force causes faster precession, while a faster-spinning rotor slows down the precession rate for the same force.
    - **Deflective Force:** A greater deflective force results in a faster rate of this turning motion.
    - **Rotor Speed:** The faster the rotor spins, the more resistant it is to tilting. This increased resistance means the same deflective force will produce a slower rate of precession. Conversely, a slower-spinning rotor will precess more quickly under the influence of the same force.

Power Sources for Gyro Instruments

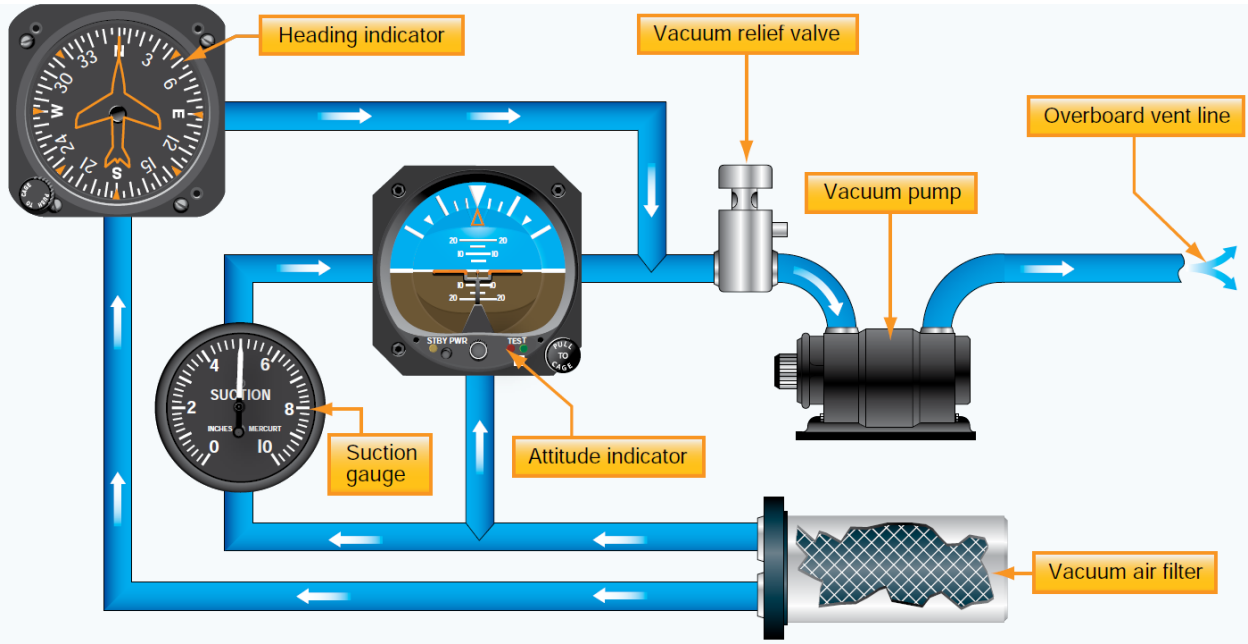
Aircraft gyroscopic instruments are powered by vacuum, pressure, or electrical systems. Typically, heading and attitude indicators use vacuum or pressure, while the turn coordinator uses electricity. Most aircraft have dual power sources to ensure bank information remains available if one fails. Vacuum or pressure systems spin the gyro by directing airflow against rotor vanes, similar to a waterwheel. Most vacuum powered instruments requiring about 4.5–5.5 "Hg for proper operation.

Engine driven vacuum system

A typical vacuum system consists of an engine-driven vacuum pump, relief valve, air filter, gauge, and tubing necessary to complete the connections. The gauge is mounted in the aircraft's instrument panel and indicates the amount of pressure in the system (vacuum is measured in inches of mercury less than ambient pressure).

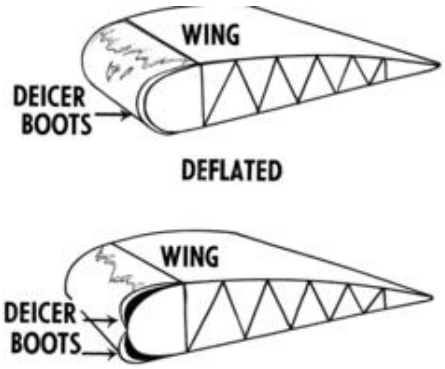
As shown in Figure below, air is drawn into the vacuum system by the engine-driven vacuum pump. It first goes through a filter, which prevents foreign matter from entering the vacuum or pressure system. The air then moves through the attitude and heading indicators where it causes the gyros to spin. A relief valve prevents the vacuum pressure, or suction, from exceeding prescribed limits. After

that, the air is expelled overboard or used in other systems, such as for inflating pneumatic deicing boots (see images below).



Deicing boots - installed on the leading edges of wings and control surfaces as these areas are most likely to accumulate ice which could severely affect the aircraft's performance. (see image below)





Monitoring vacuum pressure during flight is crucial because low suction can make attitude and heading indicators unreliable. The vacuum gauge is generally marked with a normal operating range, and some aircraft include a warning light that activates when pressure falls below acceptable levels.

When the vacuum pressure drops below the normal operating range, the gyroscopic instruments may become unstable and inaccurate. Cross-checking the instruments routinely is a good habit to develop.

Care of Gyro Instruments

The gyro is a precise instrument and requires special care. Gyro driven instruments should be caged before aerobatics to avoid tumbling the gyros which action can cause damage to the bearings. In addition, of course, the instrument is useless until the gyro is erected again.

Abrupt brake use should also be avoided, since it can impose acceleration loads on the gyro bearings.

Since airborne contaminants can damage the gyro bearings, it is good practice to periodically clean or replace the air filters of air driven units, especially if your aeroplane operates out of a dusty airport.

Gyro instruments that are driven by a Venturi system will be disabled if ice forms in the Venturi, blocking off the supply of air. Atmospheric conditions that are conducive to the formation of carburetor ice may also produce ice in the Venturi.

Gyro instruments do regrettably sometimes fail but usually give some indication in advance that things are not well with them. Any abnormal operation, such as excessive precession, sluggish response or noise, should be viewed as an indication of impending failure. Gyro instruments, however, can fail without prior warning leaving the pilot without any indication that the instrument he/she is monitoring is giving incorrect information. Gyro instruments have been developed that incorporate a warning flag that warns of aberrant operation of the gyros.

Since shock damage is a sure way to turn a gyro instrument into junk, be sure that instruments are handled with care during removal for repair.

THE HEADING INDICATOR (HI)

The heading indicator (also known as the directional gyro) is an instrument designed to indicate the heading of the aeroplane and, because it is steady and accurate, to enable the pilot to steer that heading with the least effort.



Fig.2.57 Heading Indicator

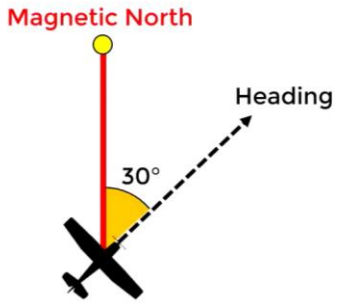
How does heading indicator works

Heading indicator indicates the current heading of the aircraft in degrees in relation to a directional reference usually the directional reference used in aviation is the Magnetic North Pole.

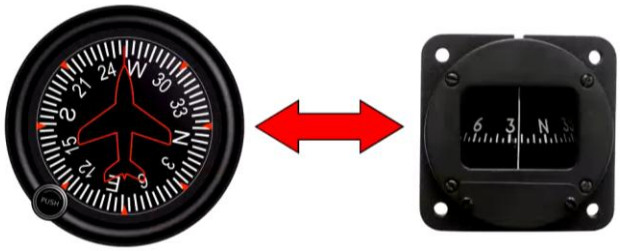
Indicates the aircraft heading in degrees in relation to a directional reference.

Magnetic North





The heading indicator has no mechanism that determines the direction of the magnetic north. Therefore, the pilot the pilot must align the heading indicator using the magnetic compass.

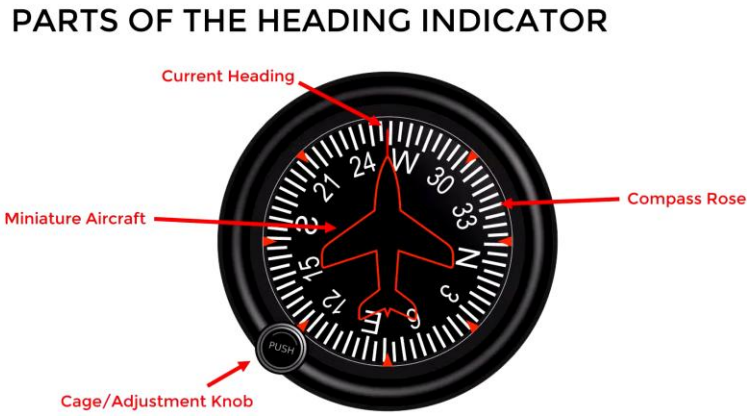


The magnetic compass does have the ability to align itself with magnetic north.

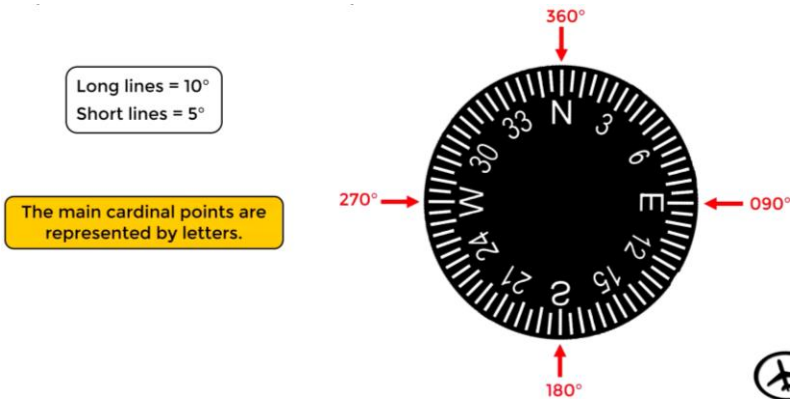
The pilot must align the heading indicator with the compass.

If we need to constantly align the heading indicator with the compass why not use the magnetic compass alone? The compass indication suffers from significant errors during turns, accelerations, turbulence.

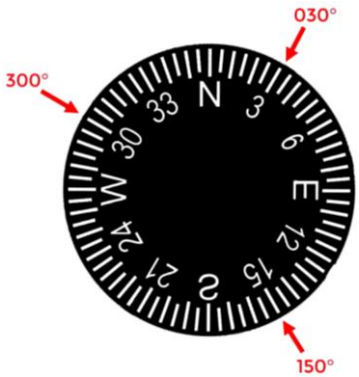
Parts of the heading indicator:



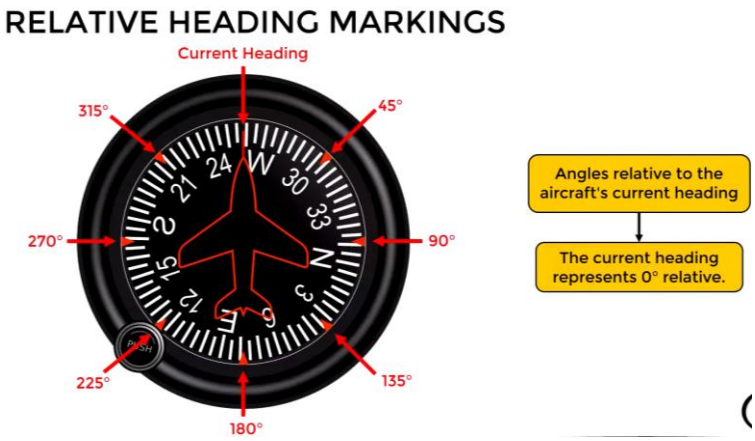
- Miniature Aircraft in the middle which will represent the orientation of the real aircraft
- The line that extends in front of the miniature aircraft represents the current heading.
- The lower left corner we have a cage/adjustment knob. The gyro is not, of its own accord, north seeking. It must, therefore, be synchronized with the magnetic compass at the beginning of a flight. The cage/adjustment knob disconnects the internal gyro from the display card, allowing the pilot to manually align the card to the magnetic compass reading.
- Compass rose is basically a graphic representation of the 360° of azimuth which is printed on the instrument card normally the long lines represent increments of 10° and the short ones represent increments of 5°.
- The main cardinal directions are represented by their corresponding letters for example
    - The letter N = North → heading 360
    - The letter E = East → heading 090
    - The letter S = South → heading 180
    - The letter W = West → heading 270



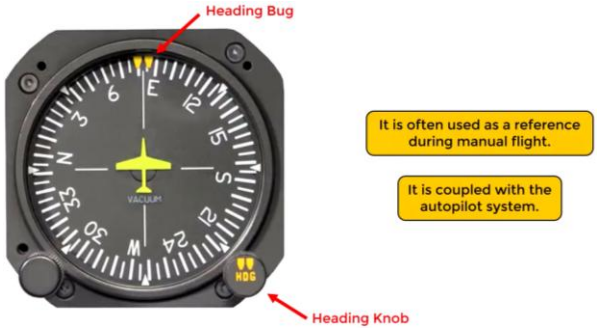
- Every 30° a number representing that heading is included omitting the last digit for example
    - The number 3 represents heading 030
    - The number 15 represents heading 150
    - The number 30 represents heading 300



- **Relative heading markings -** Reference angles relative to the aircraft's current heading. In other words, the current heading of the aircraft is used as reference. It will represent 0° relative, here we can find the markings for 45°, 90°, 135°, 180°, 225°, 270°, and 315° relative. These angles are not measured in relation to magnetic north but in relation to the current heading of the aircraft.



Some heading indicators also incorporate a heading bug and its corresponding heading knob this allows the pilot to select a particular heading in the instrument it is often used as a reference during manual flight but can also be used in conjunction with the autopilot system.



Reading the heading



Image 1



Image 2



.

Image 3



Image 4



Image 5

Image 1: Heading 0°
Image 2: Heading 060°
Image 3: Heading 320°
Image 4: Heading 090°
Image 5: Heading 285°

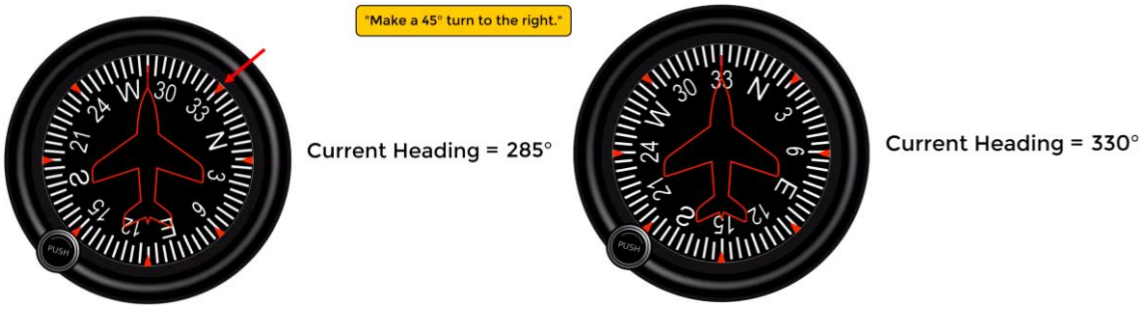
Relative Marking Benefits

Let's take a few example based on the below images.

Example 1



- Current heading of 285
- ATC tells us to make a 45° turn to the right (They are not asking us to turn to heading 045 but to turn 45° to the right in relation to our current heading)
- What will be our new heading
    - We could add 45° to heading 285 to determine the heading i.e. 285+45 = 330
    - it is easier to use the relative heading markings of theinstrument. In this case we just have to look at the 45° relative marking which indicates heading 330

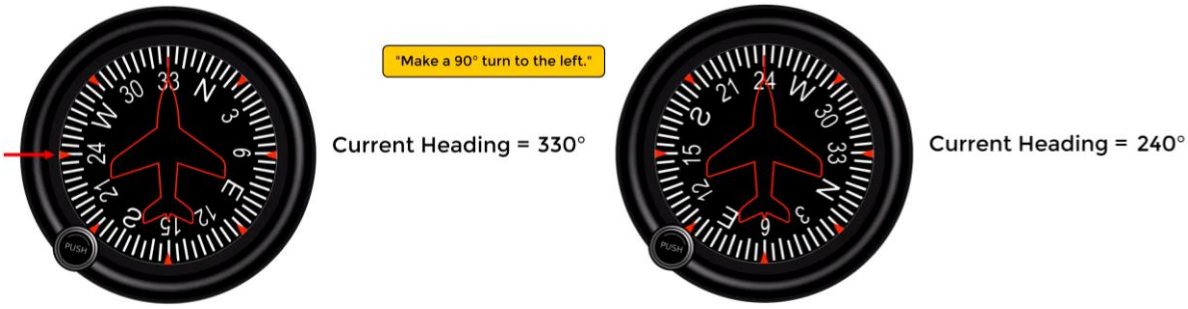


Example 2

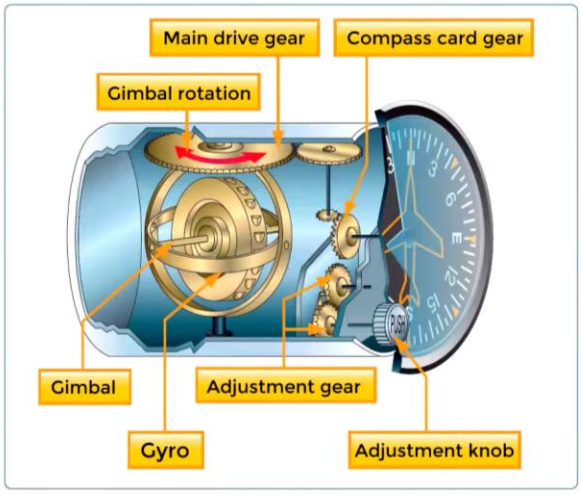
- Current heading of 330°
- ATC tells us to make a 90° turn to the left

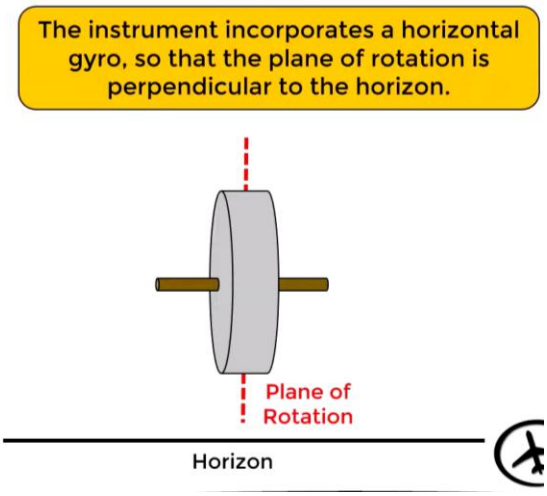


- What will be our new heading
    - We could subtract 90° to heading 330° to determine the heading i.e. 330-90 = 240
    - it is easier to use the relative heading markings of theinstrument. In this case we just have to look at the 90° relative marking which indicates heading 240



Inner working of Heading Indicator





- Incorporates a horizontal gyro so that the plane of rotation is perpendicular to the Horizon i.e. the wheel is mounted vertically.
- Gyro wheel spins about its horizontal axis at ~12,000 RPM

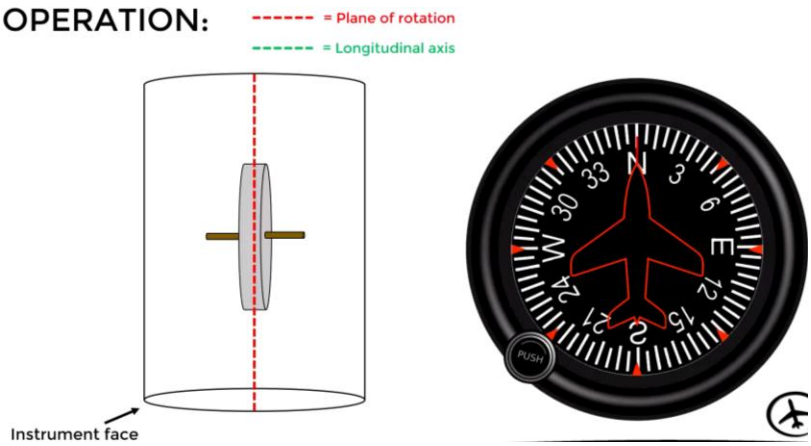
Gimbal System

- Gyro wheel is mounted in an inner gimbal ring (free to turn about horizontal axis)
- Inner gimbal is mounted inside an outer gimbal ring

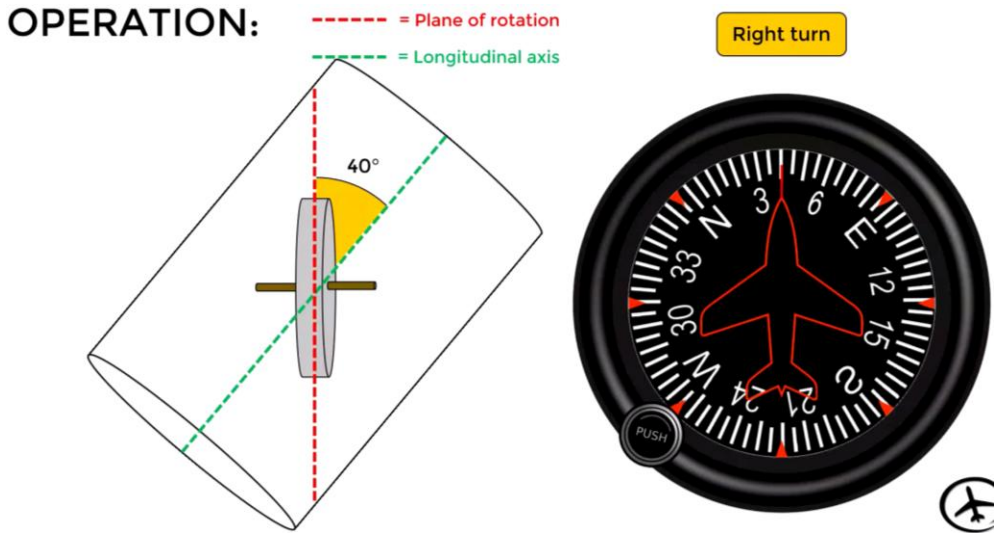
Compass Card Mechanism

- Compass rose card on the instrument face is connected to the outer gimbal ring via gears
- As the aircraft turns, the compass card rotates to indicate left or right turns

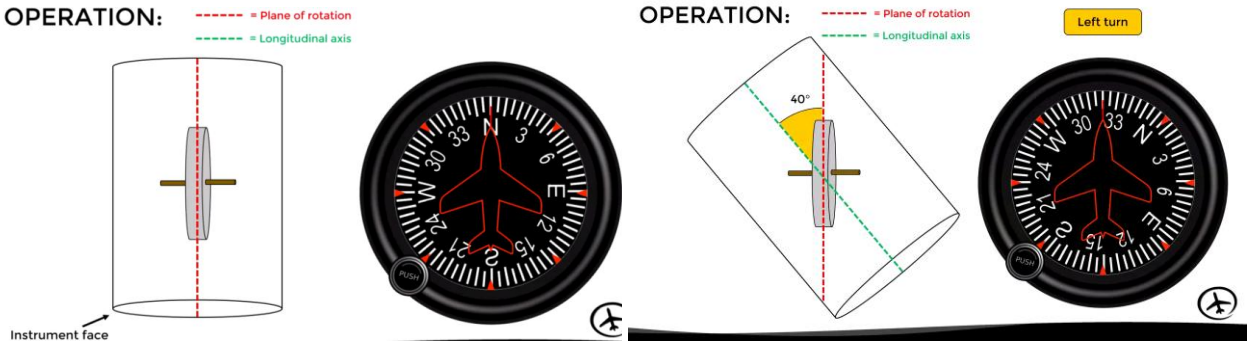
In the below images, on the right side we have the instrument as the pilot would see it, and on the left side we are looking at the instrument from above.



In this case, it is indicating a North heading. If the aircraft makes a turn to the right, both the aircraft and the instrument will turn to the right while the gyro remains rigid in space. This way the instrument can measure the angle between the directional reference and the current aircraft heading



The same happens if the aircraft turns to the left. In this case both the aircraft and the instrument will turn to the left while the gyro remains rigid in space allowing to measure the heading correctly.



For additional information on the Heading Indicator

The Heading Indicator (Directional Gyro)

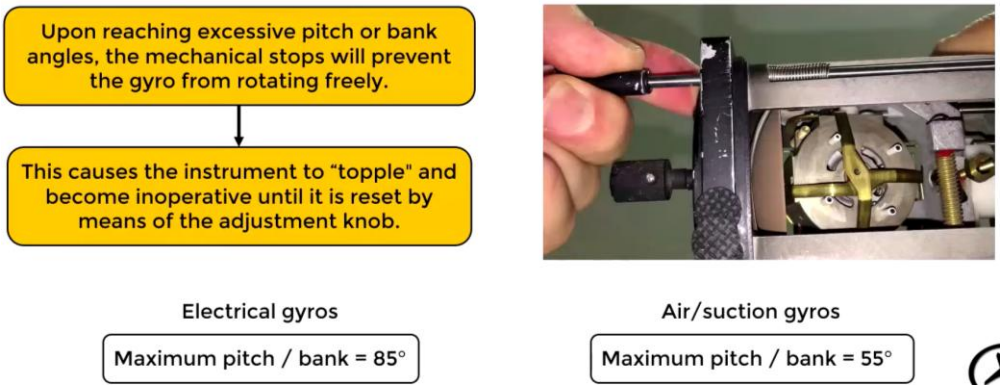
https://www.youtube.com/watch?v=q1TVR9xdl64

Limitations of the heading indicator

Gyro's degree of Freedom

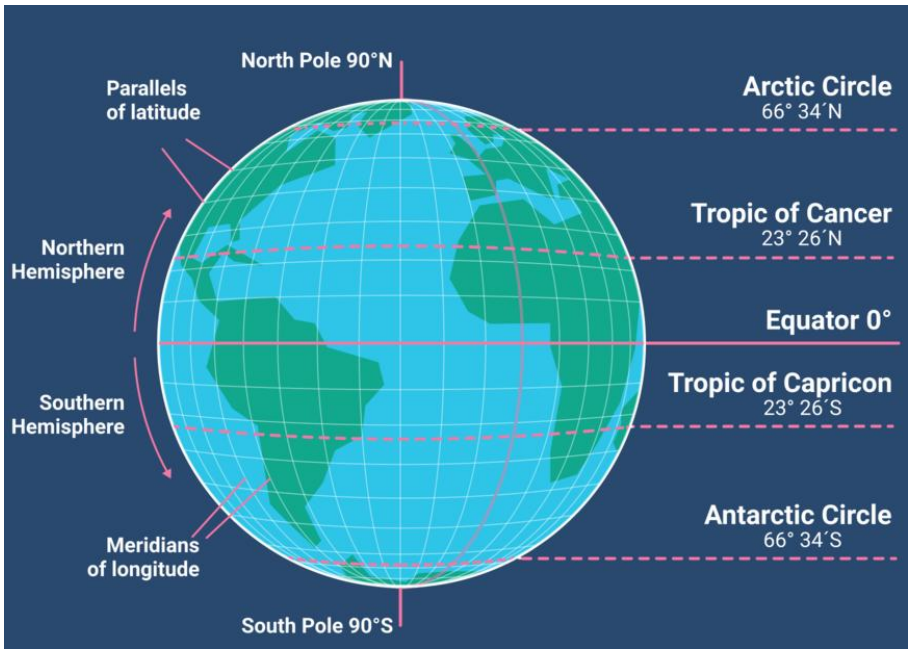
- Theoretically the instrument's gyro has three degrees of freedom.
- In practice, mechanical stops limit its full range of rotation.
- When the aircraft reaches excessive pitch or bank angles, these stops prevent free gyro movement.
- This restriction can cause the instrument to topple and become inoperative.
- The instrument must then be reset using the adjustment knob.
- Pitch and bank limits differ based on the type of heading indicator:

- Electric-driven gyros: limits are about 85°.
- Air-driven gyros: more restrictive, around 55°.



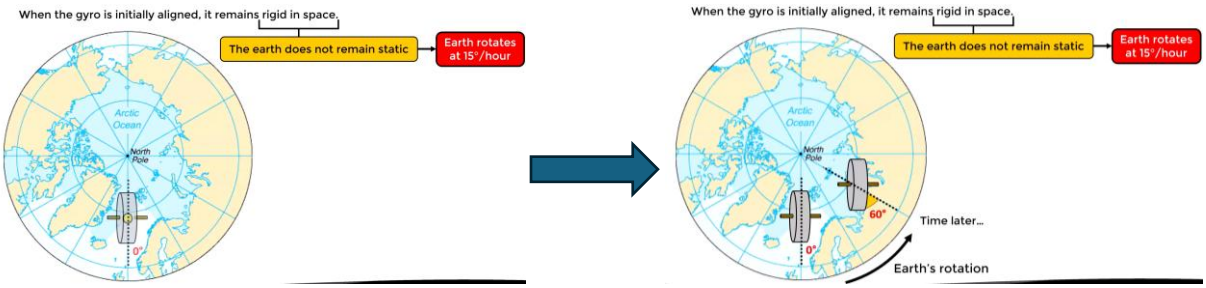
PRECESSION ERROR

Apparent precession – First let's see some basic terms.



As we know when the gyro starts spinning it remains rigid in space. However, the Earth does not remain static it rotates at about 15° hour.

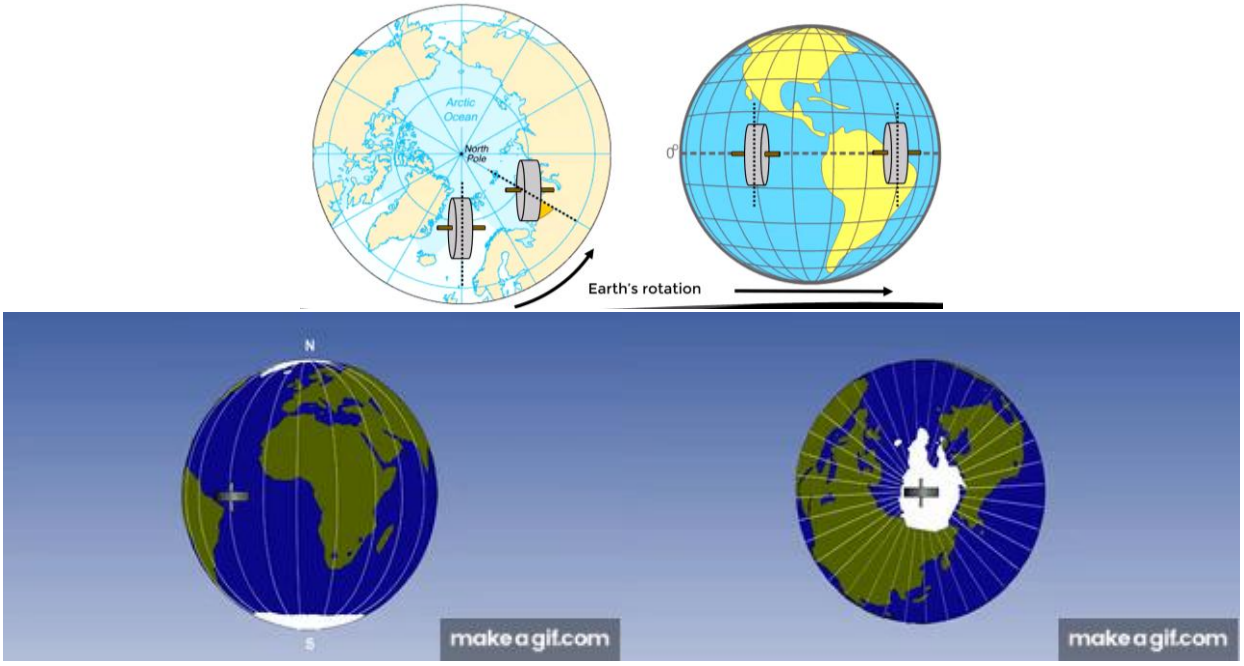
To understand how this affects the gyro here we have the Earth seen from above let's suppose we are at this yellow point, and we align the gyro of the heading indicator in such a way that it points to the north.



Now after a few hours the Earth would have rotated several degrees. Let's say that we are now at this other position. As we can see, as the gyro remains rigid in space it is no longer pointing to the north and it has now a deviation of 60°. This implies that the heading indicator will be using a wrong directional reference so the heading indication will be incorrect. This apparent wander error can be up to 15° per hour near the poles.

It is that the apparent wander is maximum at the poles and minimum at the equator.

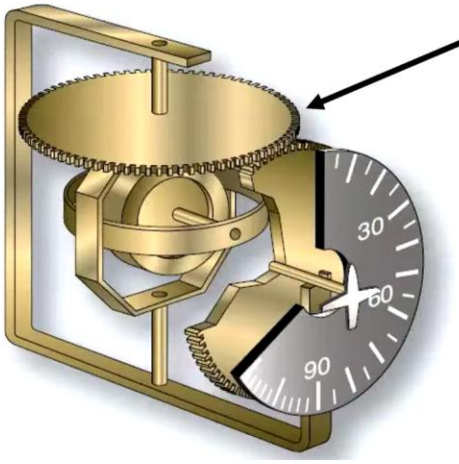
The image on the left we can see that near the poles the apparent wander is significant. However, in this image on the right we can see that at the Equator the plane of rotation of the gyroscope remains aligned with the meridians pointing North regardless of the rotation of the earth.



In reality it is not the gyroscope that drifts but the rotation of the earth that creates this apparent precession error.

Gyroscopic Wanderer - It is produced since the mechanisms and gears inside the instrument are not perfect, they suffer from wear friction and imbalance that causes the plane of rotation of the gyro to drift due to precession. However, this error is normally small in well-maintained modern Gyros. The real wander is very low around 1° per hour.

The mechanisms and gears inside the instrument are not perfect.



That causes the plane of rotation of the gyroscope to drift due to precession caused by imperfections in the gears.

In well-maintained modern gyroscopes, the real wander is very low (≈1°/hour).

Correction of errors

The cage/adjustment knob allows the gyro to be disengaged from the card, and corrections can be made.

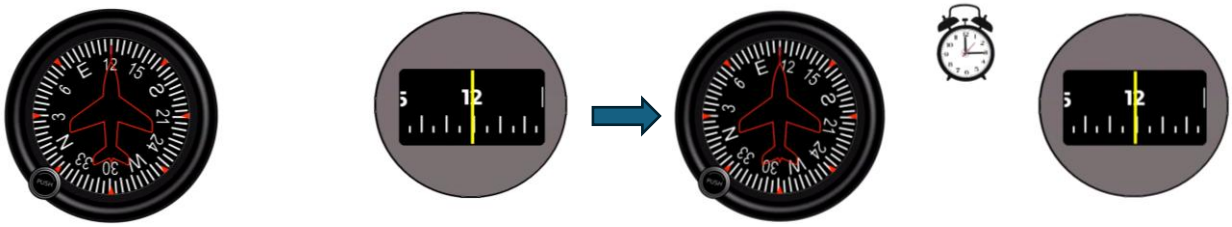


To avoid significant indication errors, the pilot should realign the heading indicator with the compass every 10 to 15 minutes.

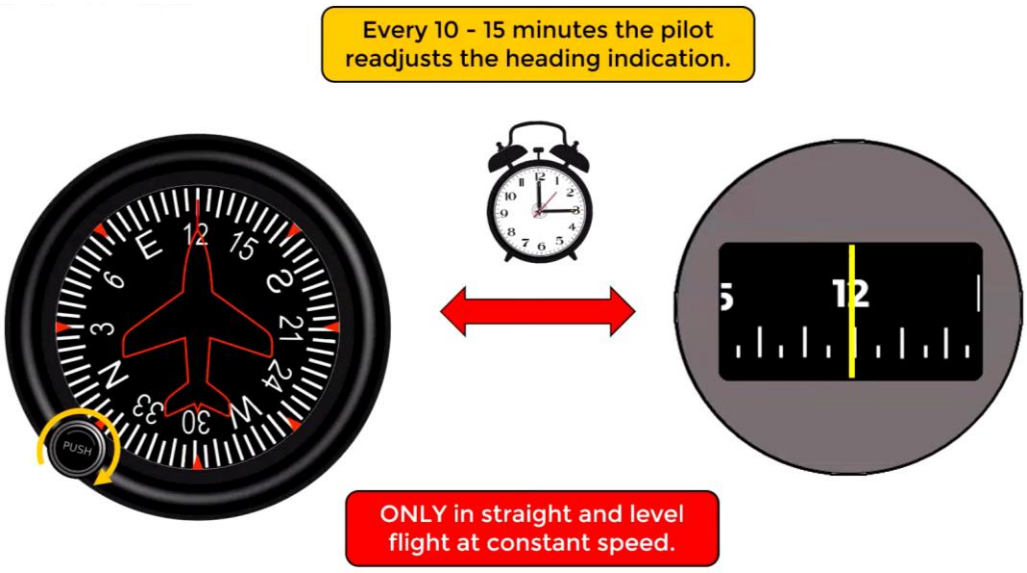
Only in straight flight at constant speed.

Precession error, both mechanical and apparent, must be corrected for at regular intervals of about 15 minutes. It is important, while synchronizing the heading indicator with the magnetic compass, to hold the aeroplane straight-and-level to ensure that there is no compass error.

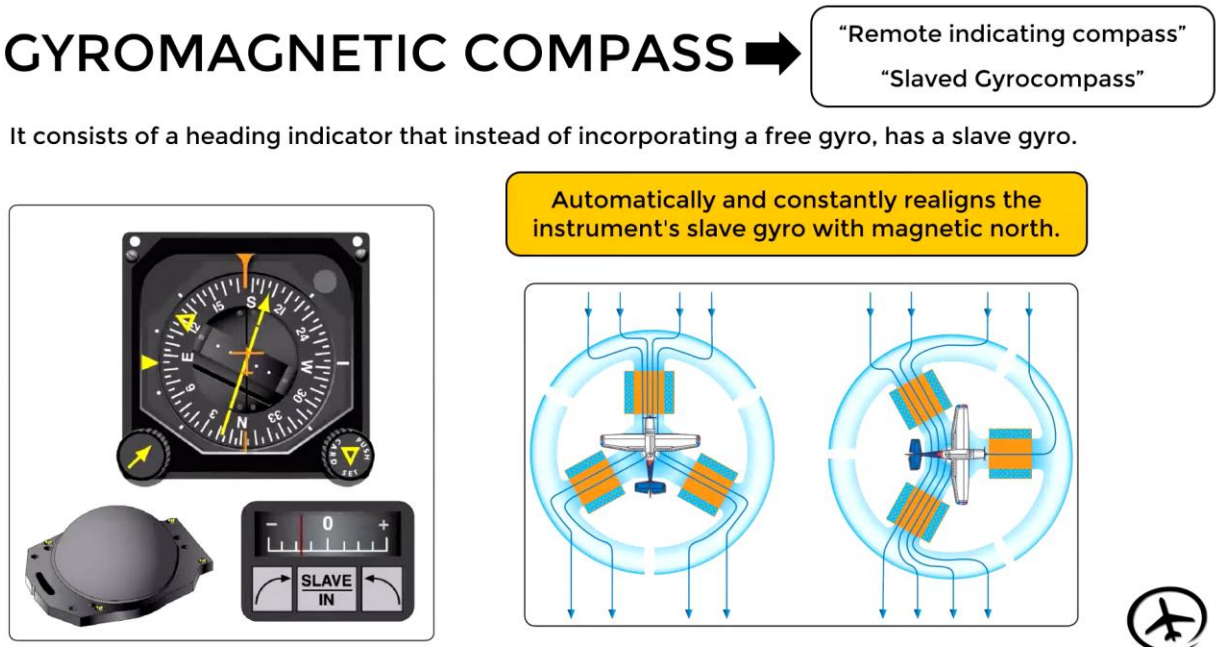
Example, we are flying straight and level with a heading of one 120 in this case both the compass and the heading indicator are indicating the same heading. However, due to the effects of real and apparent wander, the heading indicator will drift progressively with time in this case now the heading indicator is showing a heading of 110 even though the Compass continues to indicate the correct heading of 120.



The pilot will have to use the adjustment knob to reset the heading indicator according to the magnetic compass indication but again this should only be made while flying straight and at a constant speed.



Gyromagnetic compass aka "Remote indicating Compass" or the "slave Gyrocompass" basically it consists of a heading indicator that instead of incorporating a free gyro has a slave gyro. It uses a system of remote magnetometers to determine the direction of magnetic north and this way it automatically and constantly realigns the instrument slave gyro with magnetic north.

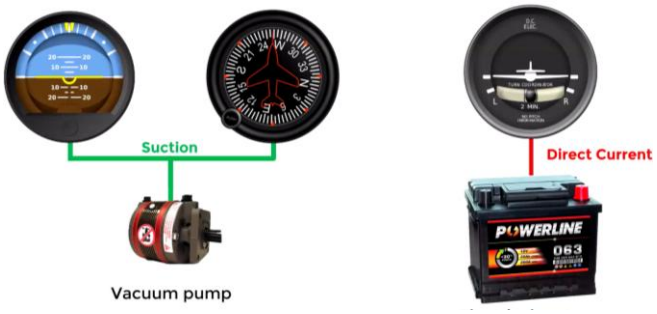


THE ATTITUDE INDICATOR (AI)

THE ATTITUDE INDICATOR

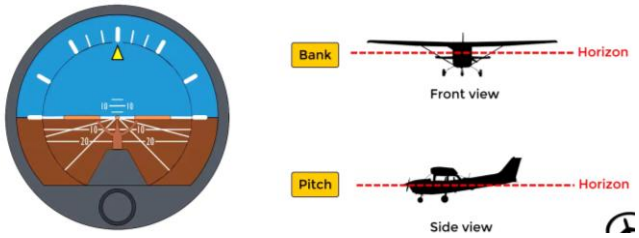
https://www.youtube.com/watch?v=ohBZHKIIA_c

- The attitude indicator is aka artificial horizon or gyro horizon.
- The vacuum pump provides suction to drive the gyros of the attitude indicator. However, there are other configurations, the attitude indicator can also be driven by electric current in some cases.

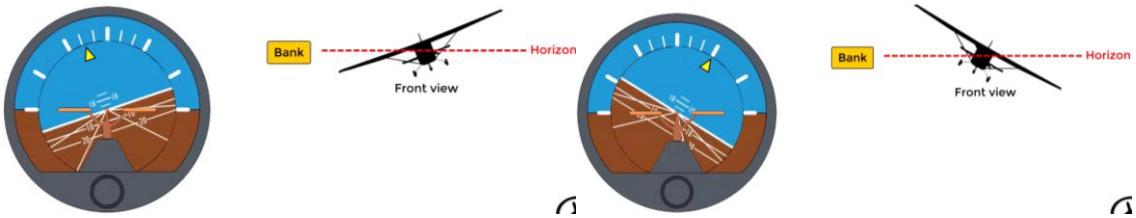


- It measures the attitude of the aircraft in relation to the horizon in terms of pitch and bank

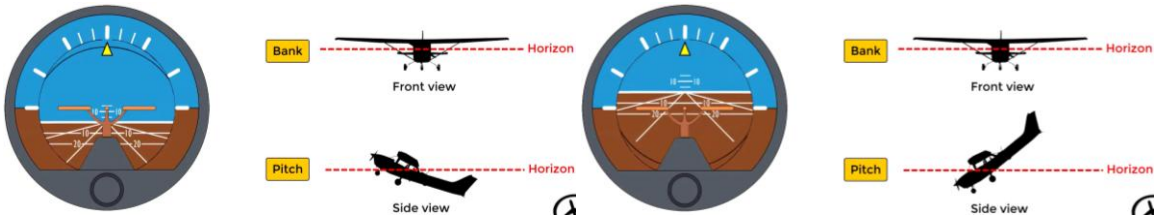
Instrument that measures the attitude (inclination) of the aircraft in relation to the horizon in terms of pitch and bank (roll).



- The bank angle is the inclination of the wings in relation to the horizon.



- The pitch angle is the inclination of the nose of the aircraft in relation to the horizon.



Parts of the attitude indicator



- Miniature aircraft - Represent the wings and the nose of the aircraft
- Artificial horizon - Represented by this white line which separates the upper part which represents the sky from the lower part which represents the ground.

Instrument markings and scales

- the banking scale it is located in the upper part of the instrument.



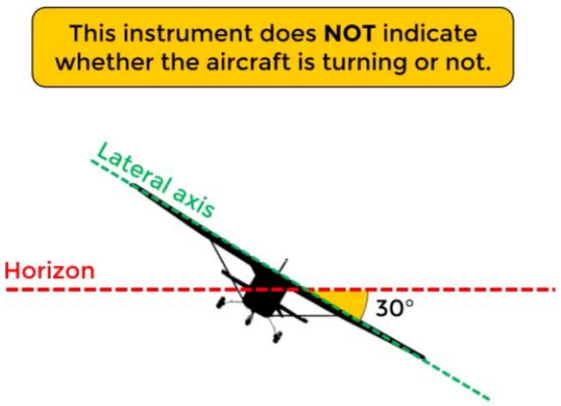
- the current bank angle of the aircraft is indicated by the **bank indicator** which in this case is this yellow bug.



- In the below image we can see that the wings of the aircraft are leveled so the bank angle indicated is 0°. Other markings to either side that represent 10,20,30,45, 60 and 90° of bank.



- This instrument indicates the bank angle only, which is the angle between the horizon and the lateral axis of the aircraft.
    - In this case the bank indicator is pointing the third marking in the right side of the banking scale which represents a 30-degree bank to the right.
    - This instrument does not indicate whether the aircraft is turning or not, it only indicates the bank angle.
    - The instrument used to determine if the aircraft is turning and how fast it is doing it is the turn coordinator.

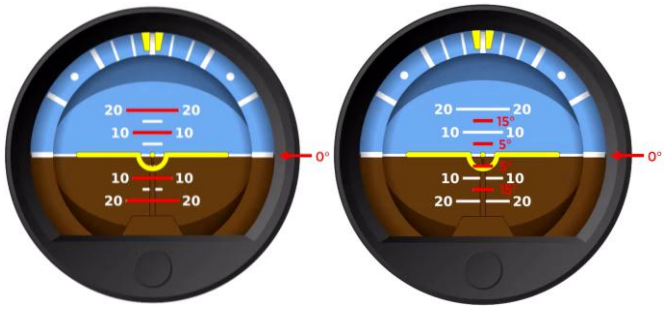




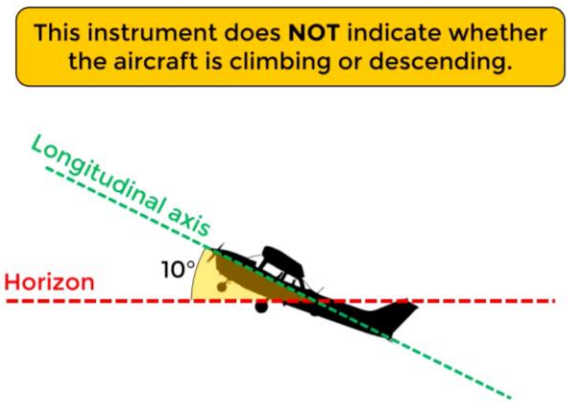
- Pitch Scale – The pitch angle is indicated by this little point (Pitch Indicator) in the middle of the miniature aircraft. In this case as we can see the point is right on the horizon line which indicates a pitch of 0 degrees on the pitch scale.



- The long lines represent increments of 10° of pitch either up or down while the short lines represent increments of 5 °.



- Attitude indicator indicates only the pitch angle i.e. the angle between the horizon and the longitudinal axis. In this case the pitch indicator is pointing 10° up in relation to the horizon.





Models and designs of the attitude indicator

There are different models and designs of the attitude indicator. Both analog and digital and although they have slight differences the information they provide is the same information.

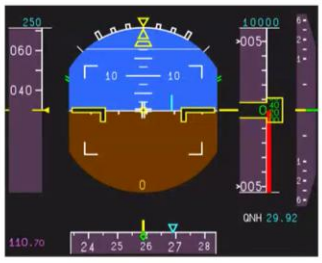
Analog







Digital







Examples of interpreting the instrument:



Pitch = 0°

Bank = 18° Left



Pitch = 8° Up

Bank = 0°



Pitch = 9° Up

Bank = 10° Right



Pitch = 9° Up

Bank = 20° Right

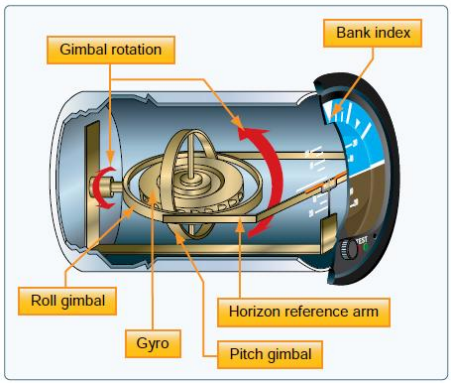


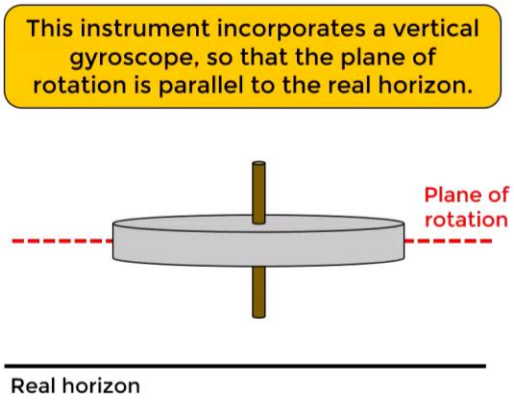
Pitch = 2° Down

Bank = 0°

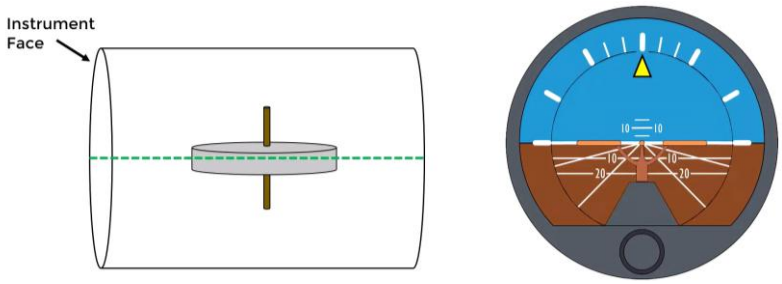
Working of Attitude Indicator

- The attitude indicator incorporates a vertical gyroscope so that the plane of rotation is parallel to the real horizon. It is mounted in a universal gimbal ring system, free about both the pitching and rolling axes of the aeroplane and is therefore able to remain spinning in a horizontal plane parallel to the true horizon, regardless of the rolling or pitching movements of the aeroplane around it.

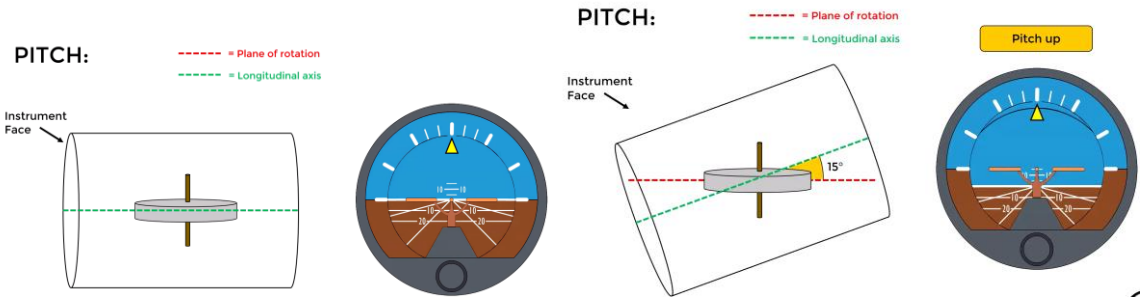




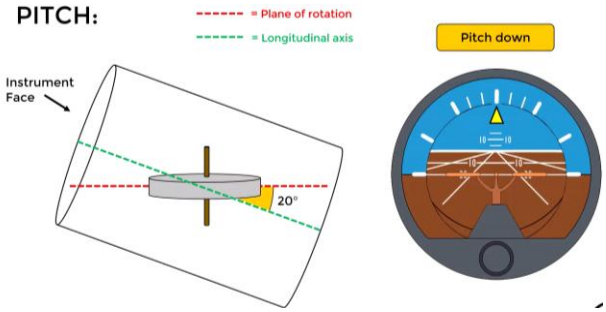
- The pitch attitude is measured using this gyro. On the right side we have the instrument as the pilot would see it, while on the left side we see the instrument from the side.
- Inside we have the gyro whose plane of rotation is parallel to the horizon. In this case the instrument is indicating a pitch attitude of 0°.



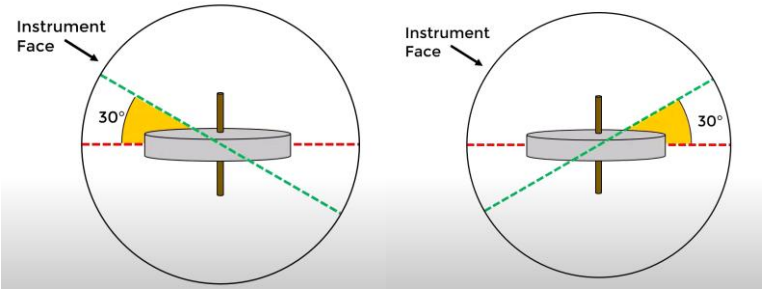
- When aircraft pitches up, both the aircraft and the instrument will pitch up. The gyro will remain rigid in space and aligned with the horizon. This way the instrument can measure and show the current pitch attitude of 15 degrees nose up.



- The opposite way, if the aircraft pitches down both the aircraft and the instrument will pitch down. While the gyro will remain rigid in space aligned with the horizon. Thus, measuring and indicating the current pitch attitude of 20 degrees nose down.



- Similarly for bank, the front of the gyro will have the following view

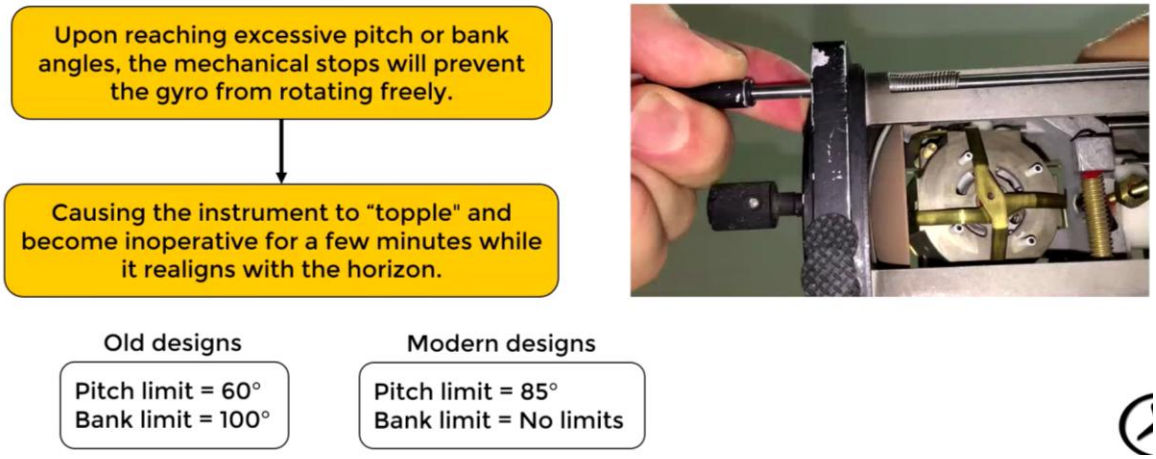


Limitations of the heading indicator

Gyro's degree of Freedom

- Theoretically the instrument's gyro has three degrees of freedom.
- In practice, mechanical stops limit its full range of rotation.

- When the aircraft reaches excessive pitch or bank angles, these stops prevent free gyro movement.
- This restriction can cause the instrument to topple and become inoperative for a few minutes while it realigns with the horizon.
- On older instruments it is common to find pitch limits of 60 degrees and bank limits of around 100 degrees. In the more modern ones we can find less restrictive limits of 85 degrees of pitch and no bank limits.



Below is an example of what happens if the pitcher bank limits are exceeded.



Vacuum driven limitation

The gyro of an attitude indicator that is driven by a vacuum system, requires 4 or more inches of mercury vacuum to operate and 5 minutes to get up to operational speed. An attitude indicator that is electrically driven is operational almost immediately.

Equipment maintenance and usage

Any force that disturbs the free rotation of the gyro will cause errors in the indications presented by the attitude indicator. Poorly balanced components, clogged filters, improperly adjusted valves, pump malfunctions, friction and worn parts will result in erratic and inaccurate indications.

Gyroscopic Apparent Wander

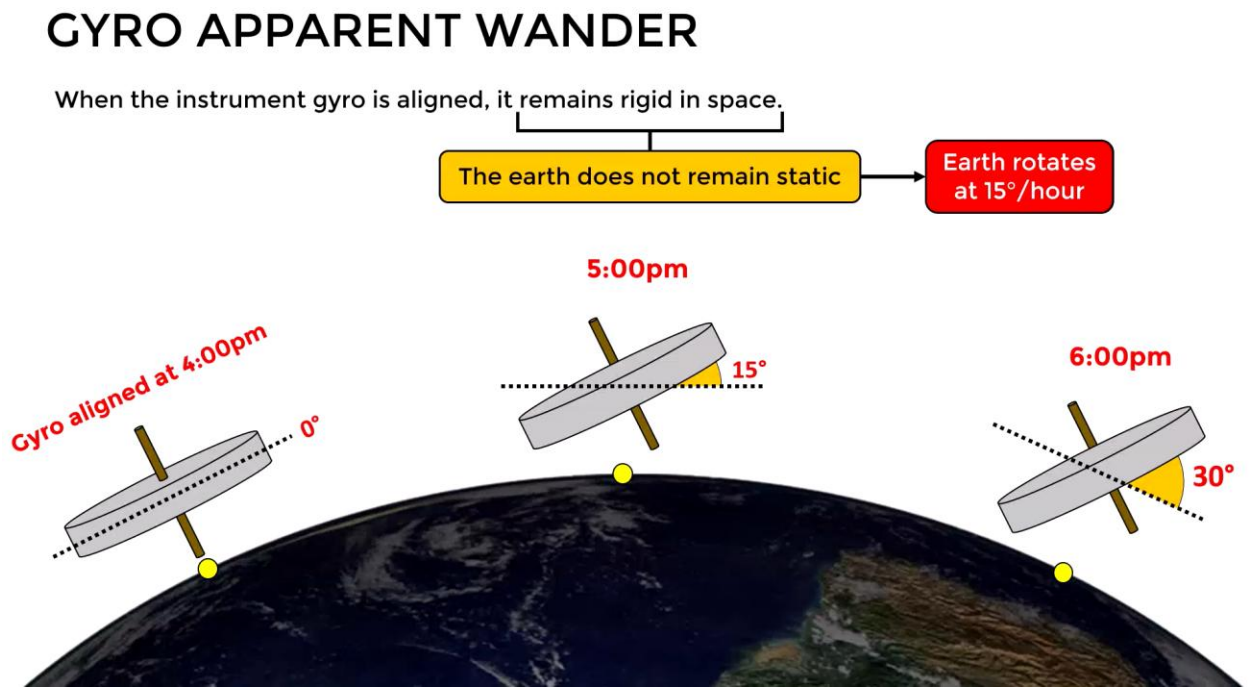
Another limitation that is present in this instrument is the gyroscopic apparent wander. When a gyroscope starts spinning it will tend to remain rigid in space. However, the earth does not remain stationary it rotates at about 15 °/ hour. This implies that the plane of rotation of the gyro will progressively drift from the real horizon.

In a vertical gyro such as the attitude indicator this effect is maximum at the equator and minimum at the poles.

The below image graphical example of why this happens. Let's say we are at the equator in this yellow point, and we turn on the instrument at 4 P.M. This way the plane of rotation of the gyro is aligned parallel to the horizon.

After one hour at 5 P.M. the earth will have rotated 15° while the gyro remains rigid in space in its original position. So, if we measure now the angle between the actual horizon and the plane of rotation of the gyro we will find that it is 15 ° which means in other words that the attitude indication of the instrument will be incorrect by 15°.

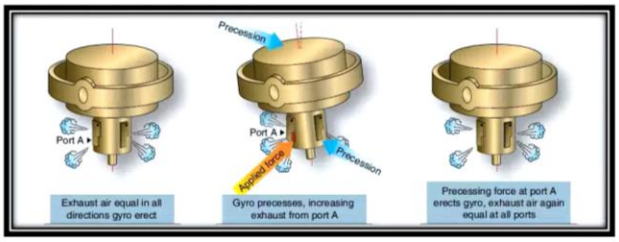
As we can see this effect worsens with time so at 6 P.M. the earth will have rotated another 15°. So, now the difference between the horizon and the plane of rotation of the gyro will be 30° and so on.



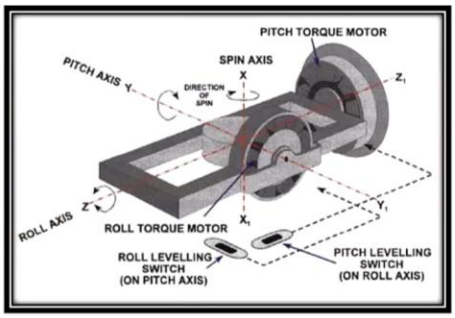
It is important that the plane of rotation of the gyro realigns itself constantly. So it remains parallel to the horizon. A control system helps that, however, that process behind this hasn't been explained.

CONTROL SYSTEM

It incorporates a mechanical control system that uses the earth's gravity to keep the plane of rotation parallel to the horizon at all times.



Pendolous Vanes



Electrical torque motor

Some electrical driven attitude indicators incorporate a caging knob. The knob allows decoupling the gyro from the instrument face this prevents the gyro from toppling during excessive attitudes or maneuvers. As the instrument will act as a free gyro without mechanical stops, this knob is also used to quickly realign the gyro with the horizon in case it is toppled. However, this procedure should only be done in straight and level flight or on the ground to ensure an adequate alignment.

CAGING KNOB

- Some attitude indicators (usually the electric ones) incorporate a knob that allows decoupling the gyro from the instrument face.
- This prevents the gyro from "toppling" during excessive attitudes or maneuvers, as it will act as a free gyro.



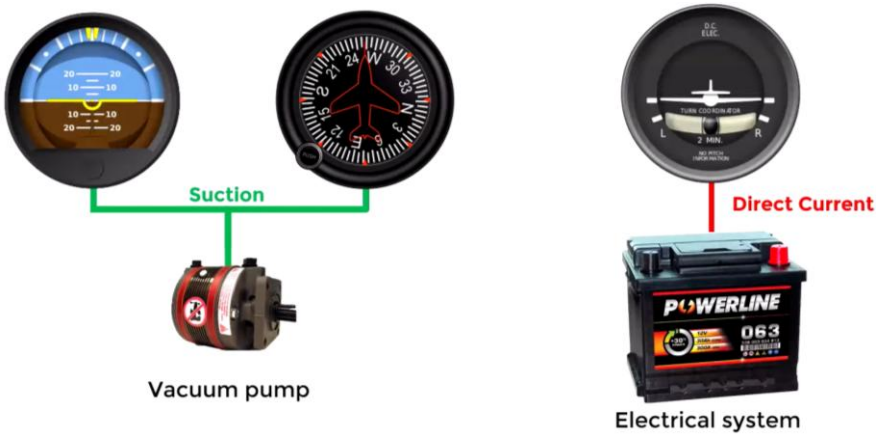
Caging knob

This knob is also used to quickly realign the gyro with the horizon in case it has "toppled".

This should only be done in straight and level flight or on the ground.

THE TURN & SLIP INDICATOR

The gyro of the turn coordinator is driven by direct current from the aircraft electrical system.



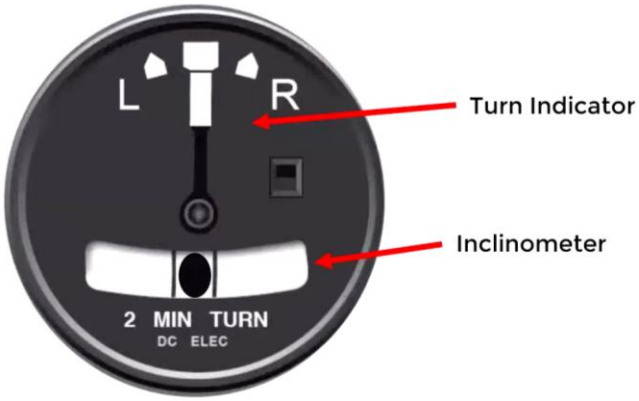
The instrument comes in two main variants:

- The turn and slip indicator
- The turn coordinator

Both have the same principle of operation, the information they provide, and the way it is presented are slightly different.

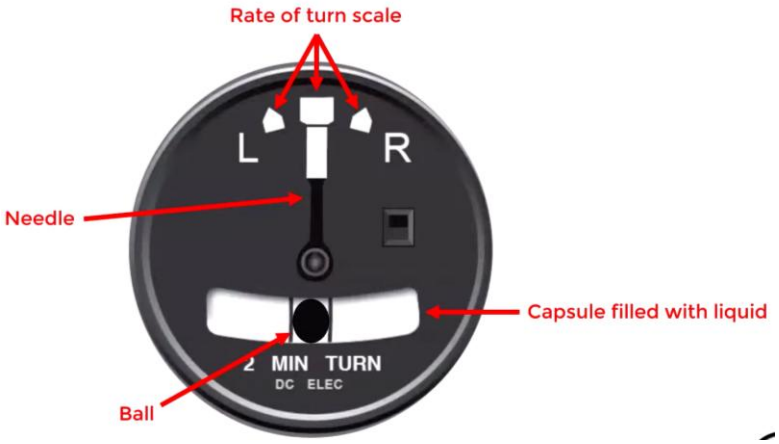
The turn and slip indicator

- Indicates whether the aircraft is making a turn i.e. direction of the turn.
- How fast the heading change is occurring which is the rate of turn.
- It does not indicate the bank angle of the aircraft.
- It also incorporates an inclinometer at the bottom; this allows to measure the quality of the turn.
- Quality of turn refers to, the aircraft is flying coordinated.



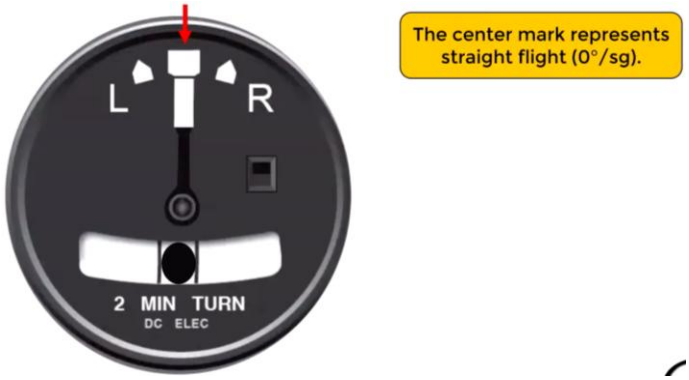
Parts of the turn and slip indicator

- In the upper part we have the rate of turn scale with its corresponding needle.
- In the lower part we have the liquid-filled capsule of the inclinometer and a solid ball which is free to move within the capsule.

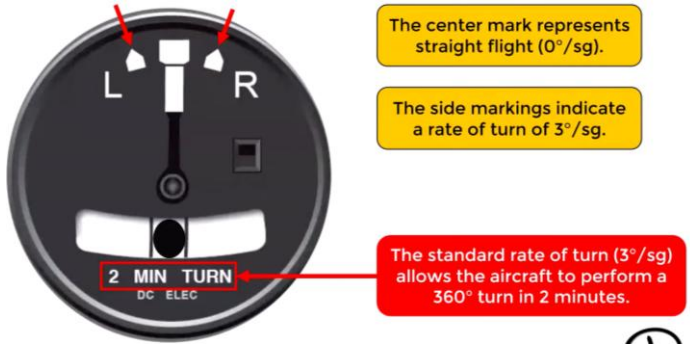


Rate of turn scale

- The center mark represents straight flight or in other words a rate of turn of 0°/sec.



- The side markings represent a rate of turn of 3° /sec, which in aviation corresponds to the standard rate of turn. It is mainly used in instrument flight procedures as well as in many maneuvers. This standard rate of turn allows the aircraft to make a 360 degree turn in two minutes, which is specified on the instrument as well.



Reading the Instrument

The needle is pointing the center mark which means that the aircraft is flying straight with a constant heading in other words the rate of turn is zero degrees per second.



Direction of the turn = Straight flight

Rate of turn = 0°/sg

If the needle moves to the right, we can see that the aircraft is turning to the right. Since, the needle is pointing exactly the side mark, this means that the rate of turn is three degrees per second.



Direction of the turn = Right

Rate of turn = 3°/sg

Similarly, if the needle points the left mark it will represent a turn to the left at 3° / second



Direction of the turn = Left

Rate of turn = 3°/sg

If the needle moves between the center and the left, we can see that the aircraft is still turning to the left. But, with a slower rate of turn in this case less than 3° / second.



Direction of the turn = Left

Rate of turn = Less than 3°/sg

if the needle moves further left to the left indicator, we can see that the aircraft is indicating a turn to the left with a rate of turn of more than 3° / second.



Direction of the turn = Left

Rate of turn = More than 3°/sg

Reading the inclinometer

- The inclinometer allows to measure the quality of the turn.
- It consists of a capsule filled with liquid and a ball that is free to move inside.
- If the ball remains centered in the capsule it will indicate a coordinated flight.



Coordinated turn to the right.

- If the ball moves to the side of the turn it will indicate a slipping turn and if the ball moves to the opposite side of the turn it will indicate a skidding turn.



Slipping turn to the right.



Skidding turn to the right.

- If one wing is permitted to drop, the ball will roll towards the side of the low wing. The needle in the image below shows the aeroplane to be flying straight, but the ball indicates it to be right wing low.



Right-wing low slip, straight flight.

- Usually in most procedures and maneuvers the objective is to maintain a coordinated flight.
- The pilot must keep the ball centered. To do so a rule of thumb is to step on the ball. This means that to correct a skidding or slipping turn the pilot must press the pedal on the side of the ball to center it.
- In a balanced turn, the ball will remain in the centre as centrifugal force offsets the pull of gravity. In a slip, there is not enough rate of turn for the amount of bank. The centrifugal force will be weak and this imbalance will be shown by the ball falling down toward the inside of the turn. In a skid, the rate of turn is too high for the amount of bank. The centrifugal force is too strong and this imbalance is indicated by the ball sliding toward the outside of the turn.

The turn coordinator

Turn coordinator is very similar to turn and slip indicator. However, the construction varies a bit. The gyro, instead of rotating about its horizontal axis as it does in the turn and slip, is canted at

approximately 35°. As a result, during movement about both the yaw and roll axes, precession causes the gyro mechanism to roll slightly.

TURN Coordinator - Basics
https://www.youtube.com/watch?v=EB-RnavOWRA

Turn and Slip VS Turn Coordinator
https://www.youtube.com/watch?v=1v4SK3dK1fA

The gyro gimbal is connected to a miniature aeroplane on the face of the instrument. This little aeroplane tilts to inform the pilot of the rate of both yaw and roll. When the aeroplane is turning left or right, the miniature aeroplane banks in the direction of the turn. When the wing of the aeroplane is aligned with one of the marks, the aeroplane is in a standard rate, or rate one, turn.

The ball indicator gives the same indications as the ball indicator of the turn and slip indicator.

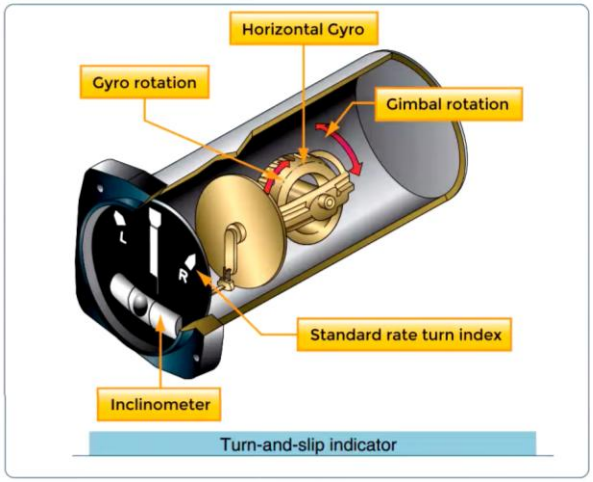
A significant advantage of the turn co-ordinator is that it can be used to keep the wings level in straight flight if the attitude indicator has failed. The wings of the turn co-ordinator will react first to roll and then to yaw. The turn co-ordinator shows rate of roll as well as rate of turn whereas the turn and slip indicator shows only rate of turn.

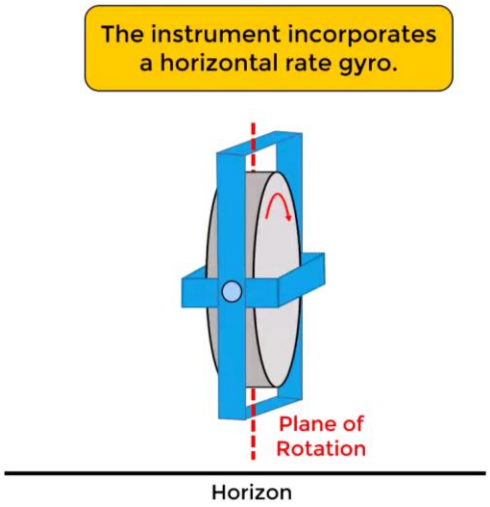
Working of the Turn and Slip Indicator

Ep. 59: How a Turn Coordinator Works | Inner Workings
https://www.youtube.com/watch?v=a4iLtZPp_-8

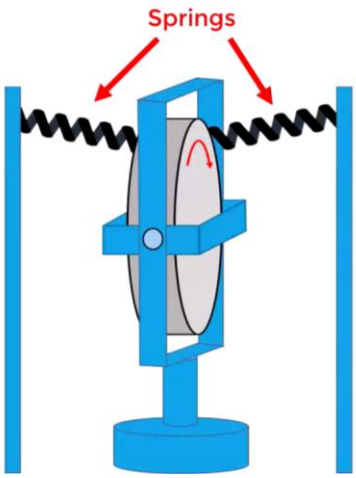
The turn needle is actuated by a gyro wheel operated either electrically or by a Venturi tube or vacuum pump. Free air entering through a nozzle impinges on the paddles of the gyro wheel, causing it to rotate at approximately 9,000 rpm. The gyro wheel is mounted vertically and rotates about its horizontal axis.

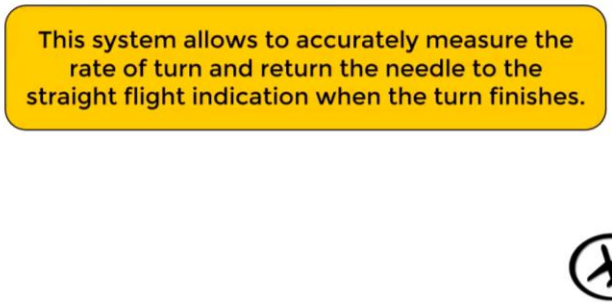
The turn and slip indicator uses the principle of gyroscopic precession to measure the rate of turn.





The spinning gyro wheel, or rotor, is mounted in a gimbal ring. When the aeroplane turns to the right or left, the gyro wheel precesses about its turning axis and rolls the gimbal ring. The rolling motion of the gimbal ring in turn rotates the turn needle on the face of the instrument. A spring returns the gyro to neutral when the aeroplane ceases to turn.





THE GYROSYN COMPASS

To be studied with compass, as par of the Air Navigation Module with The Earth's Magnetism.

2.2.3 EFIS INSTRUMENTS

An Electronic Flight Instrument System (EFIS) is a cockpit instrumentation system that uses electronic rather than electromechanical technology. EFIS normally consists of two panel-mounted display units: a primary flight display (PFD) and a multifunction display (MFD). The liquid crystal display (LCD) used for these systems has resulted in their being referred to as glass cockpits, or glss panels.

Glass cockpits have been credited with improving flight safety by enhancing situational awareness for pilots during flight. With traditional analog instruments, pilots are required to monitor six in dividual instruments, commonly referred to as the "six-pack.". Glass cockpits, by contrast, integrate all six instruments into one display rendering the pilot's monitoring of the aircraft quicker and more efficient.

THE PRIMARY FLIGHT DISPLAY (PFD)

The PFD presents information dedicated to depicting all aspects critical to flight, including airspeed, altitude, heading, attitude, vertical speed and yaw. It displays on one LCD unit what the traditional pitot static and gyro analog instruments require six individual instruments to show.



PFD systems are driven by ATTITUDE AND HEADING REFERENCE SYSTEMS (AHRS). AHRS are three-axis sensors that provide heading, attitude and yaw in formation for the aircraft.

Gyroscopes: Measure the rates of rotation around the three axes (roll, pitch, and yaw).

Accelerometers: In an Attitude and Heading Reference System (AHRS), the accelerometer acts as a long-term, stable reference for the horizon. By measuring the direction of gravity, it helps calculate the device's pitch and roll angles, correcting for a common problem in gyroscopes called "drift".

Think of an accelerometer as a ball in a box. The accelerometer measures the force of gravity, and based on which "wall" of the box the ball is pressing against, the system can tell its orientation.

- Standing still and level: When the AHRS is flat and not moving, the accelerometer senses only the force of gravity pulling straight down. The system interprets this signal as a level pitch and roll.
- Tilting up or down (pitch): If you tilt the device up or down, the force of gravity on the accelerometer's internal components will change. By measuring this change, the AHRS can calculate how far the device is tilted relative to the ground.
- Rolling side-to-side (roll): Similarly, if you roll the device left or right, the accelerometer's sensors will detect a different pull from gravity, allowing the AHRS to determine the roll angle.

Magnetometers: Provide a stable, long-term reference for magnetic heading, which helps to correct for drift in the gyroscope data.

They consist of either solid-state or gyroscopes, accelerometers and magnetometers that drive the attitude indicator and heading indicator. They replace mechanical gyroscopic flight instruments that are found in aircraft equipped with traditional analog (i.e., round dial or "steam gauge") flight instruments.

When combined with high-speed microchips that process air data, AHRS form an AIR DATA ATTITUDE AND HEADING REFERENCE SYSTEM (ADAHRS). An aircraft with ADAHRS has three rate sensors for each axis, and three accelerometers for each axis. This provides additional information such as airspeed, altitude and outside air temperature, thus enabling the PFD to provide the full representation of analog instruments in one electronic LCD display.

PFD systems typically have built-in means by which to alert pilots of potentially hazardous conditions. By changing the colour of a display, by changing its shape, or by triggering an audio alert, any condition (such as too low an airspeed) can be immediately brought to the pilot's attention. A system failure, such as the failure of an input sensor, may (depending on the type of PFD unit) drop a warning bar across data that is no longer valid on the display.

THE MULTIFUNCTION DISPLAY (MFD)

The MFD displays the constant relationship between the aircraft and its environment. Its most significant use is for displaying of navigation and weather information. It typically displays GPS-derived moving maps upon which aircrew can overlay different information. Information overlayed on the MFD may include the aircraft's current route plan, weather information, terrain and obstacle depiction, restricted airspace and aircraft traffic. The MFD can also display information about aircraft systems, such as fuel status and electrical system status. Furthermore, it can display engine monitoring data, and constantly scan for out-of-range conditions. Like the PFD, the MFD is a single multi-function display, usually in colour, and is generally identical in size and shape to the PFD. The MFD can be thought of as a cockpit management system designed, like the PFD, to reduce pilot workload and to enhance "eyes up" operations.

PFD and MFD features are accessed via softkeys and knobs that surround their respective LCD screens. Some MFD units have touchscreen functionality.



Fig.2.65 Multifunction Display (MFD) Cockpit Management

Back-Up EFIS Display

Aircraft with glass cockpits must have back-up flight instruments. Traditionally, these were mechanical analog instruments, but newer aircraft often use back-up EFIS (Electronic Flight Instrument Systems). These back-up EFIS units act as compact Primary Flight Displays (PFDs) and operate independently using emergency power, allowing pilots to safely reach an alternate or intended airport if the main system fails. Advanced back-up EFIS displays include key flight data such as attitude, altitude, airspeed, and more, and can also serve as a cross-check tool during normal operations.

The Heads-Up Display (HUD)



An extension in the development of EFIS technology is a system that projects the displayed information onto the front windscreen of the aeroplane. This heads-up display (HUD) system permits

the pilot to monitor both the instrumentation and the progress of the aeroplane, especially during the approach and landing, while always having their eyes looking forward and over the nose of the aircraft. In addition, thermal imaging systems are now developed that project real-time, real-world images of the view outside the cockpit window on a heads-up display in the cockpit. Aircraft with such advanced technology allow pilots to see through fog and darkness, providing for greater safety during all phases of very low visibility flight operations.

SMARTPHONES & TABLETS

Personal smartphones and tablets, such as iPads and Android devices, can support various cockpit applications, effectively replicating panel-mounted instrument capabilities. These devices offer features like attitude, airspeed, altitude, heading, compass, vertical speed, navigation, weather updates, and flight planning. This functionality turns them into personal backup glass cockpits, helping pilots maintain situational awareness during both normal and emergency conditions.

Like the glass panel-mounted EFIS systems used as primary instrumentation in aircraft, smartphones and tablets that provide attitude and heading reference system (AHRS) data typically use microelectromechanical system (MEMS) gyros to generate their screen information. In addition to providing information such as pitch and bank, these handheld units can also provide automatic dependent surveillance broadcast (ADS-B) information, wide area augmentation system (WAAS) GPS data, and air traffic and weather information. Airspeed data can be displayed via connectivity to the aircraft's pitot and static ports, or it can be GPS derived. Engine instrumentation data can also be integrated into the units. Airport information and flight plan filing are also served by use of smartphone and tablet technology.

Smartphone and tablet solutions for the cockpit also have the power to display synthetic vision systems (SVS) that provide a 3-D reality of the environment in which an aircraft is flying. Sometimes referred to as "highway in the sky" technology, this feature provides situational awareness by linking displays to databases of known terrain and obstacles. The navigation component of a synthetic vision system is achieved through the use of GPS and inertial reference systems.

Some avionics manufacturers enable connectivity between smartphone or tablet apps and panel-mounted EFIS displays, allowing pilots to transfer data such as flight plans from personal devices to the aircraft's avionics system. Additionally, smartphones and tablets can be configured with split-screen displays to show attitude information alongside weather or navigation data. Users can also choose between landscape and portrait orientations, depending on their preferences and the app capabilities.

Smartphones and tablets must never be used as primary flight instruments in place of panel-mounted equipment. They are best used as personal back-up instruments only, as flight support solutions.

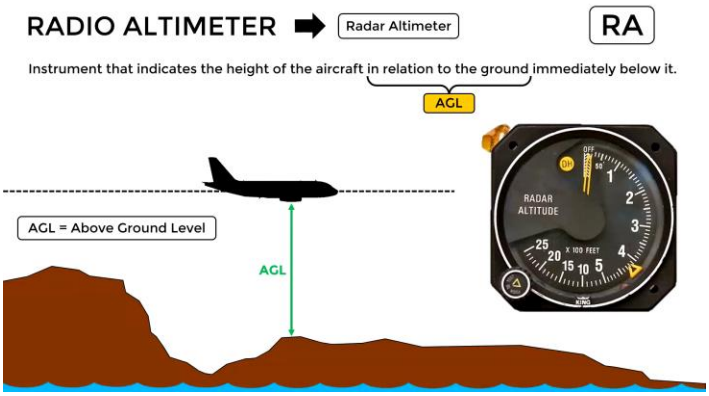
2.2.4 THE RADAR ALTIMETER

HOW A RADIO ALTIMETER WORK

https://www.youtube.com/watch?v=CGSK8wUJRqQ

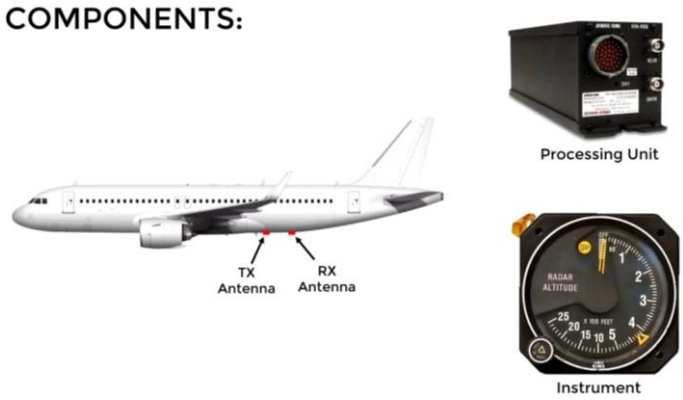
The Radar altimeter OR Radio altimeter ("RA")

An instrument that indicates the height of the aircraft in relation to the ground immediately below. In other words, it indicates the Above Ground Level ("AGL") height of the aircraft in real time.



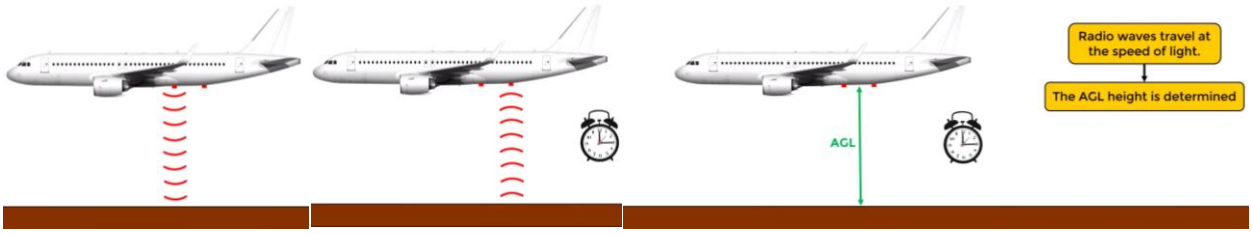
Components

- Two antennas located at the bottom of the fuselage one transmitting and the other receiving
- a central processing unit
- the instrument or indicator in the cockpit



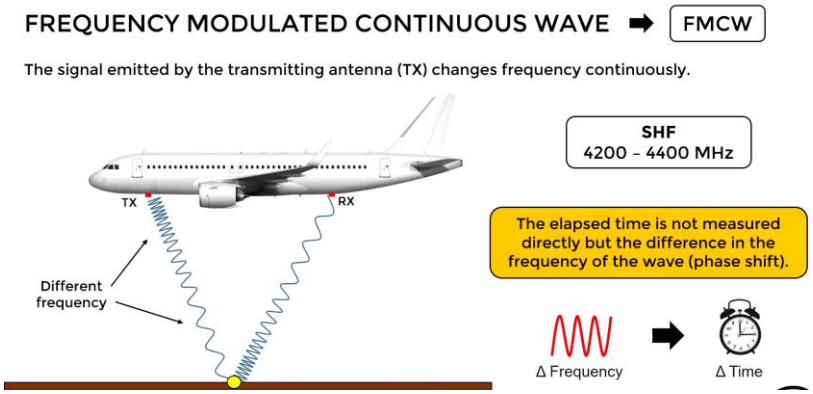
Working

Basic Working - The transmitting antenna emits a radio wave downward from the aircraft when this wave reaches the surface it bounces back to the aircraft and is received by the receiving antenna. The processing unit calculates the time it took for the wave to go and return. The radio waves travel at a known speed which is the speed of light. The system can calculate the current height of the aircraft above ground level.



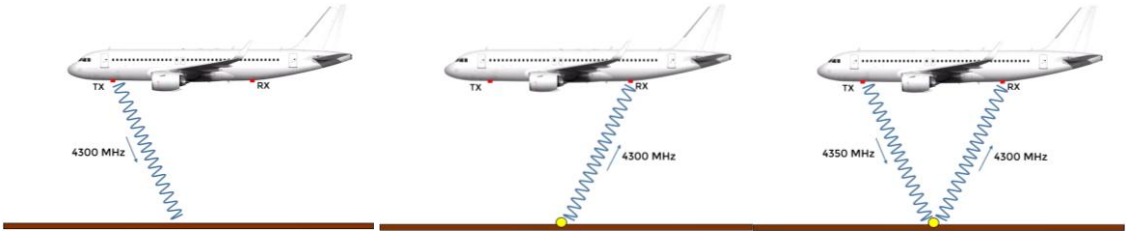
The above was the basic principle of operation modern radio altimeters. They work in a slightly different way. A FREQUENCY MODULATED CONTINUOUS WAVE (FMCW) system is used. Let's see how this system works:

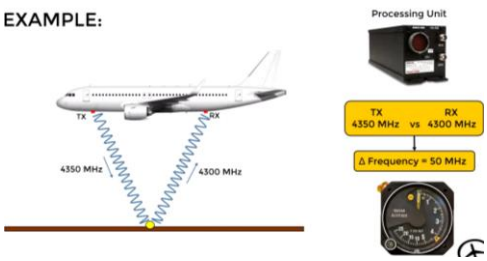
- The signal emitted by the transmitting antenna changes its frequency continuously within the range of 4200 and 4400 MHz.
- With this system the elapsed time is not measured directly but the difference in the frequency of the wave or in other words the phase shift.
- Since the rate at which the wave changes frequency is known the time it takes for the wave to go and return can be determined indirectly.



The above explanation is quite complex, let's understand this with an example.

- The transmitting antenna emits a radio wave with a frequency of 4300 MHz this wave reaches the surface bounces back to the aircraft and is received by the receiving antenna.
- By the time this antenna receives this wave the transmitting antenna is already emitting another wave with a different frequency in this case 4,350 MHz
- The system compares the frequency of the received wave with the frequency of the wave that is being emitted. The difference in frequency between these two waves allows the system to calculate the time it took for the wave to go and return, and therefore also calculate the current height of the aircraft above ground level.

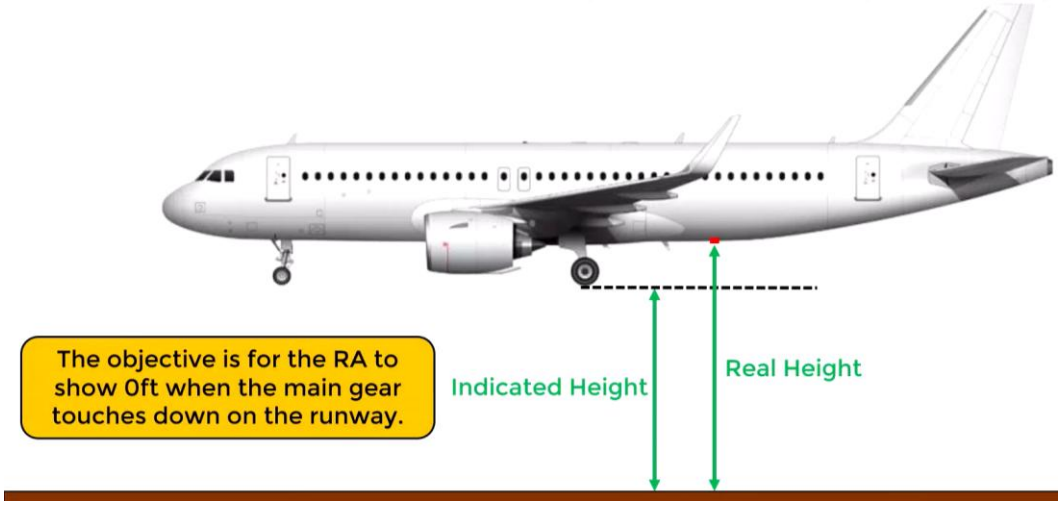




These waves travel at the speed of light so this process occurs almost instantaneously several times per second making the system highly accurate and reliable.

Residual height

The system is calibrated to show the height between the main landing gear and the ground. The difference in height between the antenna and the landing gear is known as residual height.

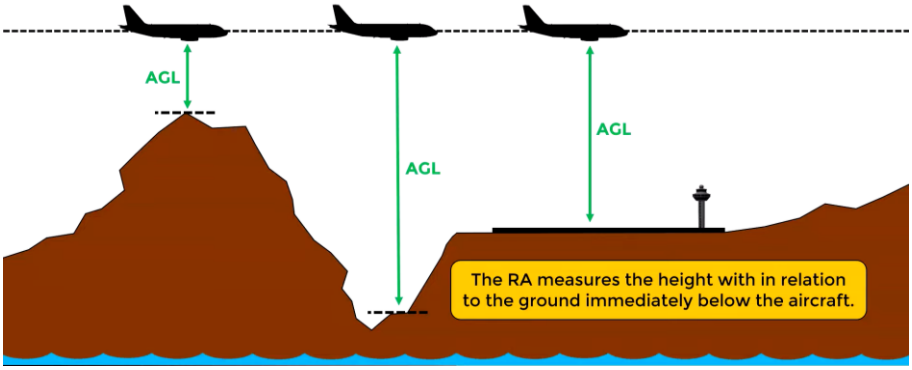


Some aircrafts that may have different residual heights depending on the design of the landing gear.

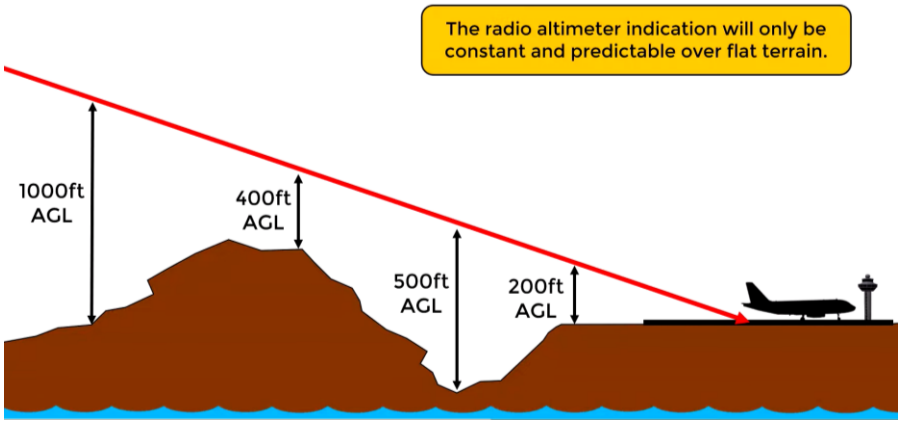
For example, the Boeing 747, in this case the main landing gear is tilted during the approach giving as a result a relatively high residual height. However, when the plane lands the main landing gear is compressed causing the residual height to decrease considerably. Since, the radio altimeter is calibrated with the residual height of the approach it may show negative values when on the ground.



When flying over irregular terrain the radio altimeter indication may vary considerably.



The radio altimeter indication will only be constant and predictable over flat terrain.



Parts of RA

- The height scale which normally is expressed in terms of hundreds of feet.
- The needle that indicates the current height.
- an adjustable height reference
- A height selection knob
- An altitude warning light



Some analog radio altimeters also incorporate a red flag which is visible when the instrument is inoperative or when the signal has too much interference thus indicating that the instrument must not be used.



Visible when the instrument is off, inoperative or when the signal has too much interference.

Reading the instrument

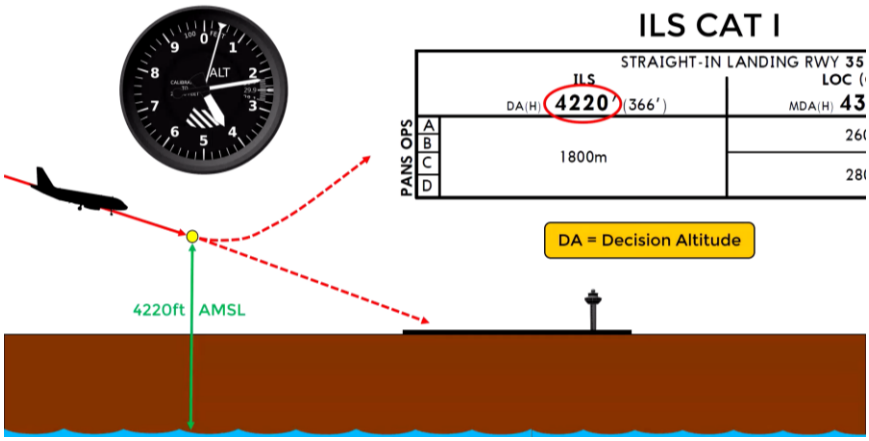
- The scale of the instrument usually goes up to 2500 feet
- When flying above 2500 feet above the ground the needle will not be visible or it my return to the zero.



There are many models developed by various manufacturers and can come in both digital and analog versions.

Use of RA in instrument approaches

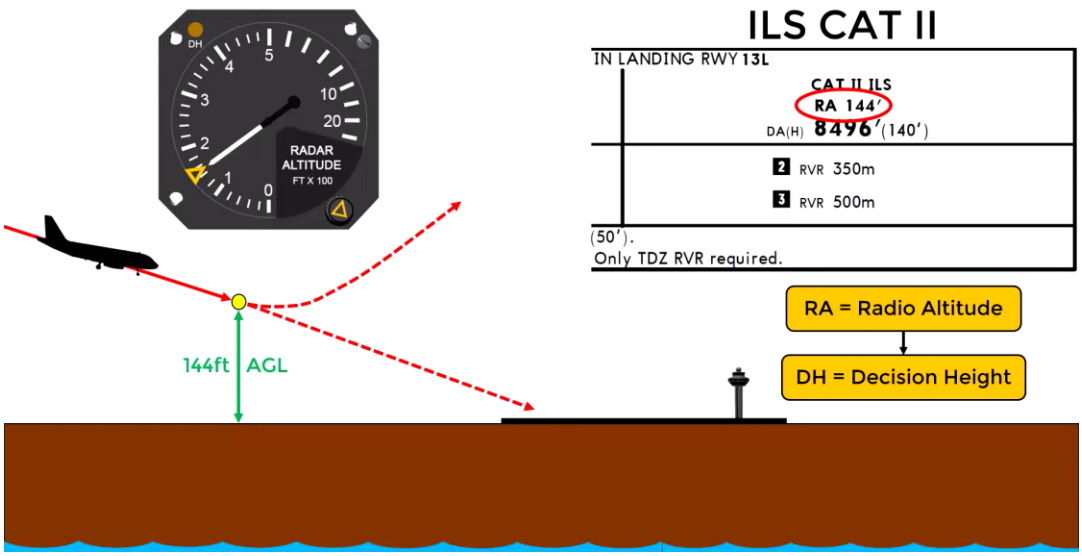
In the example below, we have an ILS category I (CAT I) procedure. In this case the minimums are specified in terms of altitude in relation to mean sea level, that's why it is called Ddecision Altitude (DA). This implies that using a barometric altimeter when the aircraft reaches 4220 feet during the procedure, the crewmust decide whether to continue the landing or perform a missed approach. So, as we could see in CAT I approaches a radio altimeter is not required and the procedure can be performed using a regular barometric altimeter.



CAT II and III approach

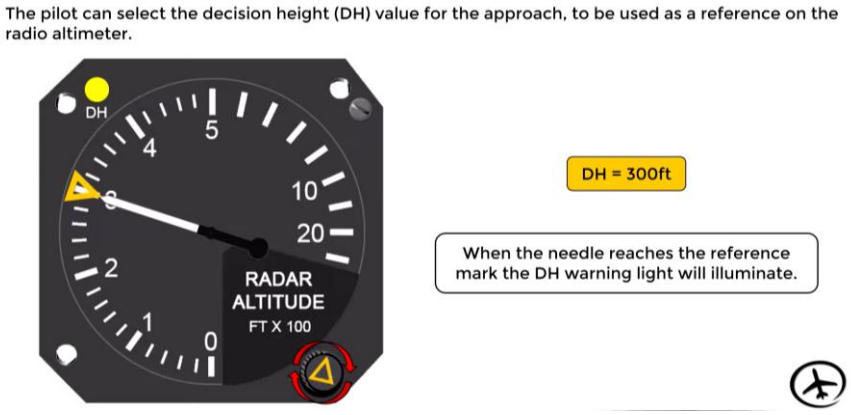
In CAT II and III approach the minimums are specified in terms of radio altitude. This means that instead of having a decision altitude expressed in relation to mean sea level there is a decision height expressed in relation to the ground. In the below procedure, when the aircraft reaches 144 feet indicated in the radio altimeter the crew must decide whether to continue the landing or perform a missed approach.

The radio altimeter is an important instrument for CAT II/III precision approaches.



Height Selection Knob

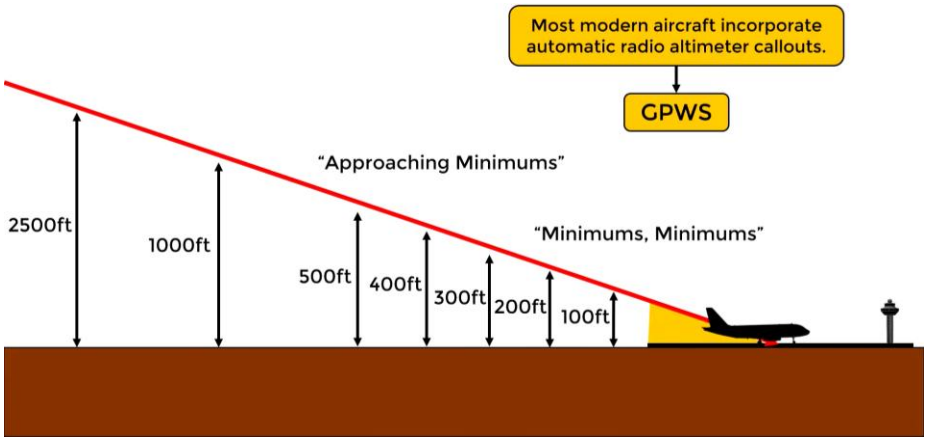
To identify more easily on the RA the exact decision height, the crew can use the height selection knob. Based on the below image, the pilot would adjust the decision height of the procedure using the height selection knob. For example, let's say that the decision height to be used is 300 feet then the pilot adjusts the reference height accordingly. Now, when the needle reaches the reference mark the warning light will illuminate and it can be accompanied by a call out like “minimums minimums” indicating to the crew that it is at this point that they must decide whether or not to continue the approach.



Radio Altimeter Callouts

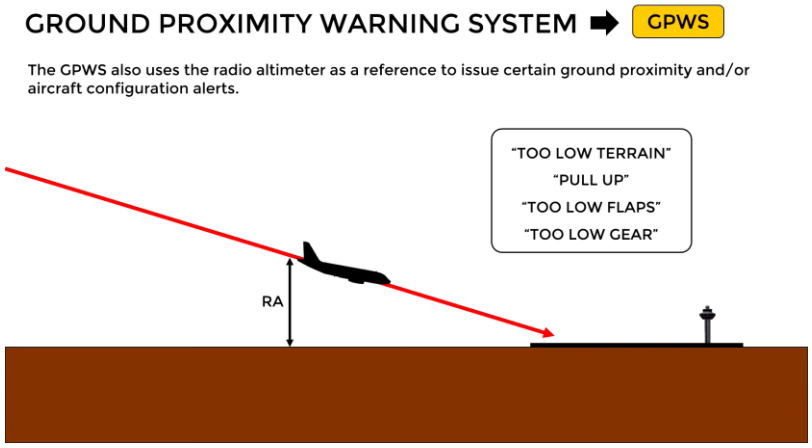
Most modern aircraft incorporate automatic radio altimeter callouts which are part of the Ground Proximity Warning System (“GPWS”). This system emits call outs at certain heights in relation to the ground such as, 2500, 1000, 500, 400, 300, 200, 150, 100, 40, 30, 20, 10….

- The main objective of these is to indicate to the pilot the current height of the aircraft above the ground without the need to look directly at the radio altimeter. During an approach normally the pilot's attention is focused in other parameters.
- Apart from these call-outs there might be others such as “*approaching minimums*”, “minimums minimums” which are directly related to the decision height selected by the pilot.



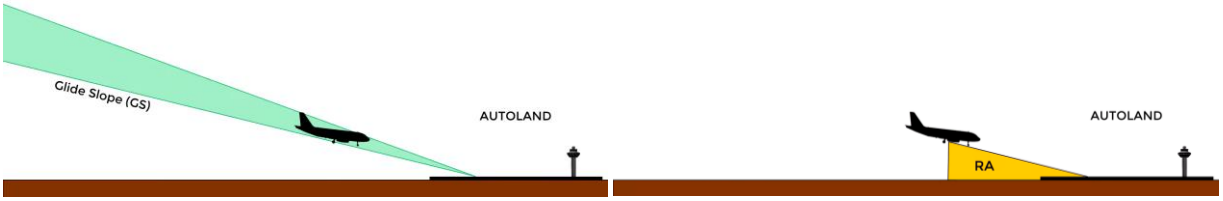
Ground Proximity Warning System (“GPWS”)

GPWS also uses the radio altimeter as a reference to issue certain ground proximity and or aircraft configuration warnings, for example, an aircraft is approaching to land but the crew forgets to extend the flaps and the landing gear. In this case when the aircraft descends below a certain height above ground level the gpws will issue call-outs like these *”too low flaps”*, *“too low gear”*, *“too low terrain pull up”*.



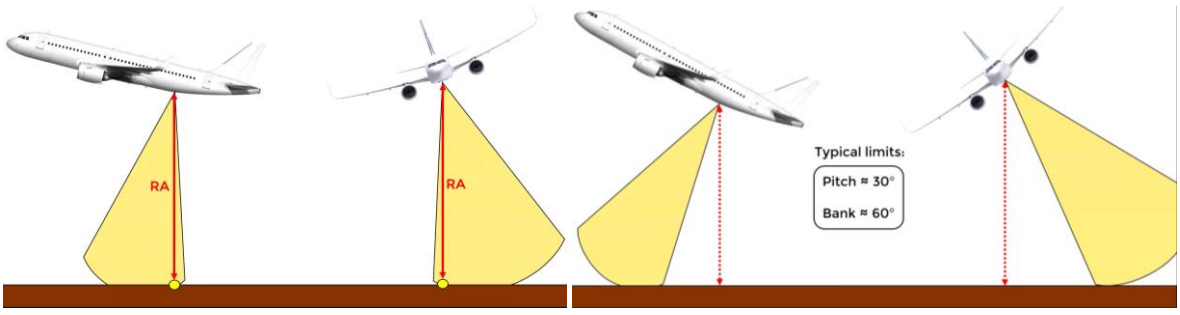
Automatic Landing System AKA Autoland

The aircraft uses the glide slope to descend in the correct approach path. However, from a certain height above the ground the system stops using the glide slope and uses only the information from the radio altimeter to complete the landing.

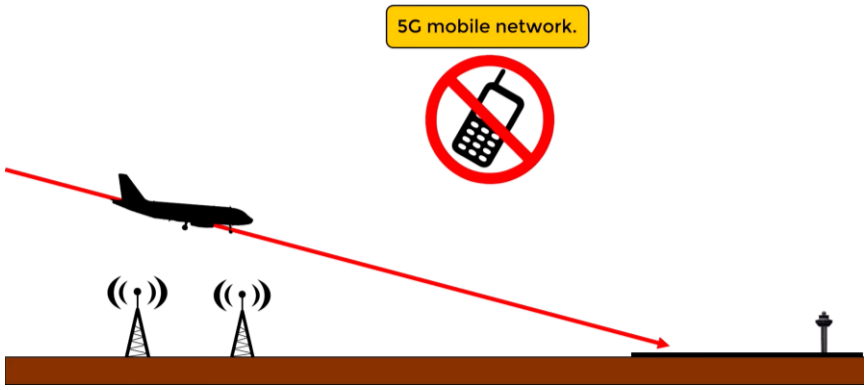


Limitation of RA

The waves are emitted downward in relation to the fuselage. There are certain pitch and bank limits. Typically the pitch limit is around 30° and the bank limit is around 60°, but this will vary depending on the design of the system. In the image below, during normal maneuvers the radio altimeter can determine correctly the height of the aircraft. However, if the pitcher bank limits are exceeded then the waves emitted by the system are not able to determine the current height of the aircraft so there will be an error in the indication.



Another problem is that when flying near antennas or when having devices on board that use the same frequency band of the radio altimeter, they may interfere with the system. A clear example of this is the 5g mobile network which uses part of the frequency band of the radio altimeters.



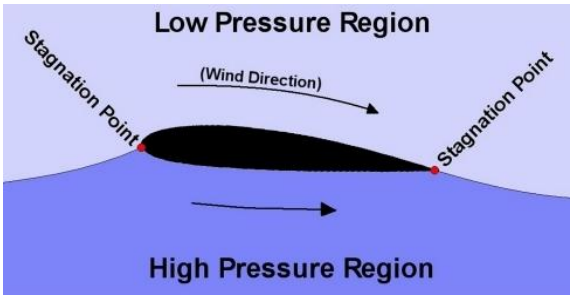
Additional benefits

- Radar altimeters are also used extensively by aeroplanes engaged in specialized types of flying, such as aerial survey, agriculture, etc., to maintain positivealtitude above the ground regardless of rolling terrain, hills or valleys.
- The radar altimeter is also useful in determining the position of pressure systems. By comparing the readings of the radar altimeter and an altimeter set to pressure altitude, current changes in atmospheric pressure may be determined. In combination with pilot's basic knowledge of the characteristics of pressure systems (i.e. highs and lows), a pilot is able to determine which direction to fly around a pressure system to take advantage of the best winds.

2.2.5 The Angle of Attack Indicator

The angle of attack indicator provides for the pilot a continuous read-out of the margin above stall, regardless of aircraft weight, C.G. location, angle of bank or any of the other factors that affect indicated stall speed.

As a wing moves forward through the air, the airflow is divided into one section that flows over the wing and one section that flows under the wing. The point at which the airflow separates is known as the stagnation point. The stagnation point moves chord wise on the undersurface of the wing as the angle of attack changes. It moves forward as angle of attack decreases and aft as angle of attack increases to a maximum aft position at which the wing stalls.



The angle of attack indicator senses the changing position of the stagnation point by means of a sensing unit, or pressure probe, that is installed on the wing slightly beneath the leading edge. Jn some angle of attack indicator designs, a springloaded vane moves up and down as the stagnation point changes position and relays this information to the cockpit display. In other angle of attack indicator designs, air pressure passes from the sensing probe through to pressure sensors co-located with an air data computer microprocessor. This processing unit translates the pressure readings into electronic signals that are sent to the cockpit indicator.

Most angle of attack indicators are designed such that green bars illuminate at the bottom of the display as the angle of attack increases. Additional green bars illuminate as the angle of attack continues to increase. If the angle of attack moves towards a critical point, the green bars will turn to yellow as a caution to the pilot. Ultimately, if the wing's angle of attack is advancing close to a stall, the display bars will switch to red.

An angle of attack indicator does not relate to airspeed and can therefore give a continuous, accurate readout of margin above stall, no matter what the flight attitude. It is an especially useful instrument during take-off, landing and steep banking maneuvers.



Fig.2.67 Example of Angle of Attack Indicator

The Angle of Attack Indicator (AOA)

https://www.youtube.com/watch?v=wjbfluqyGhc

2.2.6 MACH INDICATOR

Mach indicator shows the ratio of an aircraft's airspeed to the local speed of sound (Mach number).

It works by comparing dynamic (pitot) and static pressures. The instrument contains two aneroid capsules inside a sealed case linked to the static system:

- One capsule is connected to dynamic (pitot) pressure, responding to both static and dynamic inputs → measures airspeed.

- The other capsule is sealed, partly evacuated, and reacts only to static pressure → measures altitude.

Changes in pressure cause the capsules to expand or contract.

Their movements are mechanically linked to a pointer, which displays the Mach number on the instrument face.

How a Machmeter Work

https://www.youtube.com/watch?v=eRykFpkjy2k
